A NECESSARY AND SUFFICIENT CONDITION FOR EDGE UNIVERSALITY OF WIGNER MATRICES

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Abstract. In this paper, we prove a necessary and sufficient condition for Tracy-Widom law of Wigner matrices. Consider $N \times N$ symmetric Wigner matrices H with $H_{ij} = N^{-1/2}x_{ij}$, whose upper right entries x_{ij} $(1 \le i < j \le N)$ are *i.i.d.* random variables with distribution μ and diagonal entries x_{ii} $(1 \le i \le N)$ are *i.i.d.* random variables with distribution $\widetilde{\mu}$. The means of μ and $\widetilde{\mu}$ are zero, the variance of μ is 1, and the variance of $\widetilde{\mu}$ is finite. We prove that, for real symmetric Wigner matrices, $\lim_{s\to\infty} s^4\mathbb{P}(|x_{12}| \ge s) = 0$ is necessary and sufficient for the edge universality of the largest eigenvalue λ_N. If it holds, the distribution function of $N^{2/3}(\lambda_N - 2)$ is that of the GOE up to a shift $N^{-\delta}$ and an additive error $o(1)$ uniform in s, so that, by the theorem of Tracy and Widom [44, 45], it converges to the Tracy-Widom law F_1. If it fails, $N^{2/3}(\lambda_N - 2)$ is not tight. The proof, including the results of [15], [16], [26] that it uses, is formalized in Lean 4 [54]; the Tracy-Widom limit of the GOE is taken from [44, 45] and is not part of the formalization.

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1. Background and Main result

Since the groundbreaking work by Wigner [53], it has been conjectured and widely believed that local statistics of eigenvalues of random matrices are universal in the sense that it depends only on the symmetric class of the ensembles. The universality is one of the most important concepts in random matrix theory, and it can roughly be divided into two different types, the bulk universality and the edge universality.

Before considering the edge universality, which we will study in this paper, we roughly introduce some important results on bulk universality. The bulk universality concerns the local statistics of eigenvalues in the interior of the spectrum. In the early works of Wigner, Dyson, Gaudin, and Mehta [35, 36, 11, 12], it was proved that, after proper rescaling, the joint probability density of eigenvalues of Gaussian Unitary Ensemble(GUE) can be explicitly described by the sine kernel, and they conjectured the universality holds for more general classes of ensembles. For a very general class of invariant ensembles, the bulk universality was proved by Deift et. al. [13, 14], Bleher and Its [8], and Pastur and Shcherbina [40]. Later by Johansson [29], the bulk universality was proved for Gaussian divisible ensembles. (See also the work by Ben Arous and Péché [6].) For general Wigner matrices, a new approach was introduced to prove the bulk universality in a series of papers by Erdős, Schlein, Yau, and others in [17, 18, 19, 20, 21, 24, 25, 26, 15, 16]. The bulk universality for Wigner matrices was also obtained by Tao and Vu [50]. See the reviews [22, 23] for further discussion.

The distribution of the largest eigenvalue exhibits another type of universality, which is called the edge universality. Let λ_N be the largest eigenvalue of a Wigner matrix. For the Gaussian ensembles, the distribution function of λ_N was first identified by Tracy and Widom [44, 45]. More precisely, it is proved that

$$\lim_{N\to\infty} \mathbb{P}(N^{2/3}(\lambda_N - 2) \le s) = F_\beta(s), \tag{1.1}$$

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where the Tracy-Widom distribution functions F_β can be described by Painlevé equations, and $\beta = 1, 2, 4$ corresponds to Orthogonal/Unitary/Symplectic ensemble, respectively. The joint distribution of k largest eigenvalues can be expressed in terms of the Airy kernel, which was shown by Forrester [27]. If we denote the k largest eigenvalues by $\lambda_N, \lambda_{N-1}, \cdots, \lambda_{N-k+1}$, then for Gaussian ensembles, the joint distribution function of rescaled eigenvalues has the limit

$$\begin{aligned} &\lim_{N\to\infty} \mathbb{P}\left(N^{2/3}(\lambda_N - 2) \le s_1, N^{2/3}(\lambda_{N-1} - 2) \le s_2, \cdots, N^{2/3}(\lambda_{N-k+1} - 2) \le s_k\right) \\ &= F_{\beta,k}(s_1, s_2, \cdots, s_k), \end{aligned} \tag{1.2}$$

which will also be called the Tracy-Widom distribution.

The condition for (1.2) has been studied intensively. In the direction of sufficient condition, it has been improved as follows: The result (1.2) was first extended to general Wigner matrices by Soshnikov [47] with the condition that all odd moments of matrix entries vanish (e.g. the symmetric distribution) and with the Gaussian decay. Ruzmaikina [43] showed that the Gaussian decay can be replaced with polynomial decay faster than x^{-18}. Also, under the condition that matrix entries are symmetrically distributed, Khorunzhiy [32] proved a bound for the spectral norm for the matrices whose entries have finite $12 + o(1)$ moment. For the non-symmetric case, (1.2) is proved in [51] by Tao and Vu with the condition that matrix entries have vanishing third moment and sub-exponential decay. (Some partial results in the non-symmetric case can be found in [38] and [39].) Later, the vanishing third moment condition was removed by Erdős, Yau, and others in [26, 16], i.e., (1.2) is implied by the sub-exponential decay condition. The current best sufficient condition for (1.2), as we know, is that the matrix entries have finite $12 + o(1)$ moments, which was proved in [16]. Numerical results by Biroli, Bouchaud, and Potters [7] predicted that the Tracy-Widom distribution would appear when the $(4 + \epsilon)$-th moment is finite.

On the other hand, for the Wigner matrices whose entries have heavy tails, the necessary condition for (1.2) is studied as follows: In the case of real symmetric matrices with i.i.d. entries, it was proved by Soshnikov [48] that, when the variance of entries diverges, the largest eigenvalue has Poisson statistics. More precisely, in [48] was considered the case where the distribution of entries satisfies

$$\mathbb{P}(|h_{ij}| > x) = \frac{h(x)}{x^\alpha}, \tag{1.3}$$

where $h(x)$ is a slowly varying function and $\alpha < 2$. The case $2 \le \alpha < 4$ was later studied by Auffinger, Ben Arous, and Péché [1], which also shows the Poisson statistics. We also remark that in the case $\alpha < 2$ the Wigner semi-circle law no longer holds in the bulk. See the work by Ben Arous and Guionnet [5] for more detail. The numerical simulation results in [7] also suggest that $\alpha = 4$ in (1.3) will provide the marginal case.

The edge universality has been generalized in many directions, for example, for the sample covariance matrices [28, 31, 46, 49, 41] and for correlation matrix [4, 42]. For the deformed matrices, which are described as a finite rank perturbation of sample covariance matrices and the deformed Wigner matrices, the Tracy-Widom law also holds when the outliers are excluded [3, 9, 10, 37, 34].

In this paper, we prove the following simple criterion on this property: The necessary and sufficient condition for the rescaled largest eigenvalue of a real symmetric Wigner matrix (see Definition 1.1) to have asymptotically the same distribution as for the GOE, and hence the Tracy-Widom distribution F_1, is that the off-diagonal entry of the Wigner matrix satisfies

$$\lim_{s\to+\infty} s^4 \mathbb{P}(|x_{12}| \ge s) = 0. \tag{1.4}$$

We note that this criterion has not been predicted in any previous works.

The precise definition of the Wigner matrix we consider in this paper is as follows:

Definition 1.1. The (standard) symmetric (Hermitian) Wigner matrix of size N is a symmetric (Hermitian) matrix

$$(H_N)_{ij} = h_{ij} = \frac{1}{\sqrt{N}} x_{ij}, \quad 1 \le i, j \le N,$$

where the upper-triangle entries (x_{ij}) $(1 \le i \le j \le N)$ are independent real (complex) random variables with mean zero satisfying the following conditions:

- The upper right entries x_{ij} $(1 \le i < j \le N)$ are *i.i.d.* random variables with distribution μ, satisfying $\mathbb{E}x_{12} = 0$ and $\mathbb{E}|x_{12}|^2 = 1$.
- The diagonal entries x_{ii} $(1 \le i \le N)$ are *i.i.d.* random variables with distribution $\widetilde{\mu}$, satisfying $\mathbb{E}x_{11} = 0$ and $\mathbb{E}|x_{11}|^2 < \infty$.
- In addition, for the Hermitian case, $\mathbb{E}(x_{12})^2 = 0$.

When the random variables x_{ij} and x_{ii} are real Gaussian with $\mathbb{E}|x_{ii}|^2 = 2$, H will be called Gaussian Orthogonal Ensemble (GOE). Similarly, when x_{ij} are complex Gaussian and x_{ii} are real Gaussian with $\mathbb{E}|x_{ii}|^2 = 1$, H will be called Gaussian Unitary Ensemble (GUE). We denote by $\lambda_1 \le \lambda_2 \cdots \le \lambda_N$ the eigenvalues of H_N and by $\mathbf{u}_1, \mathbf{u}_2 \cdots \mathbf{u}_N$ the corresponding eigenvectors of H_N. Theorem 1.2 concerns the real symmetric case; the Hermitian case is recalled only for comparison (Remark 1.4).

The main result of this paper is the following theorem:

Theorem 1.2. *For any centered distributions ν and $\widetilde{\nu}$ on $\mathbb{R}$ with variance 1 and finite variance, respectively, let H_N be the real symmetric Wigner matrix of Definition 1.1 such that x_{12} and x_{11} have distributions ν and $\widetilde{\nu}$, and let λ_N be its largest eigenvalue. Then,*

- *Sufficient condition: if* (1.4) *holds, then there exist $\delta > 0$ and a sequence $\epsilon_N \to 0$, depending only on ν and $\widetilde{\nu}$, such that for all sufficiently large N and all $s \in \mathbb{R}$,*

$$\begin{aligned} \mathbb{P}^{GOE}\big(N^{2/3}(\lambda_N - 2) \le s - N^{-\delta}\big) - \epsilon_N \le \mathbb{P}\big(N^{2/3}(\lambda_N - 2) \le s\big) \\ \le \mathbb{P}^{GOE}\big(N^{2/3}(\lambda_N - 2) \le s + N^{-\delta}\big) + \epsilon_N. \end{aligned} \tag{1.5}$$

- *Necessary condition: if* (1.4) *does not hold, i.e. $s^4\mathbb{P}(|x_{12}| \ge s)$ does not tend to 0 as $s \to \infty$, then*

$$\limsup_{N\to\infty} \mathbb{P}(\lambda_N(H_N) \ge 3) > 0. \tag{1.6}$$

  Consequently $N^{2/3}(\lambda_N - 2)$ is not tight, and its distribution function converges, at the continuity points of the limit, to no probability distribution function; in particular it does not converge to the Tracy–Widom distribution F_1.

Here $\mathbb{P}^{GOE}$ is the probability under which H_N is the GOE of Definition 1.1, i.e. $x_{ij} \sim N(0,1)$ for $i < j$ and $x_{ii} \sim N(0,2)$, so that $F_N^{GOE}(s) := \mathbb{P}^{GOE}(N^{2/3}(\lambda_N - 2) \le s)$ is the distribution function of the rescaled largest eigenvalue of the GOE; in both ensembles $\lambda_N = \max \mathrm{spec}(H_N)$. The order of the quantifiers is: δ and (ϵ_N) first, then N, then s; in particular ϵ_N does not depend on s. The sequence ϵ_N is not related to the cutoff exponent ϵ of Section 4. The two consequences in the necessary condition follow from (1.6). Indeed, on the event $\lambda_N \ge 3$ we have $N^{2/3}(\lambda_N - 2) \ge N^{2/3}$, so there is $c > 0$ such that, for every K, $\mathbb{P}(N^{2/3}(\lambda_N - 2) > K) \ge c$ for infinitely many N. If the distribution functions of $N^{2/3}(\lambda_N - 2)$ converged to a distribution function F at the continuity points of F, then a continuity point K of F with $F(K) > 1 - c/2$ would give $\mathbb{P}(N^{2/3}(\lambda_N - 2) \le K) > 1 - c/2$ for all large N, a contradiction.

Remark 1.3. While any distribution with finite fourth moment satisfies the criterion (1.4), the converse is not true. For example, let x be a symmetric random variable with density $f(x) = c|x|^{-5}(\log|x|)^{-1}$ for $|x| \ge e$ and $f(x) = 0$ otherwise, where $c > 0$ is the normalising constant, and let $x_{12} := x/\sigma$ with $\sigma^2 := \mathbb{E}x^2$, so that x_{12} is centered with variance 1. Then $s^4\mathbb{P}(|x_{12}| \ge s) \sim c/(2\sigma^4 \log s) \to 0$ as $s \to \infty$, so (1.4) holds, while $\mathbb{E}x_{12}^4 = \infty$. For this law of x_{12} and $x_{11} \sim N(0,2)$, (1.5) does not hold

with $\epsilon_N = N^{-\delta}$ for any $\delta > 0$; see Remark 4.1. The existence of this particular example, however, does not contradict the result in [2], which proved that $\lim_{N\to\infty} \lambda_N(H_N) = 2$ a.s. if and only if the fourth moment exists.

Our result provides a very simple sufficient and necessary condition for the edge universality of Wigner matrices without assuming any other properties of matrix entries. This also shows the existence of four moments, which was predicted to be needed for the edge universality, is not necessary for the Tracy-Widom result, as we can see from Remark 1.3.

Our proof of the main result features two key observations.

1. If we introduce a 'cutoff' on each matrix element at $N^{-\epsilon}$, then the matrix with the cutoff can well approximate the original matrix in terms of the behavior of the largest eigenvalue if and only if the criterion (1.4) holds.

2. The Green function comparison method (e.g. Theorem 6.3 in [26]), which was first introduced in [24], can be extended to the random matrices whose entries are almost surely bounded by $N^{-\epsilon}$ for some $\epsilon > 0$. The Green function comparison method was applied on studying the distribution of the eigenvalues of the Wigner matrices, deformed Wigner matrices, covariance matrices, correlation matrices, and adjacency matrices of random graphs [24, 25, 26, 34, 41, 42, 15, 16]. It was also used in the study of the distribution of eigenvectors [33] and the determinant [52] of Wigner matrices. We believe that our new method in the present paper can be used to improve the results in these topics.

The first observation can be understood in the framework of the deformed Wigner matrix. We consider the matrix with the cutoff as the unperturbed part and the remaining part the perturbation. As studied in [3, 37, 9, 10, 34], if the perturbation is small enough, then we can predict the behaviors of the largest eigenvalues of the original matrix from the matrix with cutoff. On the other hand, if the original/perturbation matrix has an entry whose absolute value is larger than 1, then the matrix will have an eigenvalue greater than 2, hence the Tracy-Widom distribution fails. Roughly speaking, the criterion (1.4) means that each off-diagonal entry is bounded by 1 with probability $1 - o(N^{-2})$, thus the condition guarantees that no entries are larger than 1 with probability $1 - o(1)$. We remark that a similar argument was introduced in [7].

The Green function comparison part is more technical. Given the matrix with the cutoff at $N^{-\epsilon}$, we first find a 'better' matrix, in the sense that it is already known to satisfy Tracy-Widom law, whose first four moments of the off-diagonal entries and first two moments of the diagonal entries coincide with those of the given matrix. We then apply Lindeberg replacement strategy sufficiently many times (more precisely, $O(\epsilon^{-1})$ times) to compare the Green functions. The basic idea is as follows: Using Green function comparison method, one can study the difference of the functional on Green functions between the 'better' matrix and the original matrix. Instead of bounding the difference directly, however, we represent it as a new functional, which is much more complicated, on Green functions, with gaining a factor $N^{-\epsilon}$. This new functional can be easily bounded for the 'better' matrix case, but not for original matrix. To solve this issue, again we use Green function comparison method to estimate the difference of this new functional between the 'better' matrix and original matrix. Repeating this process, we obtain the desired bound. The details will be explained later.

Though the Green function comparison method has been used in previous papers, it was always required to have a good bound on Green function with high probability. This is one of the reasons that the distribution of matrix entries have been assumed to satisfy subexponential decay condition in many papers. In this paper, however, we show a way to circumvent this problem, which can be used to achieve many other results, besides the edge universality, for heavy-tailed random matrices. See, for example, the rigidity result in Theorem 3.6 that holds for the random matrices whose entries are only bounded by $N^{-\epsilon}$ for some $\epsilon > 0$. (Note: it is also an interesting result, since it shows that the locations and the fluctuations of the eigenvalues keep unchanged, even if the fluctuations of the matrix entries become very large, i.e., from $N^{-1/2}$ to $N^{-\epsilon}$.)

This paper is organized as follows. In Sections 2 and 3, we introduce the notations and collect tools we use to prove the main result. In Section 4, we prove the main result using the cutoff argument. Technical results on the Green function comparison method will be proved in Sections 5 and 6. Appendix A lists the changes with respect to the published version.

Remark 1.4. Theorem 1.2 concerns the real symmetric case and the largest eigenvalue. The Hermitian case (GUE, F_2; Definition 1.1 with $\mathbb{E}x_{12}^2 = 0$) and the joint distribution (1.2) of the k largest eigenvalues for $k \geq 2$ are expected to follow from the same method (cf. Remark 3.4), but they are not proved here.

Remark 1.5. Assume, as proved by Tracy and Widom [44, 45], that $F_N^{GOE}(s) \to F_1(s)$ for every $s \in \mathbb{R}$, where F_1 is continuous. Under (1.4), Theorem 1.2 then gives $\mathbb{P}(N^{2/3}(\lambda_N - 2) \leq s) \to F_1(s)$ for every $s \in \mathbb{R}$. Indeed, fix s and $h > 0$. For N so large that (1.5) holds and $N^{-\delta} < h$, (1.5) and the monotonicity of F_N^{GOE} give $F_N^{GOE}(s-h) - \epsilon_N \leq \mathbb{P}(N^{2/3}(\lambda_N - 2) \leq s) \leq F_N^{GOE}(s+h) + \epsilon_N$. Hence $F_1(s-h) \leq \liminf_N \mathbb{P}(N^{2/3}(\lambda_N - 2) \leq s) \leq \limsup_N \mathbb{P}(N^{2/3}(\lambda_N - 2) \leq s) \leq F_1(s+h)$, and letting $h \downarrow 0$ gives the claim by the continuity of F_1 at s. This deduction uses [44, 45] and is not part of the formalization.

Formalization. Theorem 1.2 and every result it uses are formalized in Lean 4 with Mathlib [54], with no external input: the formal statements assume only the conditions of Definition 1.1 on ν and $\widetilde{\nu}$, and (1.4) or its negation. The results of [15], [16], [26] and [25, Lemma 4.2] are used in the forms stated in this paper (Lemma 2.6, Lemma 3.1, Lemma 3.5, Lemma 5.6 and the steps of the proof of Theorem 3.6 that follow [26, Section 5]), and those forms are proved in the formalization. In particular, where Section 3 says so, the entries are bounded almost surely. The Lean statements of Theorem 1.2 are `WEdgeNS.E1`, `WEdgeNS.E1_not_tight`, `WEdgeNS.E1_no_limit` and `WEdgeNS.E2`. Remark 1.5 rests on [44, 45] and is not formalized.

2. Notations

In the proof, we will use some variations of standard Wigner matrix defined in Definition 1.1, which are defined as follows:

Definition 2.1 (Generalized symmetric Wigner matrix)**.** A symmetric matrix H_N is said to be a generalized symmetric Wigner matrix of size N if its upper-triangular entries

$$(H_N)_{ij} = h_{ij}, \quad 1 \leq i \leq j \leq N,$$

are independent real random variables with mean zero, *whose distribution may depend on* i, j *and* N, and satisfy, for some constant C_0,

$$|\mathbb{E}h_{ij}^2 - N^{-1}| \leq C_0 N^{-1} \delta_{ij}. \tag{2.1}$$

Remark 2.2. The results on generalized Wigner matrices, especially the constants in the results, may depend on C_0, but we will not emphasize it in the sequel.

As in [15, 16], we will use the following definition to characterize events of very high probability.

Definition 2.3 (High probability events)**.** Define

$$\varphi := (\log N)^{\log \log N}. \tag{2.2}$$

We say that an N-dependent event Ω holds with ζ-high probability if there exist constants $c, C > 0$, independent of N, such that

$$\mathbb{P}(\Omega) \geq 1 - N^C e^{-c\varphi^\zeta} \tag{2.3}$$

for all sufficiently large N. For simplicity, for the case $\zeta = 1$, we just say high probability.

The next condition on the distributions of the matrix entries will be used in the proof.

Definition 2.4 (Bounded support condition)**.** We say a family of random matrices $(H_N)_{ij} = (h_{ij})$ satisfies the bounded support condition with q and $c > 0$, if for all $1 \le i, j \le N$

$$\mathbb{P}\big(|h_{ij}| > q^{-1}\big) \le e^{-N^c}, \tag{2.4}$$

and for $i \ne j$

$$\mathbb{E}\big[h_{ij}^2\, \mathbf{1}(|h_{ij}| > q^{-1})\big] \le e^{-N^c}. \tag{2.4$'$}$$

We say that it satisfies the bounded support condition with q almost surely if $|h_{ij}| \le q^{-1}$ almost surely for all $1 \le i, j \le N$; this implies (2.4) and (2.4′) for every $c > 0$. Here, $q > 0$ may depend on N, and c is a fixed parameter: the constants and the threshold N_0 in the results below may depend on c. By (2.1), the variance of an off-diagonal entry is N^{-1}, so $N^{-1} \le q^{-2} + e^{-N^c}$ and hence $q \le \sqrt{2}\, N^{1/2}$ for $N \ge N_1(c)$; in the almost sure form, $q \le N^{1/2}$.

Note that the Gaussian distribution satisfies bounded support condition with any $q < N^{\phi}$ for any $\phi < 1/2$. We also remark that, when H_N satisfies the bounded support condition, the event $\{|h_{ij}| \le q^{-1}\}$ holds with 'very' high probability, i.e., it holds with ζ-high probability for any positive constant ζ. In the proof of Lemma 3.1 the extreme event $\{|h_{ij}| > q^{-1}\}$ is removed by an explicit truncation (Section 3). In the matrices compared in Sections 5 and 6 the entries are bounded almost surely, so this event does not occur there.

Definition 2.5 (Green function, semicircle, m_{sc} and m)**.** For a Wigner matrix H, we define the Green function of H by

$$G_{ij}(z) := \left(\frac{1}{H - z}\right)_{ij}, \quad z = E + i\eta, \quad E \in \mathbb{R}, \quad \eta > 0. \tag{2.5}$$

The Stieltjes transform of the empirical eigenvalue distribution of H is given by

$$m(z) = m_N(z) := \frac{1}{N}\sum_j G_{jj}(z) = \frac{1}{N}\mathrm{Tr}\ \frac{1}{H - z}\,. \tag{2.6}$$

Define $m_{sc}(z)$ as the unique solution of

$$m_{sc}(z) + \frac{1}{z + m_{sc}(z)} = 0, \tag{2.7}$$

with positive imaginary part for all z with $\mathrm{Im}\, z > 0$, i.e.,

$$m_{sc}(z) = \frac{-z + \sqrt{z^2 - 4}}{2}, \tag{2.8}$$

where the square root function is chosen with a branch cut in the segment $[-2, 2]$ so that asymptotically $\sqrt{z^2 - 4} \sim z$ at infinity. This guarantees that the imaginary part of m_{sc} is non-negative for $\eta = \mathrm{Im} z > 0$ and in the $\eta \to 0$ limit it is the Wigner semicircle distribution

$$\varrho_{sc}(E) := \lim_{\eta \to 0+0} \frac{1}{\pi}\mathrm{Im}\ m_{sc}(E + i\eta) = \frac{1}{2\pi}\sqrt{(4 - E^2)_+}. \tag{2.9}$$

We will also frequently use the notations

$$z := E + i\eta, \quad \kappa := \big|\, |E| - 2\big|.$$

The following lemma (Lemma 4.2 of [25]) collects elementary properties of the Stieltjes transform of the semicircle law. As a technical note, we use the notation $f \sim g$ for two positive functions in some domain D if there exists a positive universal constant C such that $C^{-1} \le f(z)/g(z) \le C$ holds for all $z \in D$.

Lemma 2.6. *We have for all z with* $\mathrm{Im}\, z > 0$ *that*

$$|m_{sc}(z)| = |m_{sc}(z) + z|^{-1} \le 1. \tag{2.10}$$

Let $z = E + i\eta$ with $|E| \le 5$ and $0 < \eta \le 10$. We have

$$|m_{sc}(z)| \sim 1, \qquad |1 - m_{sc}^2(z)| \sim \sqrt{\kappa + \eta} \tag{2.11}$$

and the following two bounds:

$$\operatorname{Im} m_{sc}(z) \sim \begin{cases} \dfrac{\eta}{\sqrt{\kappa+\eta}} & \text{if } \kappa \ge \eta \text{ and } |E| \ge 2, \\ \sqrt{\kappa+\eta} & \text{if } \kappa \le \eta \text{ or } |E| \le 2. \end{cases} \tag{2.12}$$

Definition 2.7 (Classical location of the eigenvalue)**.** We denote by γ_j the *classical location* of the j-th eigenvalue, i.e., γ_j is defined by

$$N \int_{-\infty}^{\gamma_j} \varrho_{sc}(x)\mathrm{d}x = j, \qquad 1 \le j \le N. \tag{2.13}$$

We take the smallest such γ_j; in particular $\gamma_N = 2$.

Remark 2.8. Throughout the paper, the notations $O(\cdot)$, $o(\cdot)$, and $\ll$ will always be with respect to the limit $N \to \infty$, where $a \ll b$ means $a = o(b)$. The constant C will denote various constants independent of N. Constants may depend on the fixed parameters (such as C_0, C_1, c, ϕ, ζ), but not on N or on the distributions of the entries within the class considered. All statements are for each N, with distributions that may depend on N.

3. Tools

In this section, we introduce some results that will be used in the proof of the main theorem. Some of them are already proved in previous papers with H.-T. Yau, L. Erdős, and A. Knowles, and we made slight changes in the statement to fit the notations and definitions in this paper. We also extend some of the known results.

Define the domain

$$\mathbf{S}(C) = \Big\{ z = E + i\eta \ : \ |E| \le 5, \quad N^{-1}\varphi^C < \eta \le 10 \Big\}. \tag{3.1}$$

Lemma 3.1 (Previous results on generalized Wigner matrices)**.** *Fix $\zeta > 0$, $C_0 \ge 0$ and $c > 0$. Let H be a generalized Wigner matrix satisfying* (2.1) *with C_0 and the bounded support condition with q and c (Definition 2.4). There exists a constant $C > 0$, depending only on (ζ, C_0, c), such that, if $q \ge \varphi^C$, then the following properties hold with ζ-high probability, where the constants of Definition 2.3 depend only on (ζ, C_0, c) and not on q:*

(1) **Local semicircle law** *(Theorem 2.8 in* [15]*):* (3.2) *and* (3.3) *hold simultaneously for all $z \in \mathbf{S}(C)$ and all i, j:*

$$\bigcap_{z \in \mathbf{S}(C)} \left\{ |m(z) - m_{sc}(z)| \le \varphi^C \left(\min\left\{ \frac{1}{\sqrt{\kappa+\eta}} \frac{1}{q^2}, \frac{1}{q} \right\} + \frac{1}{N\eta} \right) \right\} \tag{3.2}$$

$$\bigcap_{z \in \mathbf{S}(C)} \left\{ \max_{ij} |G_{ij}(z) - \delta_{ij} m_{sc}(z)| \le \varphi^C \left(\frac{1}{q} + \sqrt{\frac{\operatorname{Im} m_{sc}(z)}{N\eta}} + \frac{1}{N\eta} \right) \right\} \tag{3.3}$$

(2) **Bound on $\|H\|$** *(Lemma 4.4 in* [15]*):*

$$\|H\| \le 2 + \varphi^C (q^{-2} + N^{-2/3}) \tag{3.4}$$

(3) **Delocalization** *(Remark 2.18 in* [15]*): for every eigenvector $\mathbf{u} \in \mathbb{R}^N$ of H with $\|\mathbf{u}\|_2 = 1$,*

$$\max_i |\mathbf{u}(i)|^2 \le \frac{\varphi^C}{N} \tag{3.5}$$

Furthermore, let $\phi > 1/3$ be a constant and assume $q \geq N^{\phi}$. Then the following properties hold, where the constants of Definition 2.3 depend only on (ζ, C_0, c, ϕ), and in part 5 also on a, and not on q:

(4) **Rigidity of the eigenvalues** *(Theorem 2.13 and Remarks 2.14, 2.15 in* [15]*): there is $C > 0$, depending only on (ζ, C_0, c, ϕ), such that, with ζ-high probability,*

$$\bigcap_{j=1}^{N} \left\{ |\lambda_j - \gamma_j| \leq \varphi^C \left(\left[\min\big(j, N-j+1\big) \right]^{-1/3} N^{-2/3} + q^{-2} \right) \right\}. \tag{3.6}$$

(5) **Bound on G_{ij} out of spectrum** *(Equations (3.32), (3.58) and (4.36)-(4.46) in* [15]*): there is a constant $K > 0$, depending only on (ζ, C_0, c, ϕ), such that for every $a > 12\zeta$ there is a constant $C > 0$, depending on a (and on (ζ, C_0, c, ϕ)), such that, with ζ-high probability, the bounds* (3.8) *hold simultaneously for all E with*

$$2 + \varphi^C N^{-2/3} \leq E \leq 3, \quad \eta = \varphi^{K+a} N^{-1} \kappa^{-1/2}, \tag{3.7}$$

where $z = E + i\eta$:

$$|m(z) - m_{sc}(z)| \leq \frac{1}{\varphi^a N\eta}, \quad \operatorname{Im} m(z) \leq \frac{1}{\varphi^a N\eta}, \quad \max_{i \neq j} |G_{ij}(z)| \leq \frac{1}{\varphi^{a/3} N\eta} \tag{3.8}$$

Proof of Lemma 3.1. Throughout the proof, constants depend only on (ζ, C_0, c), and in parts 4 and 5 also on ϕ and a; they do not depend on q, and $N \geq N_0$ with N_0 depending on the same parameters. The dependence of N_0 on c is needed, since e^{-N^c} must be small. By Definition 2.4, $q \leq \sqrt{2}\, N^{1/2}$. We write $\Lambda_d := \max_i |G_{ii} - m_{sc}|$, $\Lambda_o := \max_{i \neq j} |G_{ij}|$ and $\Lambda := |m - m_{sc}|$, as in [15].

Step 1: reduction to entries bounded almost surely. Let $\chi_{ij} := \mathbf{1}(|h_{ij}| \leq q^{-1})$ and $\mu_{ij} := \mathbb{E} h_{ij} \chi_{ij}$. Set $\hat{h}_{ii} := h_{ii}\chi_{ii} - \mu_{ii}$ and, for $i \neq j$, $\hat{h}_{ij} := s_{ij}(h_{ij}\chi_{ij} - \mu_{ij})$, where $s_{ij} > 0$ is chosen so that $\mathbb{E}\hat{h}_{ij}^2 = N^{-1}$. Let $\hat{H} = (\hat{h}_{ij})$, with Green function $\hat{G}$. Since $\mathbb{E} h_{ij} = 0$,

$$|\mu_{ij}| = \big|\mathbb{E} h_{ij}(1 - \chi_{ij})\big| \leq \big(\mathbb{E} h_{ij}^2 (1 - \chi_{ij})\big)^{1/2}\, \mathbb{P}(|h_{ij}| > q^{-1})^{1/2}.$$

For $i \neq j$, (2.4) and (2.4′) give $|\mu_{ij}| \leq e^{-N^c}$, and $\mathbb{E}(h_{ij}\chi_{ij} - \mu_{ij})^2 = N^{-1} - \mathbb{E} h_{ij}^2(1-\chi_{ij}) - \mu_{ij}^2 \in [N^{-1} - 2e^{-N^c}, N^{-1}]$, so $s_{ij} = 1 + O(Ne^{-N^c})$. For $i = j$, (2.1) and (2.4) give $|\mu_{ii}| \leq ((1 + C_0)N^{-1})^{1/2} e^{-N^c/2}$. Hence:

- $\hat{H}$ is a generalized Wigner matrix satisfying (2.1) with the constant $\max(C_0, 1)$. Indeed, the off-diagonal variances are N^{-1}, and $\mathbb{E}\hat{h}_{ii}^2 \in [0, (1 + C_0)N^{-1}]$. No tail-variance bound is assumed for the diagonal entries, so $\mathbb{E}\hat{h}_{ii}^2$ may be much smaller than N^{-1}; this is why the constant $\max(C_0, 1)$ appears.
- $|\hat{h}_{ij}| \leq 2q^{-1}$ almost surely, since $q^{-1} \geq (2N)^{-1/2}$ is much larger than $e^{-N^c/4}$. So $\hat{H}$ satisfies the bounded support condition with $q/2$ almost surely.
- On the event $\Omega_T := \{|h_{ij}| \leq q^{-1} \text{ for all } i, j\}$, whose probability is at least $1 - 2N^2 e^{-N^c}$, we have $|h_{ij} - \hat{h}_{ij}| \leq e^{-N^c/4}$ for all i, j.

On Ω_T, $\|H - \hat{H}\| \leq Ne^{-N^c/4}$. Hence the eigenvalues of H and $\hat{H}$ differ by at most $Ne^{-N^c/4}$, and, by the resolvent identity, $|G_{ij}(z) - \hat{G}_{ij}(z)| \leq \|H - \hat{H}\|\eta^{-2} \leq N^3 e^{-N^c/4}$ for $\eta \geq N^{-1}$; the same bound holds for $|m - \hat{m}|$. Every right side in Lemma 3.1 is at least $\varphi^{-O(1)} N^{-1}$ at the points z considered. So these differences are absorbed by a factor φ^{-1} of room in the corresponding bound for $\hat{H}$, and $2N^2 e^{-N^c} \leq e^{-\varphi^{\zeta}}$, since $\varphi^{\zeta} = N^{o(1)}$. The conditions $q \geq \varphi^C$ and $q \geq N^{\phi}$ with $\phi > 1/3$ are used only up to constant factors, so the change from q to $q/2$ changes only the constants. Therefore it suffices to prove parts 1, 2 and 5 when $|h_{ij}| \leq q^{-1}$ almost surely and (2.1) holds with $\max(C_0, 1)$. Parts 3 and 4 follow from parts 1 and 2 (Steps 3 and 4).

Step 2: parts 1 and 2 for bounded entries. For $C_0 = 0$, (3.2), (3.3) and (3.4) are Theorem 2.8 and Lemma 4.4 of [15], with the parameter ξ of [15] chosen as $\xi = \zeta \log\log N$, so that $(\log N)^{\xi} = \varphi^{\zeta}$. The

probability of the exceptional event is at most $e^{-c'\varphi^\zeta}$ for some $c' > 0$, uniformly in $q \geq \varphi^C$. Three points of the argument of [15] are used in a form not stated there.

- *Diagonal variances.* The proofs of [15] hold, with constants depending on C_0, under $|\mathbb{E}h_{ii}^2 - N^{-1}| \leq C_0 N^{-1}$ in place of $\mathbb{E}h_{ii}^2 = N^{-1}$. No rescaling of G, λ or $\mathbf{u}$ by a factor $1 + O(N^{-1})$ is needed, since the case of a general constant C_0 in (2.1) is treated directly. (In Section 4 the entries of H^S have variances $(1 - O(N^{-1+2\epsilon}))N^{-1}$, so the factor relating them to variance N^{-1} is $1 + O(N^{-1+2\epsilon})$, not $1 + O(N^{-1})$.)
- *Large η.* The domain $\mathbf{S}(C)$ contains $3 < \eta \leq 10$, which the domain of [15] does not. The initial estimate of [15], Lemma 3.16, extends to $2 \leq \eta \leq 10$: there are constants $K_1, C_1, c_1 > 0$ such that, if $q \geq \varphi^{K_1}$, then for each z with $|E| \leq 5$ and $2 \leq \eta \leq 10$,
$$\mathbb{P}\Big(\Lambda_d(z) + \Lambda_o(z) > C_1\big(\varphi^\zeta q^{-1} + \varphi^{2\zeta} N^{-1/2}\big)\Big) \leq e^{-c_1\varphi^\zeta}.$$
The continuity argument of [15] then starts at $\eta = 10$.
- *All z at once.* The bounds (3.2) and (3.3) hold at each point of a lattice in $\mathbf{S}(C)$ with $N^{O(1)}$ points, with probability at least $1 - e^{-c'\varphi^\zeta}$ each. By a union bound they hold on the whole lattice, and they extend to all $z \in \mathbf{S}(C)$ by Lipschitz continuity in z (the derivatives of G_{ij}, m and m_{sc} are at most $\eta^{-2} \leq N^2$), using $q^2 \leq 2N$. This gives the intersection over $z \in \mathbf{S}(C)$ and all i, j.

Step 3: part 3. Let C_1 and C_2 be the constants of parts 1 and 2, and let $C \geq \max(2C_1 + 3, C_2)$. Let $H\mathbf{u} = \mu\mathbf{u}$ with $\|\mathbf{u}\|_2 = 1$. By (3.4), $|\mu| \leq 3$. Let $z = \mu + i\eta$ with $\eta = \varphi^{2C_1+2}N^{-1}$, so that $z \in \mathbf{S}(C_1)$. Let $\mathbf{u}_\beta$, $\beta = 1, \dots, N$, be an orthonormal eigenbasis of H. Then $\mathbf{u}$ is a combination of the $\mathbf{u}_\beta$ with $\lambda_\beta = \mu$, so $|\mathbf{u}(i)|^2 \leq \sum_{\beta:\lambda_\beta=\mu} |\mathbf{u}_\beta(i)|^2$ by the Cauchy–Schwarz inequality. Hence
$$|\mathbf{u}(i)|^2 \leq \sum_\beta \frac{\eta^2 |\mathbf{u}_\beta(i)|^2}{(\lambda_\beta - \mu)^2 + \eta^2} = \eta \operatorname{Im} G_{ii}(z).$$
By (3.3) with $q \geq \varphi^{C_1}$ and $N\eta = \varphi^{2C_1+2}$, together with $|m_{sc}| \leq 1$, we get $\operatorname{Im} G_{ii}(z) \leq 4 \leq \varphi$. Therefore $|\mathbf{u}(i)|^2 \leq \varphi^{2C_1+3}N^{-1} \leq \varphi^C N^{-1}$. This holds for every unit eigenvector, also when eigenvalues are repeated.

Step 4: part 4. Let $q \geq N^\phi$ with $\phi > 1/3$, so that $q \geq \varphi^C$ for the constants C of parts 1 and 2. For $1 \leq j \leq N - 1$, (3.6) follows from (3.2) and (3.4) as in Theorem 2.13 and Remarks 2.14, 2.15 of [15]; this step is deterministic. Theorem 2.13 of [15] does not cover $j = N$, which we treat separately. By (2.13), $\gamma_N = 2$ and $2 - \gamma_{N-1} \leq C' N^{-2/3}$, since $\varrho_{sc}(x) \geq c''\sqrt{2 - x}$ for $0 \leq x \leq 2$. Then (3.4) gives $\lambda_N - 2 \leq \varphi^C(q^{-2} + N^{-2/3})$. Moreover, $\lambda_N \geq \lambda_{N-1}$, and (3.6) at $j = N - 1$ gives
$$\lambda_N \geq \gamma_{N-1} - \varphi^C(N^{-2/3} + q^{-2}) \geq 2 - \varphi^{C+1}(N^{-2/3} + q^{-2}).$$
This is (3.6) at $j = N$, with $C + 1$ in place of C.

Step 5: part 5. Let $a > 12\zeta$. By Step 1 we may assume $|h_{ij}| \leq q^{-1}$ almost surely.

The first two bounds of (3.8). In (4.36)–(4.46) of [15] it is proved that, for E in the window
$$2 + \varphi^{C'}(q^{-2} + N^{-2/3}) \leq E \leq 3$$
of (4.36) of [15], one has $\Lambda \leq ((\log N)N\eta)^{-1}$ and $\operatorname{Im} m \leq ((\log N)N\eta)^{-1}$ (see (4.45) and (4.46) of [15]); here $\alpha \sim \sqrt{\kappa}$ in the notation of [15]. Run at level ζ, the same argument gives the following. There is $K_0 > 0$, depending only on (ζ, C_0), such that for every $a' > 0$ there is C' for which, with ζ-high probability,
$$\Lambda(z) \leq \frac{1}{\varphi^{a'} N\eta}, \qquad \operatorname{Im} m(z) \leq \frac{1}{\varphi^{a'} N\eta}, \qquad \eta = \varphi^{K_0+a'} N^{-1}\kappa^{-1/2},$$
for all E in this window. To see this, replace $\varphi^3(\log N)^2$ in (4.37) of [15] by $\varphi^{K_0+2a'}$, and replace $\varphi^3 \log N$ in (4.38), (4.42) and below (4.44) of [15] by $\varphi^{K_0+a'}$. The offset K_0 collects the constants of

[15] that enter at level ζ. In the original statement the offset was 3, obtained by setting $C_1 = 1$ in (2.15) of [15]; this conflates two different constants C_1 of [15], so the offset is a constant K_0 instead.

The bound holds at each fixed E with ζ-high probability. It holds simultaneously for all E in the window by a union bound over a grid of spacing N^{-10}, because $m(E + i\eta(E))$, $m_{sc}(E + i\eta(E))$ and the right sides have derivatives in E bounded by $N^{O(1)}$ on the window (there $\eta(E) \geq N^{-1}$ and $|\eta'(E)| \leq 1$).

Since $q \geq N^{\phi}$ with $\phi > 1/3$, we have $q^{-2} \leq N^{-2/3}$. Hence the window (3.7) with $C \geq C' + 1$ lies inside the window above. Set $K := K_0 + K_2 + 1$, where K_2 is the constant of the next paragraph, and apply the above with $a' = a + K - K_0$. Then η is the η of (3.7), and the bounds hold with the extra factor $\varphi^{-(K-K_0)} \leq \varphi^{-1}$, which leaves room for Step 1. This proves the first two parts of (3.8).

The third bound of (3.8). On the window (3.7) we have $\varphi^K N^{-1} < \eta \leq 1$, so $z \in \mathbf{S}(K_2)$ once $K \geq K_2$. Using (3.32) and (3.58) of [15] at level ζ, there are $K_2, C_o > 0$, depending only on (ζ, C_0), such that, if $q \geq \varphi^{K_2}$, then with ζ-high probability

$$\Lambda_o(z) \leq C_o\big(q^{-1} + \varphi^{2\zeta}\Psi(z)\big), \qquad \Psi := \sqrt{\frac{\Lambda + \operatorname{Im} m_{sc}}{N\eta}},$$

where Ψ is the control parameter of (3.26) of [15]. On the event of the first two parts, $\operatorname{Im} m_{sc} \leq \operatorname{Im} m + \Lambda \leq 2(\varphi^a N\eta)^{-1}$, so $\Psi \leq \sqrt{3}\,\varphi^{-a/2}(N\eta)^{-1}$. Renaming $\sqrt{3}C_o$ as C_o, we get, with ζ-high probability,

$$\max_{i \neq j} |G_{ij}| \leq C_o\Big(\frac{1}{q} + \frac{\varphi^{2\zeta}}{\varphi^{a/2}N\eta}\Big). \tag{3.9}$$

For the second term, $C_o\varphi^{2\zeta - a/2} \leq \frac{1}{3}\varphi^{-a/3}$ holds for $N \geq N_0$ if $a/6 - 2\zeta > 0$, that is, $a > 12\zeta$. At $\zeta = 1$ this is the condition $a > 12$. For the first term, (3.7) gives $\kappa = E - 2 \geq \varphi^C N^{-2/3}$, hence

$$N\eta = \varphi^{K+a}\kappa^{-1/2} \leq \varphi^{K+a-C/2}N^{1/3},$$

and so $\varphi^{a/3}N\eta\, C_o q^{-1} \leq C_o\varphi^{K+4a/3}N^{1/3-\phi} \leq \frac{1}{3}$ for $N \geq N_0$, since $\phi > 1/3$. Hence $\max_{i\neq j}|G_{ij}| \leq \frac{2}{3}(\varphi^{a/3}N\eta)^{-1}$ at each E with ζ-high probability. The grid argument above makes this simultaneous in E, and the remaining factor $\frac{2}{3}$ leaves room for Step 1. This gives the third part of (3.8) and completes the proof.

□

Remark 3.2. Lemma 3.1 holds for every fixed $\zeta > 0$, with a constant $C = C_\zeta$; it is stated in this form. In this paper we use $\zeta \in \{1, 3, 4\}$; the case $\zeta = 4$ is used for (5.19).

Theorem 3.3 (Edge universality on generalized Wigner matrix: [16, Theorem 2.7 and Proposition 6.4, (6.15) at $f = 0$])**.** *Let $H^{\mathbf{w}}$ be a GOE and $H^{\mathbf{v}}$ a generalized symmetric Wigner matrix with*

$$\mathbb{E}^{\mathbf{v}}|h_{ij}|^2 = N^{-1} + \delta_{ij}N^{-1}.$$

Assume that $H^{\mathbf{v}}$ satisfies the bounded support condition with $q = N^{\phi}$, for some constant $1/3 < \phi \leq 1/2$. Then, there exist constants $\delta > 0$ and N_0 such that, for any $N \geq N_0$ and any $s \in \mathbb{R}$, we have

$$\mathbb{P}^{\mathbf{w}}\Big(N^{2/3}(\lambda_N - 2) \leq s - N^{-\delta}\Big) - N^{-\delta} \leq \mathbb{P}^{\mathbf{v}}\big(N^{2/3}(\lambda_N - 2) \leq s\big) \leq \mathbb{P}^{\mathbf{w}}\Big(N^{2/3}(\lambda_N - 2) \leq s + N^{-\delta}\Big) + N^{-\delta}. \tag{3.10}$$

Here, $\mathbb{P}^{\mathbf{v}}$ and $\mathbb{P}^{\mathbf{w}}$ denote the laws of the ensembles $H^{\mathbf{v}}$ and $H^{\mathbf{w}}$, respectively, and δ and N_0 do not depend on s.

The diagonal variance here and for the GOE is $2/N$, while it is $1/N$ in [16, Definition 2.1]. For entries bounded by $N^{-\phi}$ almost surely, Theorem 3.3 is the case $C_0 = 1$ of Lemma 3.5 below. In the form stated above, with the bounded support condition of Definition 2.4, it is cited from [16] and is not used separately in this paper.

Remark 3.4. As in [26] and [16], Theorem 3.3, as well as Lemma 3.5 and Theorem 3.7 below, can be extended to finite correlation functions of extreme eigenvalues. For example, we have the following extension:

$$\begin{aligned}&\mathbb{P}^{\mathbf{w}}\Big(N^{2/3}(\lambda_N-2)\le s_1-N^{-\epsilon},\dots,N^{2/3}(\lambda_{N-k}-2)\le s_{k+1}-N^{-\epsilon}\Big)-N^{-\delta}\\&\le\mathbb{P}^{\mathbf{v}}\Big(N^{2/3}(\lambda_N-2)\le s_1,\dots,N^{2/3}(\lambda_{N-k}-2)\le s_{k+1}\Big)\\&\le\mathbb{P}^{\mathbf{w}}\Big(N^{2/3}(\lambda_N-2)\le s_1+N^{-\epsilon},\dots,N^{2/3}(\lambda_{N-k}-2)\le s_{k+1}+N^{-\epsilon}\Big)+N^{-\delta}\end{aligned}\tag{3.11}$$

for all k fixed and N sufficiently large. The proof of (3.11) is similar to that of (3.10) except that it uses the general form of the Green function comparison theorem. This remark is not used in this paper (see (4.26)), and it is not formalized.

We slightly extend this result as follows:

Lemma 3.5. *Let $H^{\mathbf{w}}$ be a GOE and $H^{\mathbf{v}}$ a generalized symmetric Wigner matrix satisfying* (2.1) *with a constant $C_0\ge 0$, whose entries satisfy $|h_{ij}|\le N^{-\phi}$ almost surely for some $1/3<\phi\le 1/2$. Then there are $\delta>0$ and N_0, depending only on (C_0,ϕ), such that* (3.10) *holds for all $N\ge N_0$ and all $s\in\mathbb{R}$.*

This is the almost sure form of the bounded support condition, the only case used here (Theorem 3.7 applies it to $\widetilde{H}$ of Lemma 5.1); the probabilistic form is not proved here. We postpone the proof of this lemma to the end of this section.

To prove our main result, we claim the following three lemmas, which extend the previous results to the Wigner matrix with bounded support condition of small q. First, we improve the previous result on rigidity. We define the *normalized empirical counting function* by

$$\mathfrak{n}(E):=\frac{1}{N}\#\{\lambda_j\le E\}.$$

Let

$$n_{sc}(E):=\int_{-\infty}^{E}\varrho_{sc}(x)\mathrm{d}x$$

be the distribution function of the semicircle law.

Theorem 3.6 (Rigidity of eigenvalues: small q case)**.** *Let H be a generalized symmetric Wigner matrix with some constant C_1 such that for any $1\le i<j\le N$*

$$\mathbb{E}|h_{ii}|^2\le C_1/N,\quad \mathbb{E}|h_{ij}|^2=1/N,\quad \mathbb{E}|h_{ij}|^3\le C_1N^{-3/2},\quad \mathbb{E}|h_{ij}|^4\le C_1(\log N)N^{-2}$$

and $|h_{ij}|\le N^{-\phi}$ almost surely for all $1\le i,j\le N$, for some constant $\phi>0$. Then, there exist constants $C>0$ and N_0, depending only on C_1 and ϕ, such that with high probability we have

$$\bigcap_j\left\{|\lambda_j-\gamma_j|\le\varphi^C\Big[\min\big(j,N-j+1\big)\Big]^{-1/3}N^{-2/3}\right\}\tag{3.12}$$

and

$$\sup_{E\in\mathbb{R}}\big|\mathfrak{n}(E)-n_{sc}(E)\big|\le\frac{\varphi^C}{N}\tag{3.13}$$

for any $N>N_0$.

Theorem 3.6 is the special case, with entries bounded almost surely, of the corresponding statement under the bounded support condition of Definition 2.4 with $q=N^{\phi}$. That statement is not proved here: under Definition 2.4 alone the moment bound (6.78) can fail. Section 4 uses only the almost sure case, since $\widetilde{H}^S$ is bounded almost surely.

Next theorem shows that the edge universality holds under the assumptions in Theorem 3.6.

Theorem 3.7 (Edge universality: small q case)**.** *Let $H^{\mathbf{w}}$ be a GOE and $H^{\mathbf{v}}$ be a generalized symmetric Wigner matrix satisfying the conditions for H in Theorem 3.6 (in particular $|h_{ij}| \le N^{-\phi}$ almost surely). Then, there exist constants $\delta > 0$ and N_0, depending only on C_1 and ϕ, such that for all $N \ge N_0$, all such $H^{\mathbf{v}}$ and all $s \in \mathbb{R}$, we have*

$$\mathbb{P}^{\mathbf{w}}\Big(N^{2/3}(\lambda_N - 2) \le s - N^{-\delta}\Big) - N^{-\delta} \le \mathbb{P}^{\mathbf{v}}\big(N^{2/3}(\lambda_N - 2) \le s\big) \le \mathbb{P}^{\mathbf{w}}\Big(N^{2/3}(\lambda_N - 2) \le s + N^{-\delta}\Big) + N^{-\delta}. \tag{3.14}$$

Here, $\mathbb{P}^{\mathbf{v}}$ and $\mathbb{P}^{\mathbf{w}}$ denote the laws of the ensembles $H^{\mathbf{v}}$ and $H^{\mathbf{w}}$, respectively.

Finally, we show a weak bound on G_{ij} $(i \ne j)$ of H satisfying the conditions in Theorem 3.6.

Lemma 3.8 (Bounds on G_{ij}: small q case)**.** *Let H be a generalized symmetric Wigner matrix satisfying the conditions for H in Theorem 3.6 (in particular $|h_{ij}| \le N^{-\phi}$ almost surely). Then, for any $0 < c < 1$, there are constants $C > 0$ and N_0, depending only on (C_1, ϕ, c), such that for all $N \ge N_0$, all $z = E + i\eta$ with $|E| \le 5$ and $\eta \ge N^{-1+c}$, and all $i \ne j$, we have the following weak bound on G_{ij}:*

$$\mathbb{E}|G_{ij}(z)|^2 \le \varphi^C \left(\frac{\operatorname{Im} m_{sc}(z)}{N\eta} + \frac{1}{(N\eta)^2} \right). \tag{3.15}$$

In the remainder of this section, we give the proof of Lemma 3.5.

Proof of Lemma 3.5. Throughout the proof, $q := N^{\phi}$ and $\kappa := \phi - 1/3 > 0$. We derive the lemma from Lemma 5.6, applied to the pairs $(H^{\mathbf{v}}, H^{\mathbf{w}})$ of Lemma 3.5 with the rate $N^{-\kappa+C\epsilon}$ in (5.15). Lemma 5.6 is the common structure of the proofs of [26, Theorem 2.4] and [16, Proposition 6.4]; its proof uses neither Lemma 3.5 nor Theorem 3.7. Its hypotheses are checked as follows.

(i) The matrix $H^{\mathbf{v}}$ satisfies (2.1) with C_0 and, since $|h_{ij}| \le q^{-1}$ almost surely, the bounded support condition with q and every $c > 0$, in particular $c = 1$. The GOE satisfies (2.1) with $C_0 = 1$ and, by the Gaussian tail, the bounded support condition with $N^{2/5}$ in place of q and some $c > 0$. For each of them, Lemma 3.1 with $\zeta = 1$ gives, for every sufficiently large C, with high probability and with constants depending only on (C_0, ϕ): $\|H\| \le 2 + \varphi^C(q^{-2} + N^{-2/3}) \le 3$ by (3.4); (5.24) for all $z \in \mathbf{S}(C)$ with $\eta \le N^{-2/3}$, since by (3.2) $|m - m_{sc}| \le \varphi^C(q^{-2}\eta^{-1/2} + (N\eta)^{-1})$ and $q^{-2}\eta^{-1/2} = (N\eta)^{-1}N^{1-2\phi}\eta^{1/2} \le (N\eta)^{-1}N^{2/3-2\phi} \le (N\eta)^{-1}$; (5.25) by the rigidity bound of Lemma 3.1 (4) at $j = N$, since $\gamma_N = 2$ and $q^{-2} \le N^{-2/3}$; and (5.26) by the same rigidity bound and (2.13), since $\gamma_j \ge 2 - 3\varphi^C N^{-2/3}$ holds for at most $C'\varphi^{3C/2}$ indices j. (For the GOE, q is replaced by $N^{2/5}$ in these bounds.)

(ii) The comparison hypothesis (5.15) for the pair, with test functions having four derivatives and right side $N^{-\kappa+C\epsilon}$, is (3.19) of the following Claim, since $N^{1/3+C\epsilon}q^{-1} = N^{-\kappa+C\epsilon}$.

Lemma 5.6 then gives $\delta > 0$ and N_0, depending only on (C_0, ϕ), such that (3.10) holds for all $N \ge N_0$, all $H^{\mathbf{v}}$ as in Lemma 3.5 and all $s \in \mathbb{R}$. It remains to prove the following claim, which corresponds to Proposition 6.6 of [16] with $f = 0$.

Claim. Let $C_F > 0$. There exist $\widetilde{\epsilon} > 0$ and $C > 0$, depending only on (C_F, C_0, ϕ), such that for every $0 < \epsilon < \widetilde{\epsilon}$ there is N_0, depending only on $(C_F, C_0, \phi, \epsilon)$, with the following property. Let $N \ge N_0$, let $H^{\mathbf{v}}$ be as in Lemma 3.5, and let $F \in C^4(\mathbb{R})$ satisfy

$$\sup_x |F^{(n)}(x)|(1 + |x|)^{-C_F} \le C_F, \quad n = 1, 2, 3, 4. \tag{3.16}$$

Set

$$I_\epsilon := \{x : |x - 2| \le N^{-2/3+\epsilon}\}, \qquad \eta := N^{-2/3-\epsilon}. \tag{3.17}$$

Then, for every $E \in I_\epsilon$,

$$\left| \mathbb{E}^{\mathbf{v}} F\left(N\eta \operatorname{Im} m(E + i\eta)\right) - \mathbb{E}^{\mathbf{w}} F\left(N\eta \operatorname{Im} m(E + i\eta)\right) \right| \le N^{1/3+C\epsilon}q^{-1} \tag{3.18}$$

and, for all $E_1, E_2 \in I_\epsilon$ (in either order; the integral is oriented),

$$\left| \mathbb{E}^{\mathbf{v}} F\left(N \int_{E_1}^{E_2} dy \operatorname{Im} m(y + i\eta)\right) - \mathbb{E}^{\mathbf{w}} F\left(N \int_{E_1}^{E_2} dy \operatorname{Im} m(y + i\eta)\right) \right| \le N^{1/3+C\epsilon}q^{-1}. \tag{3.19}$$

Only (3.19) enters Lemma 5.6.

The differences in (3.18)–(3.19) and below do not change if F is replaced by $F-F(0)$, so we assume $F(0)=0$; then $|F(x)|\le C_F|x|(1+|x|)^{C_F}$. By (3.16), for every fixed $a>0$ and $n\in\{1,2\}$,

$$|F^{(n)}(x)|\le N^{(aC_F+1)\epsilon}\quad \text{for } |x|\le 2N^{a\epsilon} \text{ and } N\ge N_0(C_F,a,\epsilon).$$

For a real symmetric matrix H with Green function G, the spectral decomposition gives the Ward identity $\sum_j |G_{ij}(z)|^2=\eta^{-1}\operatorname{Im} G_{ii}(z)$, hence

$$N\eta\operatorname{Im} m(z)=\eta^2\sum_{i,j}|G_{ij}(z)|^2,\qquad N\int_{E_1}^{E_2}\operatorname{Im} m(y+i\eta)\,dy=\eta\int_{E_1}^{E_2}\sum_{i,j}|G_{ij}(y+i\eta)|^2\,dy.$$

We first prove the corresponding bounds for the off-diagonal parts:

$$\left|\mathbb{E}F\left(\eta^2\sum_{i\ne j}G^{\mathbf{v}}_{ij}\overline{G^{\mathbf{v}}_{ij}}\right)-\mathbb{E}F\left(\eta^2\sum_{i\ne j}G^{\mathbf{w}}_{ij}\overline{G^{\mathbf{w}}_{ij}}\right)\right|\le N^{1/3+C\epsilon}q^{-1},\tag{3.20}$$

with G evaluated at $E+i\eta$, and

$$\left|\mathbb{E}F\left(\eta\int_{E_1}^{E_2}\sum_{i\ne j}G^{\mathbf{v}}_{ij}\overline{G^{\mathbf{v}}_{ij}}(y+i\eta)\,dy\right)-\mathbb{E}F\left(\eta\int_{E_1}^{E_2}\sum_{i\ne j}G^{\mathbf{w}}_{ij}\overline{G^{\mathbf{w}}_{ij}}(y+i\eta)\,dy\right)\right|\le N^{1/3+C\epsilon}q^{-1}.\tag{3.20$'$}$$

The removed diagonal parts are estimated at the end of the proof.

Interpolation. Fix a bijective ordering map on the index set of the independent matrix elements,

$$\Phi:\{(k,\ell):1\le k\le\ell\le N\}\to\{1,\cdots,\gamma_{\max}=\frac{N(N+1)}{2}\},\tag{3.21}$$

and let H_γ be the Wigner matrix whose matrix elements $h_{k\ell}$ $(k\le\ell)$ are independent and follow the distribution of $h^{\mathbf{v}}_{k\ell}$ if $\Phi(k,\ell)\le\gamma$ and the distribution of $h^{\mathbf{w}}_{k\ell}$ otherwise. In particular, $H_0=H^{\mathbf{w}}$ and $H_{\gamma_{\max}}=H^{\mathbf{v}}$. (Note that the index used here is slightly different from previous papers.) For simplicity, let

$$G^\gamma_{ij}:=\left(\frac{1}{H_\gamma-z}\right)_{ij}.\tag{3.22}$$

The matrices H_γ and $H_{\gamma-1}$ differ only at the (a,b) and (b,a) elements, where $\Phi(a,b)=\gamma$. Let $v:=v_{ab}:=h^{\mathbf{v}}_{ab}$ and $w:=w_{ab}:=h^{\mathbf{w}}_{ab}$. We define matrices V and W by

$$V_{k\ell}:=\begin{cases}(\delta_{ak}\delta_{b\ell}+\delta_{a\ell}\delta_{bk})\,v_{ab}, & a\ne b,\\ \delta_{ak}\delta_{a\ell}\,v_{aa}, & a=b,\end{cases}\qquad W_{k\ell}:=\begin{cases}(\delta_{ak}\delta_{b\ell}+\delta_{a\ell}\delta_{bk})\,w_{ab}, & a\ne b,\\ \delta_{ak}\delta_{a\ell}\,w_{aa}, & a=b,\end{cases}\tag{3.23}$$

so that we can rewrite H_γ and $H_{\gamma-1}$ as

$$H_\gamma=Q+V,\quad H_{\gamma-1}=Q+W,\tag{3.24}$$

with a matrix Q satisfying $Q_{ab}=Q_{ba}=0$. The matrix Q is independent of v and w, and its law is the same in both identities. Define Green's functions

$$R:=\frac{1}{Q-z},\quad S:=\frac{1}{H_\gamma-z},\quad T:=\frac{1}{H_{\gamma-1}-z}.\tag{3.25}$$

Local law for H_γ. Each H_γ is a generalized Wigner matrix satisfying (2.1) with the constant $\max(C_0,1)$, since each of its diagonal entries comes from $H^{\mathbf{v}}$ or from the GOE. Let $\phi':=(1/3+\phi)/2$, so that $1/3<\phi'<\phi\le 1/2$, and $q':=N^{\phi'}$. The entries of $H^{\mathbf{v}}$ satisfy $|h|\le N^{-\phi}\le q'^{-1}$ almost surely, and a centred Gaussian entry h with $\mathbb{E}h^2\le 2/N$ satisfies $\mathbb{P}(|h|>q'^{-1})\le 2\exp(-N^{1-2\phi'}/4)$. Hence every H_γ satisfies the bounded support condition with q' and $c:=(1-2\phi')/2$ for $N\ge N_1(\phi)$, and Lemma 3.1 applies to every H_γ with q', with constants depending only on (C_0,ϕ) and not on γ. (In general H_γ does not satisfy the bounded support condition with q itself: for $\phi=1/2$ and a Gaussian entry h of variance $1/N$, $\mathbb{P}(|h|>N^{-1/2})=\mathbb{P}(|g|>1)\approx 0.32$ for a standard Gaussian g.)

We claim that, with high probability, uniformly in γ and in $H^{\mathbf{v}}$,

$$\max_{k,\ell}\max_{E\in I_\epsilon}|S_{k\ell}(E+i\eta)-\delta_{k\ell}m_{sc}(E+i\eta)|\le N^{-1/3+2\epsilon}. \tag{3.26}$$

Indeed, for $E\in I_\epsilon$ and N large the point $E+i\eta$ lies in $\mathbf{S}(C)$, and (3.3) holds simultaneously for all $z\in\mathbf{S}(C)$ by part (1) of Lemma 3.1. By (2.12), $\operatorname{Im} m_{sc}(E+i\eta)\le C\sqrt{||E|-2|+\eta}\le CN^{-1/3+\epsilon/2}$, so the right side of (3.3) is at most $\varphi^C(N^{-\phi'}+CN^{-1/3+3\epsilon/4}+N^{-1/3+\epsilon})\le N^{-1/3+2\epsilon}$ for $N\ge N_0(C_0,\phi,\epsilon)$. In particular, since $|m_{sc}|\le 1$ by (2.10), $|S_{kk}|\le 2$ and $|S_{k\ell}|\le N^{-1/3+2\epsilon}$ for $k\ne\ell$ on this event.

Let $B:=N^\epsilon q^{-1}=N^{\epsilon-\phi}$. With high probability, uniformly in γ, all entries of H_γ are bounded by B in absolute value: almost surely for the entries from $H^{\mathbf{v}}$, and for the Gaussian entries because the GOE satisfies the bounded support condition with $N^{1/2-\epsilon/2}$ in place of q and $c=\epsilon/2$ for $N\ge\lceil 8^{2/\epsilon}\rceil$, while $N^{-1/2+\epsilon/2}\le N^{\epsilon-\phi}$ as $\phi\le 1/2$. This bound replaces $|v_{ab}|\le q^{-1}$, which fails for Gaussian entries when $\phi=1/2$. Let Ω be the intersection of the event of (3.26) for H_γ and the event that all entries of H_γ are bounded by B; then $\mathbb{P}(\Omega^c)\le N^Ce^{-c\varphi}$. Let $P:=N^{-1/3+3\epsilon}$. We claim that, on Ω,

$$\max_{k,\ell}\max_{E\in I_\epsilon}|R_{k\ell}(E+i\eta)-\delta_{k\ell}m_{sc}(E+i\eta)|\le N^{-1/3+3\epsilon}=P. \tag{3.27}$$

To see (3.27), we expand R using the resolvent identity $R=S+SVR$ five times:

$$R=S+SVS+(SV)^2S+(SV)^3S+(SV)^4S+(SV)^5R. \tag{3.28}$$

Since V has at most two non-zero entries, each bounded by B on Ω, $|S_{k\ell}|\le 2$ on Ω, and $|R_{k\ell}|\le\eta^{-1}$ always, we have $|((SV)^mS)_{k\ell}|\le 2^{2m+1}B^m$ for $1\le m\le 4$ and $|((SV)^5R)_{k\ell}|\le 2^{10}B^5\eta^{-1}$. Hence $|R_{k\ell}-S_{k\ell}|\le 16N^{\epsilon-\phi}+2^{10}N^{2/3+6\epsilon-5\phi}$, and both terms are at most $P/3$ for $N\ge N_0(\phi,\epsilon)$, since $\phi>1/3$ and $5\phi>1+3\epsilon$. (The five terms of (3.28) suffice because $\phi>1/5+3\epsilon/5$.) Together with (3.26) and $N^{-1/3+2\epsilon}\le P/3$, this gives (3.27). In particular, on Ω, $|R_{kk}|\le 2$ and $|R_{k\ell}|\le P$ for $k\ne\ell$. The bounds (3.26), (3.27) and $|v|\le B$ hold on Ω simultaneously for all $E\in I_\epsilon$.

The case $a\ne b$. Let Δ be the matrix with entries $\Delta_{k\ell}=\delta_{ak}\delta_{b\ell}+\delta_{a\ell}\delta_{bk}$, so that $V=v\Delta$. We follow the proof of Lemmas 6.7 and 6.8 of [16] with $f=0$ ([16], (6.41)–(6.97)), with the following inputs and changes. The variables v and w are centred, $\mathbb{E}v^2=\mathbb{E}w^2=N^{-1}$, and every entry h of every H_γ satisfies

$$\mathbb{E}|h|^k\le C_*N^{-1}q^{2-k},\qquad 2\le k\le 6,\qquad C_*:=\max(1+C_0,120),$$

which is [16, (6.42)]: for the entries from $H^{\mathbf{v}}$ it follows from $|h|\le q^{-1}$, and for the Gaussian entries from their moments and $q\le N^{1/2}$. The a priori bounds used in [16] are (3.26), (3.27) and $|v|\le B$ on Ω. The resolvent expansion [16, (6.58)] reads $S=\sum_{m=0}^5(-v)^m(R\Delta)^mR+(-v)^6(R\Delta)^6S$, and on Ω the entries of $(R\Delta)^mR$ obey [16, (6.59)–(6.61)] with the constants 2^{2m+1}. The functionals of (3.20) contain $S_{ij}\overline{S_{ij}}$; [16, (6.46), (6.56), (6.66)] write $S_{ij}S_{ji}$, which we read as $S_{ij}\overline{S_{ij}}$, expanding the second factor by complex conjugation. Since v is real, it factors out of the coefficients

$$K_k^{(s,t)}:=\eta^2\sum_{i\ne j,\ |\{i,j\}\cap\{a,b\}|=k}\big((R\Delta)^sR\big)_{ij}\overline{\big((R\Delta)^tR\big)_{ij}},\qquad k=0,1,2,$$

which are functions of Q alone. A Taylor expansion of F to third order at $y^R:=\eta^2\sum_{i\ne j}|R_{ij}|^2$ ([16, (6.63)–(6.76)]), with the bad event Ω^c treated as in (3.36) below, gives

$$\Big|\mathbb{E}F\Big(\eta^2\sum_{i\ne j}S_{ij}\overline{S_{ij}}\Big)-\mathbb{E}F(y^R)-\mathbb{E}g(Q)\,\mathbb{E}v^2-\mathbb{E}g_3(Q)\,\mathbb{E}(-v)^3\Big|\le N^{-5/3+C\epsilon}q^{-1},$$

where

$$g(Q):=F'(y^R)\sum_{k=0,1}\big(K_k^{(2,0)}+K_k^{(1,1)}+K_k^{(0,2)}\big)+\frac{1}{2}F''(y^R)\big(K_0^{(1,0)}+K_0^{(0,1)}\big)^2,$$

$$g_3(Q):=F'(y^R)\big(K_0^{(3,0)}+K_0^{(0,3)}\big).$$

Here g is the function of Q in [16, (6.75)], whose second line should read $Q_0^{(1)}$ in place of $Q_1^{(0)}$, as in its first line and in (6.74). The first-order term has expectation zero, since Q is independent of v and $\mathbb{E}v = 0$. The same holds with T and w in place of S and v. Since $\mathbb{E}v^2 = \mathbb{E}w^2 = N^{-1}$ and Q has the same law in both cases, the second-order terms $\mathbb{E}g(Q)\,\mathbb{E}v^2$ and $\mathbb{E}g(Q)\,\mathbb{E}w^2$ are equal and cancel. For the third-order term, $|\mathbb{E}(-v)^3| \le C_* N^{-1} q^{-1}$, and $|\mathbb{E}g_3(Q)| \le N^{-2/3+C\epsilon}$ by [16, (6.77)–(6.86)]: the main part of $((R\Delta)^3 R)_{ij}\overline{R_{ij}}$ for $i,j \notin \{a,b\}$ is $m_{sc}^2(R_{ia}R_{bj} + R_{ib}R_{aj})\overline{R_{ij}}$, and for i,j,a,b distinct, [16, Lemma 6.8] with $f = 0$ gives

$$\left|\mathbb{E}F'(y^R)R_{ia}R_{bj}\overline{R_{ij}}\right| \le N^{-4/3+C\epsilon}.$$

In [16] the replaced entry is (b,d), and the assumption of Lemma 6.8 there should read "b,d,i,j are all distinct". In its proof we use [16, Lemma 6.10] for R in the one-sided form $R_{ia} = -R_{aa}\sum_k Q_{ak}R_{ki}^{(a)}$ $(i \ne a)$, and we replace the large deviation bound after [16, (6.97)] by the second moment bound $\mathbb{E}|\sum_k Q_{ak}c_k|^2 \le N^{-1}\sum_k \mathbb{E}|c_k|^2$ for coefficients c_k independent of the row a of Q with $c_a = 0$, since that row contains both bounded and Gaussian entries. Altogether, when $a \ne b$,

$$\left|\mathbb{E}F\left(\eta^2\sum_{i\ne j}S_{ij}\overline{S_{ij}}\right) - \mathbb{E}F\left(\eta^2\sum_{i\ne j}T_{ij}\overline{T_{ij}}\right)\right| \le N^{-5/3+C\epsilon}q^{-1}. \tag{3.29}$$

The case $a = b$. Then $V = v\Delta^{(a)}$ with $v = v_{aa}$, where $\Delta^{(a)}_{k\ell} := \delta_{ak}\delta_{a\ell}$, and the resolvent identity $S = R - RVS$, applied twice, gives

$$S = R - RVR + (RV)^2 S, \qquad \text{i.e.} \qquad S_{ij} = R_{ij} - vR_{ia}R_{aj} + v^2 R_{ia}R_{aa}S_{aj}. \tag{3.30}$$

Hence, for $i \ne j$,

$$S_{ij}\overline{S_{ij}} = R_{ij}\overline{R_{ij}} - 2v\,\mathrm{Re}\left(\overline{R_{ij}}R_{ia}R_{aj}\right) + 2v^2\,\mathrm{Re}\left(\overline{R_{ij}}R_{ia}R_{aa}S_{aj}\right) + \left|vR_{ia}R_{aj} - v^2R_{ia}R_{aa}S_{aj}\right|^2. \tag{3.31}$$

On Ω we have $|v| \le B \le 1/2$, $|R_{kk}|, |S_{kk}| \le 2$ and $|R_{k\ell}|, |S_{k\ell}| \le P \le 1$ for $k \ne \ell$. Hence, for $i \ne j$, on Ω,

$$\begin{aligned}\left|S_{ij}\overline{S_{ij}} - R_{ij}\overline{R_{ij}} + 2v\,\mathrm{Re}\left(\overline{R_{ij}}R_{ia}R_{aj}\right)\right| &\le \begin{cases} 8v^2P^3, & a \notin \{i,j\},\\ 24v^2P^2, & a \in \{i,j\},\end{cases}\\ \left|S_{ij}\overline{S_{ij}} - R_{ij}\overline{R_{ij}}\right| &\le \begin{cases} 6|v|P^3, & a \notin \{i,j\},\\ 16|v|P^2, & a \in \{i,j\}.\end{cases}\end{aligned} \tag{3.32}$$

Let

$$y^S := \eta^2\sum_{i\ne j}S_{ij}\overline{S_{ij}}, \quad y^R := \eta^2\sum_{i\ne j}R_{ij}\overline{R_{ij}}, \quad \ell(R) := -2\eta^2\sum_{i\ne j}\mathrm{Re}\left(\overline{R_{ij}}R_{ia}R_{aj}\right). \tag{3.33}$$

There are at most N^2 pairs with $a \notin \{i,j\}$ and at most $2N$ pairs with $a \in \{i,j\}$, and $NP \ge 2$. Summing the previous bounds, we obtain on Ω

$$|y^S - y^R| \le D_1 := 32B\eta^2N^2P^3, \qquad |y^S - y^R - v\,\ell(R)| \le D_2 := 32B^2\eta^2N^2P^3, \tag{3.34}$$

where $\eta^2N^2P^3 = N^{-1/3+7\epsilon}$, so that $D_1 = 32N^{-1/3+8\epsilon}q^{-1}$ and $D_2 = 32N^{-1/3+9\epsilon}q^{-2}$.

Consider the Taylor expansion

$$F(y^S) - F(y^R) = F'(y^R)(y^S - y^R) + \frac{1}{2}F''(\zeta)(y^S - y^R)^2 \tag{3.35}$$

for some ζ between y^S and y^R. On Ω, $0 \le y^S \le \eta^2N^2N^{-2/3+4\epsilon} = N^{2\epsilon}$ and $|y^R| \le N^{2\epsilon} + D_1 \le 2N^{2\epsilon}$, so $|F'(y^R)|, |F''(\zeta)| \le N^{(2C_F+1)\epsilon}$, and by the bounds on $y^S - y^R$,

$$\left|F(y^S) - F(y^R) - F'(y^R)\,v\,\ell(R)\right| \le N^{(2C_F+1)\epsilon}\Big(D_2 + \frac{D_1^2}{2}\Big) \qquad \text{on } \Omega.$$

The linear term $F'(y^R)\,v\,\ell(R)$ has expectation zero on the whole probability space: y^R and $\ell(R)$ are bounded functions of Q, which is independent of v, and $\mathbb{E}v = 0$. Off Ω we use the deterministic

bounds $|R_{k\ell}|, |S_{k\ell}| \le \eta^{-1} \le N$, which give $|y^S|, |y^R| \le N^2$ and $|\ell(R)| \le 2N^3$, hence $|F(y^S) - F(y^R) - F'(y^R)v\ell(R)| \le \Phi_N(1+|v|)$ with $\Phi_N := 2C_F N^3(1+N^2)^{C_F}$. Since $|v| \le T + v^2/T$ for every $T > 0$ and $\mathbb{E}v^2 \le C'N^{-1}$ with $C' := 1 + \max(C_0, 1)$, we obtain

$$\big|\mathbb{E}F(y^S) - \mathbb{E}F(y^R)\big| \le N^{(2C_F+1)\epsilon}\Big(D_2 + \frac{D_1^2}{2}\Big) + \Phi_N\Big((1+T)\,\mathbb{P}(\Omega^c) + \frac{C'}{NT}\Big). \tag{3.36}$$

With $T := \Phi_N N^2$ and $\mathbb{P}(\Omega^c) \le N^C e^{-c\varphi}$, the last term is at most $N^{-2} \le N^{-1/3}q^{-2}$ for N large. Since $D_1^2/2 \le D_2$ for N large, we obtain

$$\big|\mathbb{E}\big(F(y^S) - F(y^R)\big)\big| \le N^{-1/3+C\epsilon}q^{-2}. \tag{3.37}$$

Only the upper bound $\mathbb{E}v^2 \le C'N^{-1}$ was used, not the value of $\mathbb{E}v^2$: on the diagonal the second moments of $H^{\mathbf{v}}$ and of the GOE need not match (for $C_0 = 0$, for instance, $\mathbb{E}(h^{\mathbf{v}}_{aa})^2 = N^{-1}$, while the GOE has $2N^{-1}$). The same argument applies to $H_{\gamma-1} = Q + W$, with $w \sim \mathcal{N}(0, 2/N)$ and the event Ω for $H_{\gamma-1}$. Since y^R is the same in both cases, we find that

$$\left|\mathbb{E}F\left(\eta^2\sum_{i\ne j} S_{ij}\overline{S_{ij}}\right) - \mathbb{E}F\left(\eta^2\sum_{i\ne j} T_{ij}\overline{T_{ij}}\right)\right| \le N^{-1/3+C\epsilon}q^{-2}. \tag{3.38}$$

The integrated functional. The same proof gives (3.29) and (3.38) for $x^S := \eta\int_{E_1}^{E_2}\sum_{i\ne j}|S_{ij}(y+i\eta)|^2dy$ in place of y^S (and likewise for R and T), for all $E_1, E_2 \in I_\epsilon$, with three changes. First, the bounds on Ω hold simultaneously for all $y \in I_\epsilon$, so the pointwise bounds above hold at every $y+i\eta$ with y between E_1 and E_2; after integration in y the weight η^2 is replaced by $\eta|E_2-E_1| \le 2N^{-4/3} = 2N^{2\epsilon}\eta^2$, which costs a factor $2N^{2\epsilon}$, and on Ω the functionals are bounded by $2N^{5\epsilon}$. Second, v does not depend on y and leaves the y-integral, so the linear term is v times a function of Q; its expectation vanishes by Fubini's theorem, since the integrand is continuous in (H_γ, y) and bounded by $2N^2\eta^{-3}$. Third, off Ω the deterministic bounds are multiplied by $\eta^{-1}|E_2 - E_1| \le N$, which only changes Φ_N. In the case $a \ne b$, the same changes apply to [16, (6.41)–(6.97)], where the integrated functional is (6.56).

Telescoping. We write the quantity in the left hand side of (3.20) as a telescopic sum,

$$\mathbb{E}F\left(\eta^2\sum_{i\ne j} G^{\mathbf{v}}_{ij}\overline{G^{\mathbf{v}}_{ij}}\right) - \mathbb{E}F\left(\eta^2\sum_{i\ne j} G^{\mathbf{w}}_{ij}\overline{G^{\mathbf{w}}_{ij}}\right) = \sum_{\gamma=1}^{\gamma_{\max}}\left(\mathbb{E}F\left(\eta^2\sum_{i\ne j} S_{ij}\overline{S_{ij}}\right) - \mathbb{E}F\left(\eta^2\sum_{i\ne j} T_{ij}\overline{T_{ij}}\right)\right), \tag{3.39}$$

where S and T depend on γ. Among the summands, N have $a = b$ and $N(N-1)/2 \le N^2$ have $a \ne b$. By (3.29) and (3.38), the absolute value of (3.39) is at most

$$N \cdot N^{-1/3+C\epsilon}q^{-2} + N^2 \cdot N^{-5/3+C\epsilon}q^{-1} \le 2N^{1/3+C\epsilon}q^{-1},$$

since $N^{2/3}q^{-2} = N^{1/3}q^{-1}N^{1/3-\phi} \le N^{1/3}q^{-1}$. This gives (3.20) after replacing C by $C+1$ (for $N^\epsilon \ge 2$), and the same telescoping sum for the integrated functional gives (3.20′).

The diagonal parts. Let H be $H^{\mathbf{v}} = H_{\gamma_{\max}}$ or $H^{\mathbf{w}} = H_0$, with Green function G, and let Ω_0 be the event of (3.26) for H. By the Ward identity, $N\eta\,\mathrm{Im}\,m(E+i\eta) - \eta^2\sum_{i\ne j}|G_{ij}|^2 = \eta^2\sum_i|G_{ii}|^2$. On Ω_0, $|G_{ii}| \le 2$ for all i and all $E \in I_\epsilon$, so this difference lies in $[0, 4N\eta^2] = [0, 4N^{-1/3-2\epsilon}]$, and $0 \le \eta^2\sum_{i\ne j}|G_{ij}|^2 \le N^{2\epsilon}$. Everywhere, $0 \le \eta^2\sum_{i\ne j}|G_{ij}|^2 \le N\eta\,\mathrm{Im}\,m \le N$, so both values of F are at most $C_F N(1+N)^{C_F}$ in absolute value. Hence

$$\begin{aligned}\left|\mathbb{E}F\big(N\eta\,\mathrm{Im}\,m(E+i\eta)\big) - \mathbb{E}F\Big(\eta^2\sum_{i\ne j}|G_{ij}|^2\Big)\right| &\le 4N^{(2C_F+1)\epsilon}N^{-1/3-2\epsilon} + 2C_F N(1+N)^{C_F}\,\mathbb{P}(\Omega_0^c)\\ &\le N^{-1/3+(2C_F+1)\epsilon}.\end{aligned}$$

For the integrated functional, $N\int_{E_1}^{E_2}\mathrm{Im}\,m(y+i\eta)\,dy - \eta\int_{E_1}^{E_2}\sum_{i\ne j}|G_{ij}|^2dy = \eta\int_{E_1}^{E_2}\sum_i|G_{ii}|^2dy$ is at most $4N\eta|E_2 - E_1| \le 8N^{-1/3}$ in absolute value on Ω_0, both functionals are bounded by $2N^{5\epsilon}$ on

Ω_0 and by $2N^2$ everywhere, and the same argument gives the error $N^{-1/3+(5C_F+2)\epsilon}$. Since $\phi \le 1/2$, $N^{-1/3} \le N^{1/3}q^{-1}$. Combining these errors for $H^{\mathbf{v}}$ and $H^{\mathbf{w}}$ with (3.20) and (3.20′), we obtain (3.18) and (3.19) with the bound $3N^{1/3+C\epsilon}q^{-1} \le N^{1/3+(C+1)\epsilon}q^{-1}$ for $N^\epsilon \ge 3$. This proves the claim, which implies the desired lemma. □

Remark 3.9 (Corrections to the cited works). The formalization of the results of [15] and [16] in the forms of Lemma 3.1 and Lemma 3.5 required the following corrections. None of them changes a statement used in this paper. The corrections to [26] are given in the proof of Lemma 5.6.

(1) The bound [15, (3.22)] holds as printed only for $q \le N^{1/4}$; with the Hilbert–Schmidt term $(N^{-2}\sum_{i,j}|B_{ij}|^2)^{1/2}$ added to its right side it holds in general, and this is the form used for [15, (3.32)].
(2) [15, (3.36)] and the proof of [15, Lemma 3.15] omit the term $j = i$ of A_i; with it, $|A_i| \le 5/(4N) + 22\Lambda_o^2$.
(3) In the proof of [15, Lemma 3.16], "$(2-1-O(\cdot))^2 \ge 3/2$" should read "$(2-1-O(\cdot))\cdot 2 \ge 3/2$".
(4) In the proofs of [15, Lemmas 5.7 and 5.11], the constant 1/2 in the induction should be 1/4.
(5) In the proof of [15, Lemma 5.13], the quantity X in (5.95) needs the additional term $(N\eta)^{-1}$, and for $U = \emptyset$ the Hilbert–Schmidt term in (5.105) is controlled through $\operatorname{Im} m^{(Si)}$.
(6) [15, (A.4)] (Lemma A.1 (ii)) holds for $n \le N$ random variables and $q \ge 1$, which is the case used; without these conditions it fails (for instance for $q < 1$ and $n = N = 2$, or for $n = N^2$ and $q = 1$).
(7) In [16], the display after (6.19) and the argument of K in (6.17) miss a factor π: $N\int_{E_1}^{E_2} \operatorname{Im} m(y+i\eta)\,dy$ is approximately π times the number of eigenvalues in $[E_1, E_2]$, so K is to be evaluated at this quantity divided by π.
(8) In the statement of [16, Lemma 6.8], "a, b, i, j are all distinct" should read "b, d, i, j are all distinct".
(9) In the second line of [16, (6.75)], $Q_1^{(0)}$ should read $Q_0^{(1)}$, as in its first line and in (6.74).

4. Proof of the main result

In this section, we prove the main result, Theorem 1.2. Let H_N be a Wigner matrix defined as in Definition 1.1 such that x_{12} and x_{11} have distributions ν and $\widetilde{\nu}$, respectively. We begin by proving the second part of the main result.

Proof of the main result: Necessary condition. Assume that $s^4\mathbb{P}(|x_{12}| \ge s)$ does not tend to 0 as $s \to \infty$. We note that there exists a constant $0 < c_1 < 1/2$ and a sequence $r_1, r_2, \cdots$ such that $r_k \ge 4$, $r_k \to \infty$ as $k \to \infty$ and

$$\mathbb{P}(|x_{12}| \ge r_k) \ge c_1 r_k^{-4}. \tag{4.1}$$

Consider an event

$$\Gamma_N := \{\text{ There exist } i \text{ and } j,\ 1 \le i < j \le N,\ \text{such that } |h_{ij}| \ge 4, |h_{ii}| \le 1,\ \text{and } |h_{jj}| \le 1\}. \tag{4.2}$$

We first show that, when Γ_N holds, $\lambda_N(H_N) \ge 3$. Define $\mathbf{u} \in \mathbb{R}^N$ through

$$\mathbf{u}(m) = \begin{cases} 1/\sqrt{2} & \text{if } m = i, \\ (1/\sqrt{2})\cdot \operatorname{sgn}(h_{ij}) & \text{if } m = j, \\ 0 & \text{otherwise .} \end{cases} \tag{4.3}$$

Here, $\operatorname{sgn}(h_{ij}) := |h_{ij}|/h_{ij}$. Since $\|\mathbf{u}\|_2 = 1$, it can be easily seen that

$$\lambda_N(H_N) \ge \langle \mathbf{u}, H_N\mathbf{u}\rangle = h_{ii}|\mathbf{u}_i|^2 + h_{jj}|\mathbf{u}_j|^2 + h_{ij}\mathbf{u}_i\mathbf{u}_j + h_{ji}\mathbf{u}_j\mathbf{u}_i = |h_{ij}| + \frac{h_{ii}+h_{jj}}{2} \ge 4 - 1 = 3. \tag{4.4}$$

We now prove that there exists a constant $c_0 > 0$, independent of N, such that $\mathbb{P}(\Gamma_N) \geq c_0$ for any $N \in \{\lfloor (r_k/4)^2 \rfloor : k \in \mathbb{N}\}$ with $N \geq \max(2, 2\mathbb{E}|x_{11}|^2)$. Since $r_k \to \infty$, there are infinitely many such N, so this implies (1.6). Define an event

$$\Gamma'_N := \{ \text{ There exist } i \text{ and } j,\ 1 \leq i < j \leq N, \text{ such that } |h_{ij}| \geq 4\}. \tag{4.5}$$

If $N = \lfloor (r_k/4)^2 \rfloor$, then $4\sqrt{N} \leq r_k$ and $r_k^2 < 16(N+1) \leq 32N$, hence $r_k^{-4} \geq N^{-2}/1024$, and we have that

$$\begin{aligned} 1 - \mathbb{P}(\Gamma'_N) &= \big(1 - \mathbb{P}(|h_{12}| \geq 4)\big)^{N(N-1)/2} \leq \big(1 - \mathbb{P}(|x_{12}| \geq r_k)\big)^{N(N-1)/2} \\ &\leq \big(1 - c_1 r_k^{-4}\big)^{N(N-1)/2} \leq (1 - c_2 N^{-2})^{N(N-1)/2} \end{aligned} \tag{4.6}$$

with $c_2 := c_1/1024$. Since $N(N-1)/2 \geq N^2/4$ for $N \geq 2$, we have $(1-c_2N^{-2})^{N(N-1)/2} \leq e^{-c_2/4} =: c_3$, and $0 < c_3 < 1$ is independent of N. Hence $\mathbb{P}(\Gamma'_N) \geq 1 - c_3$. From Markov inequality, we have for every i

$$\mathbb{P}(|h_{ii}| > 1) = \mathbb{P}(|x_{ii}| > \sqrt{N}) \leq N^{-1}\mathbb{E}|x_{ii}|^2 \leq \frac{1}{2}. \tag{4.7}$$

On Γ'_N, let (i', j') be the first pair (i, j), $i < j$, in the lexicographic order such that $|h_{ij}| \geq 4$. It is a function of the off-diagonal entries alone, hence independent of the diagonal entries, and the diagonal entries are independent of each other. Since the events $\{(i', j') = (i, j)\}$ are disjoint, we find that

$$\mathbb{P}(\Gamma_N) \geq \sum_{i<j} \mathbb{P}\big((i', j') = (i, j)\big)\, \mathbb{P}(|h_{ii}| \leq 1)\, \mathbb{P}(|h_{jj}| \leq 1) \geq \frac{1}{4}\mathbb{P}(\Gamma'_N) \geq \frac{1}{4}(1 - c_3). \tag{4.8}$$

This completes the proof with $c_0 := (1 - c_3)/4$. □

Next, we prove that, if (1.4) holds, then there exist $\delta > 0$ and a sequence $\epsilon_N \to 0$ such that, for all $N \geq N_0$ and all $s \in \mathbb{R}$,

$$\mathbb{P}\Big(N^{2/3}(\widetilde{\lambda}_N - 2) \leq s - N^{-\delta}\Big) - \epsilon_N \leq \mathbb{P}\big(N^{2/3}(\lambda_N - 2) \leq s\big) \leq \mathbb{P}\Big(N^{2/3}(\widetilde{\lambda}_N - 2) \leq s + N^{-\delta}\Big) + \epsilon_N, \tag{4.9}$$

where λ_N is the largest eigenvalue of H_N and $\widetilde{\lambda}_N$ denotes the largest eigenvalue of GOE. This is (1.5). The constants δ, N_0 and the sequence (ϵ_N) depend only on ν and $\widetilde{\nu}$, and not on s.

Remark 4.1. The additive error ϵ_N in (4.9) cannot in general be taken to be $N^{-\delta}$. Let ν be the law of Remark 1.3 and $\widetilde{\nu}$ the centered Gaussian law with variance 2. Since $\mathbb{P}(|x_{12}| \geq r) \geq cr^{-4}(\log r)^{-1}$ for large r, the independence of the $N(N-1)/2$ entries h_{ij}, $i < j$, gives $\mathbb{P}(\exists\, i < j : |h_{ij}| \geq 4) \geq c_\nu/\log N$ for large N, with a constant $c_\nu > 0$ depending only on ν. Since h_{kk} is Gaussian with variance $2/N$, $\mathbb{P}(\max_k |h_{kk}| \geq 1/2) \leq 2Ne^{-N/16}$. On the intersection of the events $\{\exists\, i < j : |h_{ij}| \geq 4\}$ and $\{\max_k |h_{kk}| < 1/2\}$, the argument of (4.4) gives $\lambda_N > 7/2$, hence $\mathbb{P}(N^{2/3}(\lambda_N - 2) \leq N^{2/3}) \leq 1 - c_\nu/\log N + 2Ne^{-N/16}$. For the GOE, $\widetilde{\lambda}_N$ is a $\sqrt{2/N}$-Lipschitz function of independent standard Gaussian variables and $\mathbb{E}\widetilde{\lambda}_N \leq 2$, so $\mathbb{P}(\widetilde{\lambda}_N \leq 5/2) \geq 1 - e^{-N/16}$ by Gaussian concentration. Hence the lower half of (4.9) with $\epsilon_N = N^{-\delta}$ at $s = N^{2/3}$ would give $1 - e^{-N/16} - N^{-\delta} \leq 1 - c_\nu/\log N + 2Ne^{-N/16}$, which fails for every $\delta > 0$ and all large N. This example is not formalized. Under the additional conditions $\mathbb{E}|x_{12}|^{4+a} < \infty$ and $\mathbb{E}|x_{11}|^{2+b} < \infty$ for some $a, b > 0$, one expects (4.9) to hold with $\epsilon_N = N^{-\delta'}$ for some $\delta' > 0$; this is not proved here.

If H_N satisfies the assumption in Theorem 3.6, then (4.9) indeed holds, with $\epsilon_N = N^{-\delta}$, as we have seen from Theorem 3.7. Thus, we construct from H_N a random matrix $\widetilde{H}^S$ whose entries are bounded almost surely by $N^{-\phi}$ for some $\phi > 0$ and which satisfies the moment conditions in Theorem 3.6. Comparing the largest eigenvalue of $\widetilde{H}^S$ with that of H_N, we will show that the difference between them will be negligible with probability $1 - o(1)$.

Proof of the main result: Sufficient condition. Fix $0 < \epsilon < 1/30$. (The cut-off steps below need $\epsilon < 1/2$, (4.24) needs $\epsilon < 1/8$, the lower bound in (4.47) needs $\epsilon < 1/20$, and (4.39) and (4.46) need $\epsilon < 1/30$.) For any N, let $T := N^{1/2-\epsilon}$ and define

$$\alpha := \alpha_N := \mathbb{P}\left(|x_{12}| > N^{1/2-\epsilon}\right), \quad \widetilde{\alpha} := \widetilde{\alpha}_N := \mathbb{P}\left(|x_{11}| > N^{1/2-\epsilon}\right), \tag{4.10}$$

$$\beta := \beta_N := \mathbb{E}\left[\mathbf{1}\left(|x_{12}| > N^{1/2-\epsilon}\right) x_{12}\right], \quad \widetilde{\beta} := \widetilde{\beta}_N := \mathbb{E}\left[\mathbf{1}\left(|x_{11}| > N^{1/2-\epsilon}\right) x_{11}\right]. \tag{4.11}$$

By (1.4), $\mathbb{E}|x_{11}|^2 < \infty$ and integration by parts, we have, as $N \to \infty$,

$$\alpha = o(N^{-2+4\epsilon}), \quad \widetilde{\alpha} = o(N^{-1+2\epsilon}), \tag{4.12}$$

$$\beta = o(N^{-3/2+3\epsilon}), \quad \widetilde{\beta} = o(N^{-1/2+\epsilon}). \tag{4.13}$$

Indeed, $T^4\alpha \le \sup_{r\ge T} r^4\mathbb{P}(|x_{12}| \ge r) \to 0$; $|\beta| \le T\alpha + \int_T^\infty \mathbb{P}(|x_{12}| > r)\,\mathrm{d}r$, so $T^3|\beta| \to 0$; and $T^2\widetilde{\alpha} + T|\widetilde{\beta}| \le 2\mathbb{E}[|x_{11}|^2\mathbf{1}(|x_{11}| > T)] \to 0$. In the same way, $\mathbb{E}[|x_{12}|^2\mathbf{1}(|x_{12}| > T)] = o(N^{-1+2\epsilon})$. In particular $\alpha, \widetilde{\alpha} < 1$ for all large N, and we consider only such N. Let

$$m := \mathbb{E}\big[x_{12} \,\big|\, |x_{12}| \le T\big] = -\frac{\beta}{1-\alpha}, \qquad \widetilde{m} := \mathbb{E}\big[x_{11} \,\big|\, |x_{11}| \le T\big] = -\frac{\widetilde{\beta}}{1-\widetilde{\alpha}},$$

where the identities use $\mathbb{E}x_{12} = \mathbb{E}x_{11} = 0$. Then $m = o(N^{-3/2+3\epsilon})$ and $\widetilde{m} = o(N^{-1/2+\epsilon})$.

Let ν_S, ν_L be the following distributions:

$$\nu_S := \text{the law of } x_{12} - m \text{ given } |x_{12}| \le T, \quad \nu_L := \begin{cases} \text{the law of } x_{12} - m \text{ given } |x_{12}| > T & \text{if } \alpha > 0, \\ \delta_0 & \text{if } \alpha = 0, \end{cases} \tag{4.14}$$

and let $\widetilde{\nu}_S$, $\widetilde{\nu}_L$ be defined in the same way from x_{11}, $\widetilde{m}$ and $\widetilde{\alpha}$:

$$\widetilde{\nu}_S := \text{the law of } x_{11} - \widetilde{m} \text{ given } |x_{11}| \le T, \quad \widetilde{\nu}_L := \begin{cases} \text{the law of } x_{11} - \widetilde{m} \text{ given } |x_{11}| > T & \text{if } \widetilde{\alpha} > 0, \\ \delta_0 & \text{if } \widetilde{\alpha} = 0. \end{cases} \tag{4.15}$$

Here, the subindices S and L are for small and large, respectively. Let ν_c, $\widetilde{\nu}_c$ be the distribution such that $x = 1$ with probability α and $\widetilde{\alpha}$, otherwise $x = 0$.

On a product probability space, let H^S, H^L and H^C be the random symmetric matrices such that $H^S_{ij} = N^{-1/2}x^S_{ij}$, $H^L_{ij} = N^{-1/2}x^L_{ij}$, and $H^C_{ij} = c_{ij}$, where x^S_{ij}, x^L_{ij}, and c_{ij} $(i \le j)$ are independent random variables such that:

(1) the entries x^S_{ij} have distribution ν_S if $i \ne j$ and x^S_{ii} have distribution $\widetilde{\nu}_S$,
(2) the entries x^L_{ij} have distribution ν_L if $i \ne j$ and x^L_{ii} have distribution $\widetilde{\nu}_L$,
(3) the entries c_{ij} have distribution ν_c if $i \ne j$ and c_{ii} have distribution $\widetilde{\nu}_c$.

For one entry $i < j$ and a Borel set A, $\mathbb{P}\big((1-c_{ij})x^S_{ij} + c_{ij}x^L_{ij} + m \in A\big) = (1-\alpha)\,\mathbb{P}(x_{12} \in A \,|\, |x_{12}| \le T) + \alpha\,\mathbb{P}(x_{12} \in A \,|\, |x_{12}| > T) = \mathbb{P}(x_{12} \in A)$, where the second term is 0 if $\alpha = 0$; the diagonal entries are treated in the same way. Since the triples $(x^S_{ij}, x^L_{ij}, c_{ij})$, $i \le j$, are independent, the random matrix H defined by

$$H_{ij} := H^S_{ij}(1 - H^C_{ij}) + H^L_{ij}H^C_{ij} + M_{ij}, \qquad M_{ij} := N^{-1/2}m + N^{-1/2}(\widetilde{m} - m)\delta_{ij}, \tag{4.16}$$

has the same law as H_N, for every N with $\alpha, \widetilde{\alpha} < 1$. From now on, we work on this product space and with H in place of H_N. Writing $M = N^{-1/2}mJ + N^{-1/2}(\widetilde{m} - m)I$, where J is the matrix whose entries are all 1, the scalar part shifts the largest eigenvalue exactly and $\|N^{-1/2}mJ\| = N^{1/2}|m|$. Hence, for every real symmetric matrix A and all large N,

$$\big|\lambda_N(A + M) - \lambda_N(A)\big| \le N^{1/2}|m| + N^{-1/2}\big(|m| + |\widetilde{m}|\big) \le N^{-1+4\epsilon}, \tag{4.17}$$

since the middle term is $o(N^{-1+3\epsilon}) + o(N^{-1+\epsilon})$; the same bound holds for $\|M\|$. By (1.4) and integration by parts, we have for $i \neq j$,

$$\mathbb{E}(x_{ij}^S) = 0, \quad \mathbb{E}|x_{ij}^S|^2 = \frac{1}{1-\alpha}\int_{|x|\le T}(x-m)^2\,\mathrm{d}\nu(x) = 1 - o(N^{-1+2\epsilon}), \tag{4.18}$$

$$\mathbb{E}|x_{ij}^S|^3 = O(1), \quad \mathbb{E}|x_{ij}^S|^4 = O(\log N),$$

and

$$\mathbb{E}x_{ii}^S = 0, \quad \mathbb{E}|x_{ii}^S|^2 = \frac{1}{1-\widetilde{\alpha}}\int_{|x|\le T}(x-\widetilde{m})^2\,\mathrm{d}\widetilde{\nu}(x) \le \frac{\mathbb{E}|x_{11}|^2}{1-\widetilde{\alpha}} \le C_1. \tag{4.19}$$

Here the second moment in (4.18) is $(1-\alpha)^{-1}(1-\mathbb{E}[|x_{12}|^2\mathbf{1}(|x_{12}|>T)]) - m^2$. Let $\sigma := (\mathbb{E}|x_{12}^S|^2)^{1/2}$. We note that the matrix

$$\widetilde{H}^S := \sigma^{-1}H^S$$

is a generalized symmetric Wigner matrix whose entries satisfy $|\widetilde{H}_{ij}^S| \le N^{-\phi_s}$ almost surely, with $\phi_s := 3\epsilon/4 \in (\epsilon/2, \epsilon)$, for all large N: indeed $|x_{ij}^S + m| \le T$, so $|x_{ij}^S| \le 2T$ and $|\widetilde{H}_{ij}^S| \le 2\sigma^{-1}N^{-\epsilon}$. By (4.18) and (4.19), $\widetilde{H}^S$ satisfies the conditions for H in Theorem 3.6 with $\phi = \phi_s$ and a constant C_1 depending only on ν, $\widetilde{\nu}$ and ϵ, for all large N. Hence, by Theorem 3.7, there exist $\delta_0 > 0$ and N_0, depending only on C_1 and ϕ_s, such that, for all $N \ge N_0$ and any $s \in \mathbb{R}$, we have

$$\begin{aligned}\mathbb{P}^{GOE}\Big(N^{2/3}(\lambda_N - 2) \le s - N^{-\delta_0}\Big) - N^{-\delta_0} &\le \mathbb{P}^{\widetilde{S}}\big(N^{2/3}(\lambda_N - 2) \le s\big) \\ &\le \mathbb{P}^{GOE}\Big(N^{2/3}(\lambda_N - 2) \le s + N^{-\delta_0}\Big) + N^{-\delta_0},\end{aligned} \tag{4.20}$$

where $\mathbb{P}^{\widetilde{S}}$ is the law of $\widetilde{H}^S$. We write the first two terms in the right hand side of (4.16) as follows:

$$H_{ij}^S(1 - H_{ij}^C) + H_{ij}^L H_{ij}^C = H_{ij}^S + \widetilde{H}_{ij}H_{ij}^C, \quad \widetilde{H}_{ij} = H_{ij}^L - H_{ij}^S.$$

We can see that H_{ij}^C is independent of H_{ij}^S and $\widetilde{H}_{ij}$. Though $\widetilde{H}_{ij}$ depends on H_{ij}^S, from the condition (1.4) we know that for any i, j

$$\mathbb{P}(|\widetilde{H}_{ij}H_{ij}^C| > 3/4) \le \mathbb{P}(|x_{ij}| > (1/2)N^{1/2}) \le o(N^{-2}) + o(N^{-1})\delta_{ij}, \tag{4.21}$$

where $x_{ij} := N^{1/2}H_{ij}$ are the entries of (4.16). Indeed, if $H_{ij}^C = 1$ and $|\widetilde{H}_{ij}| > 3/4$, then $x_{ij} = x_{ij}^L + m$ (or $x_{ii}^L + \widetilde{m}$), and since $|x_{ij}^S + m| \le T$, we get $|x_{ij}| \ge N^{1/2}|\widetilde{H}_{ij}| - T > N^{1/2}/2$ as soon as $N^\epsilon \ge 4$. For the last inequality, we used (1.4) and that $\mathbb{E}|x_{ii}|^2 < \infty$. Because of this reason, instead of $H_{ij}^S + \widetilde{H}_{ij}H_{ij}^C$, we only need to study the matrix whose entries have a cutoff on $\widetilde{H}_{ij}H_{ij}^C$ as follows:

$$H_{ij}^S + \widetilde{D}_{ij}H_{ij}^C, \quad \widetilde{D}_{ij} = \mathbf{1}_{|\widetilde{H}_{ij}H_{ij}^C| \le 3/4}\widetilde{H}_{ij}. \tag{4.22}$$

By a union bound over the $N(N-1)/2$ off-diagonal and N diagonal positions, we note that

$$\mathbb{P}\left(\max_{i,j}|\widetilde{D}_{ij}H_{ij}^C - \widetilde{H}_{ij}H_{ij}^C| = 0\right) = 1 - o(1). \tag{4.23}$$

Furthermore, we introduce a cutoff on the matrix H^C. Let SelGood be the event that $\#\{(i,j) : H_{ij}^C \neq 0\} \le N^{5\epsilon}$ and that, whenever $H_{ij}^C \neq 0$ and $H_{kl}^C \neq 0$, either $\{i,j\} = \{k,l\}$ or $\{i,j\} \cap \{k,l\} = \emptyset$. It depends only on H^C. Define $\widetilde{H}^C := H^C\,\mathbf{1}_{\mathrm{SelGood}}$. Then:

- The matrix with the cutoff, $\widetilde{H}^C$ coincides with H^C with probability higher than $1 - o(1)$, i.e.,

$$\mathbb{P}(\widetilde{H}^C = H^C) \ge 1 - o(1). \tag{4.24}$$

  Indeed, $\mathbb{E}\#\{(i,j) : H_{ij}^C \neq 0\} = N(N-1)\alpha + N\widetilde{\alpha} = o(N^{4\epsilon})$, so by Markov's inequality the count exceeds $N^{5\epsilon}$ with probability $o(N^{-\epsilon})$. Two distinct index pairs that share an index are either two off-diagonal pairs $\{i,j\}, \{i,k\}$, of probability α^2 and at most N^3 choices, or an off-diagonal pair $\{i,j\}$ and a diagonal position $\{i\}$, of probability $\alpha\widetilde{\alpha}$ and at most N^2 choices. Hence

$\mathbb{P}(\mathrm{SelGood}^c) \le N^{-5\epsilon}\mathbb{E}\#\{(i,j) : H^C_{ij} \neq 0\} + N^3\alpha^2 + N^2\alpha\widetilde{\alpha} = o(N^{-\epsilon}) + o(N^{-1+8\epsilon}) + o(N^{-1+6\epsilon})$, which tends to 0 since $\epsilon < 1/8$.

- The number of non-zero entries are bounded by
$$\#\{(i,j) : \widetilde{H}^C_{ij} \neq 0\} \le N^{5\epsilon}. \tag{4.25}$$
- If $\widetilde{H}^C_{ij} \neq 0$ and $\widetilde{H}^C_{kl} \neq 0$, then either $\{i,j\} = \{k,l\}$ or $\{i,j\} \cap \{k,l\} = \emptyset$.

By (4.23) and (4.24), on an event of probability $1 - o(1)$ we have

$$H = H^S + E + M \quad \text{where} \quad E_{ij} := \widetilde{D}_{ij}\widetilde{H}^C_{ij}.$$

We note that E is symmetric, $\max_{ij}|E_{ij}| \le 3/4$, E has at most $N^{5\epsilon}$ non-zero entries, the non-zero entries of E satisfy the third property above, and the rank of E is at most $N^{5\epsilon}$.

Let λ_N and μ_N be the largest eigenvalue of $\widetilde{H}^S$ and $\widetilde{H}^S + \sigma^{-1}E$, respectively. We claim that

$$\mathbb{P}\left(|\lambda_N - \mu_N| \le N^{-3/4}\right) = 1 - o(1). \tag{4.26}$$

We first show that the claim (4.26) together with (4.20), (4.24), (4.23) and (4.17) implies (4.9). Let $X_H := N^{2/3}(\lambda_N(H) - 2)$ and $X_{\widetilde{S}} := N^{2/3}(\lambda_N(\widetilde{H}^S) - 2)$. Since $H^S + E = \sigma(\widetilde{H}^S + \sigma^{-1}E)$, on the event where $H = H^S + E + M$ we have

$$X_H - X_{\widetilde{S}} = N^{2/3}\big[\lambda_N(H^S + E + M) - \lambda_N(H^S + E)\big] + \sigma N^{2/3}\big[\mu_N - \lambda_N\big] + N^{2/3}(\sigma - 1)\lambda_N.$$

By (4.17), the first term is at most $N^{-1/3+4\epsilon}$. On the event of (4.26), the second term is at most $\sigma N^{-1/12}$. By (3.4) for $\widetilde{H}^S$, $|\lambda_N| \le 3$ with high probability, and then the third term is at most $3N^{2/3}|\sigma^2 - 1| = o(N^{-1/3+2\epsilon})$ by (4.18). Since $\epsilon < 1/30$, all three exponents are smaller than $-1/24$. Hence, by a union bound over the complements of the events in (4.23), (4.24), (4.26) and $\{\|\widetilde{H}^S\| \le 3\}$, we obtain

$$p_N := \mathbb{P}\big(|X_H - X_{\widetilde{S}}| > N^{-1/24}\big) \to 0.$$

Here p_N is a probability on the product space, defined for the N with $\alpha, \widetilde{\alpha} < 1$. Since H has the law of H_N and $\widetilde{H}^S$ has the law $\mathbb{P}^{\widetilde{S}}$, (4.20) at $s + N^{-1/24}$ gives

$$\begin{aligned}\mathbb{P}\big(N^{2/3}(\lambda_N(H_N) - 2) \le s\big) &\le \mathbb{P}\big(X_{\widetilde{S}} \le s + N^{-1/24}\big) + p_N \\ &\le \mathbb{P}^{GOE}\big(N^{2/3}(\lambda_N - 2) \le s + N^{-1/24} + N^{-\delta_0}\big) + N^{-\delta_0} + p_N.\end{aligned}$$

Let $\delta := \frac{1}{2}\min(\delta_0, 1/24)$ and $\epsilon_N := N^{-\delta_0} + p_N$. Then $N^{-1/24} + N^{-\delta_0} \le 2N^{-2\delta} \le N^{-\delta}$ for $N \ge 2^{1/\delta}$, and the monotonicity of the distribution function gives the upper half of (4.9). The lower half follows in the same way from $\{X_{\widetilde{S}} \le s - N^{-1/24}\} \subset \{X_H \le s\} \cup \{|X_H - X_{\widetilde{S}}| > N^{-1/24}\}$ and the lower half of (4.20) at $s - N^{-1/24}$. Neither δ nor ϵ_N depends on s, and since ϵ is fixed, they depend only on ν and $\widetilde{\nu}$. This proves (4.9), i.e., (1.5).

Now we prove (4.26). Let $\widetilde{E} := \sigma^{-1}E$, so that $\max_{ij}|\widetilde{E}_{ij}| \le \theta := 3/(4\sigma)$, and let $G^S(z) := (\widetilde{H}^S - z)^{-1}$; we denote by λ_α and $\mathbf{u}_\alpha$ the eigenvalues and the eigenvectors of $\widetilde{H}^S$. Fix an outcome, and let s be the number of pairs $\{a,b\}$, $a < b$, with $E_{ab} \neq 0$ and t the number of indices a with $E_{aa} \neq 0$. By the third property of $\widetilde{H}^C$, these pairs and indices are pairwise disjoint, and $2s + t \le N^{5\epsilon}$. Relabelling the indices by one permutation applied to both $\widetilde{H}^S$ and $\widetilde{E}$, which changes no eigenvalue, we can assume that among the matrix entries of $\widetilde{E}$, only

$$\widetilde{E}_{12}, \quad \widetilde{E}_{21}, \quad \widetilde{E}_{34}, \quad \widetilde{E}_{43} \cdots, \widetilde{E}_{2s-1,2s}, \quad \widetilde{E}_{2s,2s-1}, \quad \widetilde{E}_{2s+1,2s+1}, \quad \widetilde{E}_{2s+2,2s+2}, \cdots \widetilde{E}_{2s+t,2s+t}$$

are non-zero. The bounds (4.38) and (4.39) below hold for all indices touched by $\widetilde{H}^C$, so they are not affected by this relabelling. Then, we can decompose $\widetilde{E}$ as

$$\widetilde{E} = VDV^* \tag{4.27}$$

where D is a $(2s+t)\times(2s+t)$ diagonal matrix and V is $N\times(2s+t)$ with orthonormal columns. Furthermore,

$$D=\mathrm{diag}\{\widetilde{E}_{12},-\widetilde{E}_{12},\widetilde{E}_{34},-\widetilde{E}_{34},\cdots,\widetilde{E}_{2s-1,2s},-\widetilde{E}_{2s-1,2s},\widetilde{E}_{2s+1,2s+1},\widetilde{E}_{2s+2,2s+2},\cdots\widetilde{E}_{2s+t,2s+t}\} \tag{4.28}$$

is invertible. If $j=2n-1$, $n\le s$, then

$$V_{ij}=\delta_{i,2n-1}\frac{1}{\sqrt{2}}+\delta_{i,2n}\frac{1}{\sqrt{2}} \tag{4.29}$$

If $j=2n$, $n\le s$, then

$$V_{ij}=\delta_{i,2n-1}\frac{1}{\sqrt{2}}-\delta_{i,2n}\frac{1}{\sqrt{2}} \tag{4.30}$$

If $j=2s+n$, $1\le n\le t$, then

$$V_{ij}=\delta_{i,2s+n} \tag{4.31}$$

Note that $\widetilde{E}$ is a symmetric matrix. We use the following form of the matrix determinant lemma: for $\mu\notin\mathrm{spec}(\widetilde{H}^S)$ and $\gamma\neq 0$,

$$\mu\in\mathrm{spec}(\widetilde{H}^S+\gamma\widetilde{E})\iff\det\big(V^*G^S(\mu)V+(\gamma D)^{-1}\big)=0,\quad\text{where}\quad G^S(\mu)=\frac{1}{\widetilde{H}^S-\mu}. \tag{4.32}$$

Indeed, $\widetilde{H}^S+\gamma VDV^*-\mu=(\widetilde{H}^S-\mu)\big(I+G^S(\mu)V\gamma DV^*\big)$ and $\det(I_N+XY)=\det(I_{2s+t}+YX)$, so that

$$\det\Big(\widetilde{H}^S+\gamma\widetilde{E}-\mu\Big)=\det\Big(\widetilde{H}^S-\mu\Big)\det(\gamma D)\det\big(V^*G^S(\mu)V+(\gamma D)^{-1}\big), \tag{4.33}$$

and the first two factors on the right are non-zero.

Define $(2s+t)\times(2s+t)$ matrix F by

$$F^\gamma:=F^\gamma(\mu):=V^*G^S(\mu)V+(\gamma D)^{-1}. \tag{4.34}$$

Then, we have for the following 2×2 blocks of F^γ,

$$\begin{aligned}&\begin{pmatrix}F^\gamma_{2\alpha-1,2\beta-1}&F^\gamma_{2\alpha-1,2\beta}\\F^\gamma_{2\alpha,2\beta-1}&F^\gamma_{2\alpha,2\beta}\end{pmatrix}\\&=\frac{1}{2}\begin{pmatrix}1&1\\1&-1\end{pmatrix}\begin{pmatrix}G^S_{2\alpha-1,2\beta-1}&G^S_{2\alpha-1,2\beta}\\G^S_{2\alpha,2\beta-1}&G^S_{2\alpha,2\beta}\end{pmatrix}\begin{pmatrix}1&1\\1&-1\end{pmatrix}+\delta_{\alpha,\beta}\begin{pmatrix}(\gamma\widetilde{E}_{2\alpha-1,2\alpha})^{-1}&0\\0&-(\gamma\widetilde{E}_{2\alpha-1,2\alpha})^{-1}\end{pmatrix}\end{aligned} \tag{4.35}$$

where $1\le\alpha,\beta\le s$. For the following 2×1 blocks of F, we get

$$\begin{pmatrix}F^\gamma_{2\alpha-1,\,\beta}\\F^\gamma_{2\alpha,\,\beta}\end{pmatrix}=\frac{1}{\sqrt{2}}\begin{pmatrix}1&1\\1&-1\end{pmatrix}\begin{pmatrix}G^S_{2\alpha-1,\,\beta}\\G^S_{2\alpha,\,\beta}\end{pmatrix} \tag{4.36}$$

where $1\le\alpha\le s$, $2s+1\le\beta\le 2s+t$. Finally, for the following 1×1 blocks of F,

$$\big(F^\gamma_{\alpha,\,\alpha}\big)=\big(G^S_{\alpha,\,\alpha}\big)+(\gamma\widetilde{E}_{\alpha,\alpha})^{-1} \tag{4.37}$$

where $2s+1\le\alpha\le 2s+t$.

Let $\widetilde{z}=2+iN^{-2/3}$. From (3.3) for $\widetilde{H}^S$ with $q=N^{\phi_s}$, and (2.11), (2.12), which give $|m_{sc}(\widetilde{z})+1|\le CN^{-1/3}$ and $\mathrm{Im}\,m_{sc}(\widetilde{z})\le CN^{-1/3}$, with high probability we have

$$\max_{1\le i\le N}|G^S_{ii}(\widetilde{z})+1|\le\varphi^C\big(N^{-\phi_s}+N^{-1/3}\big)\le N^{-\epsilon/2}, \tag{4.38}$$

where we used $\phi_s>\epsilon/2$ and $1/3>\epsilon/2$. For off-diagonal terms, let I be the set of indices i such that $\widetilde{H}^C_{ik}\neq 0$ for some k. It is a function of H^C alone, hence independent of $\widetilde{H}^S$, and $|I|\le N^{5\epsilon}$; the set of indices touched by E is contained in I, but $\mathrm{supp}(E)$ itself depends on H^S through $\widetilde{D}$. Conditionally on H^C, for fixed $i\neq j$, (3.15) with $\eta=N^{-2/3}=N^{-1+1/3}$ and (2.12) give $\mathbb{E}|G^S_{ij}(\widetilde{z})|^2\le\varphi^CN^{-2/3}$,

hence $\mathbb{P}(|G^S_{ij}(\widetilde{z})| > N^{-1/6}) \le \varphi^C N^{-1/3}$ by Markov's inequality. Summing over the at most $N^{10\epsilon}$ pairs $i, j \in I$ and integrating over H^C (Fubini), we find that

$$\max_{i\neq j,\ i,j\in I} |G^S_{ij}(\widetilde{z})| \le N^{-1/6} \tag{4.39}$$

fails with probability at most $\varphi^C N^{10\epsilon-1/3}$, so that (4.39) holds with probability $1-o(1)$, since $\epsilon < 1/30$. Define

$$\widetilde{\mu} := \lambda_N + N^{-3/4}. \tag{4.40}$$

Since λ_N is the largest eigenvalue of $\widetilde{H}^S$, we have, for every α,

$$\lambda_\alpha \le \lambda_N = \widetilde{\mu} - N^{-3/4}, \tag{4.41}$$

thus, deterministically,

$$\min_\alpha |\lambda_\alpha - \widetilde{\mu}| \ge N^{-3/4}. \tag{4.42}$$

In particular, $\widetilde{\mu} \notin \operatorname{spec}(\widetilde{H}^S)$. From the rigidity (3.12) for $\widetilde{H}^S$ at $j = N$, where $\gamma_N = 2$, we also have with high probability that

$$|\widetilde{\mu} - 2| \le N^{-2/3}\varphi^C. \tag{4.43}$$

Consider the identity

$$\begin{aligned} \big|G^S_{ij}(\widetilde{z}) - G^S_{ij}(\widetilde{\mu})\big| &= \Big|\sum_\alpha u_\alpha(i)u_\alpha(j)\Big(\frac{1}{\lambda_\alpha - \widetilde{z}} - \frac{1}{\lambda_\alpha - \widetilde{\mu}}\Big)\Big| \\ &\le \sum_\alpha |u_\alpha(i)u_\alpha(j)| \left(\frac{|\widetilde{\mu} - \widetilde{z}|}{|\lambda_\alpha - \widetilde{z}||\lambda_\alpha - \widetilde{\mu}|}\right). \end{aligned} \tag{4.44}$$

Here $|\widetilde{\mu} - \widetilde{z}| \le 2\varphi^C N^{-2/3}$ by (4.43), and $|u_\alpha(i)u_\alpha(j)| \le \varphi^C/N$ by the delocalization (3.5). By the rigidity (3.12), $|\lambda_\alpha - 2| \ge c(N-\alpha+1)^{2/3}N^{-2/3}$ for all α with $|\lambda_\alpha - 2| > 2\varphi^C N^{-2/3}$, and there are at most φ^{2C} indices α with $|\lambda_\alpha - 2| \le 2\varphi^C N^{-2/3}$. For the latter, we use $|\lambda_\alpha - \widetilde{z}| \ge N^{-2/3}$ and (4.42), and each term is at most $2\varphi^{2C}N^{-1/4}$. For the former, $|\lambda_\alpha - \widetilde{\mu}| \ge |\lambda_\alpha - 2|/2$ and $|\lambda_\alpha - \widetilde{z}| \ge |\lambda_\alpha - 2|$, and their sum is at most $C\varphi^{2C}N^{-5/3}\sum_{k\ge1} N^{4/3}k^{-4/3} \le C\varphi^{2C}N^{-1/3}$. Hence, from (4.42), (4.43), the delocalization in Lemma 3.1, and the rigidity result in Theorem 3.6, we have with probability $1-o(1)$ that

$$\max_{i,j} |G^S_{ij}(\widetilde{z}) - G^S_{ij}(\widetilde{\mu})| \le \varphi^C N^{-1/4}. \tag{4.45}$$

Combining (4.38), (4.39), and (4.45), we find that, with probability $1-o(1)$, $|G^S_{ii}(\widetilde{\mu}) + 1| \le a$ for all i and $|G^S_{ij}(\widetilde{\mu})| \le b$ for all $i \neq j$ in I, where $a := N^{-\epsilon/2} + \varphi^C N^{-1/4}$ and $b := N^{-1/6} + \varphi^C N^{-1/4}$. From that $\max_{i,j} |\widetilde{E}_{ij}| \le \theta$, together with (4.35)-(4.37), we have for any $0 < \gamma \le 1$ that

$$\min_i |F^\gamma_{ii}(\widetilde{\mu})| \ge \theta^{-1} - 1 - a - b, \quad \max_i \sum_{j\neq i} |F^\gamma_{ij}(\widetilde{\mu})| \le a + 2(2s+t)\,b \tag{4.46}$$

holds with probability $1-o(1)$. Indeed, in a 2×2 block of (4.35) with $\alpha = \beta$, the diagonal entries are $\pm(\gamma\widetilde{E}_{2\alpha-1,2\alpha})^{-1} + \frac{1}{2}(G^S_{2\alpha-1,2\alpha-1} + G^S_{2\alpha,2\alpha}) \pm G^S_{2\alpha-1,2\alpha}$ and the off-diagonal entries are $\frac{1}{2}(G^S_{2\alpha-1,2\alpha-1} - G^S_{2\alpha,2\alpha})$, while every other entry of F^γ is a combination of at most four off-diagonal entries of G^S with coefficients of absolute value at most 1, the sum of their absolute values being at most 2. Since $\theta^{-1} - 1 = 4\sigma/3 - 1 \to 1/3$, $a \to 0$ and $(2s+t)b \le N^{5\epsilon}(N^{-1/6} + \varphi^C N^{-1/4}) \to 0$ for $\epsilon < 1/30$, the matrix $F^\gamma(\widetilde{\mu})$ is strictly diagonally dominant for large N, and by the Gershgorin theorem

$$\det F^\gamma(\widetilde{\mu}) \neq 0, \qquad 0 < \gamma \le 1.$$

Recall (4.32), then we know that the following event holds with probability $1-o(1)$: $\widetilde{\mu}$ is not an eigenvalue of $\widetilde{H}^S + \gamma\widetilde{E}$ for any $0 < \gamma \le 1$.

If we let μ^γ_N be the largest eigenvalue of $\widetilde{H}^S + \gamma\widetilde{E}$ for $0 \le \gamma \le 1$, then by definition $\lambda_N = \mu^0_N < \widetilde{\mu}$, since λ_N is the largest eigenvalue of $\widetilde{H}^S$. The function $\gamma \mapsto \mu^\gamma_N$ is continuous, and it does not take

the value $\widetilde{\mu}$ on $(0,1]$, hence $\mu_N = \mu_N^1 < \widetilde{\mu} = \lambda_N + N^{-3/4}$. For the lower bound, we use the Rayleigh quotient with the eigenvector $\mathbf{u}_N$ of $\widetilde{H}^S$ for λ_N, the delocalization (3.5) and (4.25):

$$\mu_N \geq \langle \mathbf{u}_N, (\widetilde{H}^S + \widetilde{E})\mathbf{u}_N \rangle = \lambda_N + \sum_{i,j} \widetilde{E}_{ij} u_N(i) u_N(j) \geq \lambda_N - \theta \, \#\{(i,j) : \widetilde{E}_{ij} \neq 0\} \, \frac{\varphi^C}{N} \geq \lambda_N - \varphi^C N^{-1+5\epsilon}.$$

Hence, with probability $1 - o(1)$,

$$\mu_N \in \left[\lambda_N - \varphi^C N^{-1+5\epsilon}, \lambda_N + N^{-3/4}\right), \tag{4.47}$$

and $\varphi^C N^{-1+5\epsilon} \leq N^{-3/4}$ for large N, since $\epsilon < 1/20$. Thus, we have proved (4.26), which implies the desired result (4.9). Neither the lower bound nor the upper bound uses a gap between the largest eigenvalues of $\widetilde{H}^S$ or Remark 3.4. □

5. Basic ideas for Theorem 3.6 and Theorem 3.7

The basic idea of proving Theorem 3.6 and Theorem 3.7 is Green function comparison method, as mentioned in the introduction. To apply the comparison results, first we show that for any H in Theorem 3.6, there exists a matrix $\widetilde{H}$ whose off-diagonal entries have the same first four moments and whose diagonal entries have the same first two moments as those of H, and whose entries are bounded almost surely by $\widetilde{q}^{-1}$, $\widetilde{q} = N^{1/2}/((2C_1+4)C_1 \log N)$. Roughly speaking, the $\widetilde{H}$ has the properties that we need to prove for H and we will use Green function comparison method to show that H has the same properties, since the moments of the entries of H and $\widetilde{H}$ match in this sense.

Lemma 5.1. *Let $N \geq 3$ and let H be a real symmetric random matrix whose entries h_{ij}, $1 \leq i \leq j \leq N$, are independent, have mean zero and satisfy the moment conditions of Theorem 3.6 with a constant C_1 (the bounded support condition is not needed). Then there exists a real symmetric random matrix $\widetilde{H}$ with independent entries $\widetilde{h}_{ij}$, $1 \leq i \leq j \leq N$, such that $|\widetilde{h}_{ij}| \leq \widetilde{q}^{-1}$ almost surely for all i, j, where*

$$\widetilde{q} := \frac{N^{1/2}}{(2C_1+4)\, C_1 \log N},$$

the first four moments of the off-diagonal entries of H and $\widetilde{H}$ match, i.e.,

$$\mathbb{E} h_{ij}^k = \mathbb{E} \widetilde{h}_{ij}^k \ (i \neq j), \quad k = 1, 2, 3, 4, \tag{5.1}$$

and the first two moments of the diagonal entries of H and $\widetilde{H}$ match, i.e., $\mathbb{E} h_{ii}^k = \mathbb{E} \widetilde{h}_{ii}^k$, $k = 1, 2$. In particular, $\widetilde{H}$ is a generalized Wigner matrix satisfying (2.1) with $C_0 = \max(C_1, 1)$.

Proof of Lemma 5.1. We choose the entries $\widetilde{h}_{ij}$, $i \leq j$, independently. First, $C_1 \geq 1$: for $i < j$, by the Lyapunov inequality, $N^{-3/2} = (\mathbb{E} h_{ij}^2)^{3/2} \leq \mathbb{E}|h_{ij}|^3 \leq C_1 N^{-3/2}$. Since $N \geq 3$, also $\log N > 1$, so $\widetilde{q}^{-1} \geq 6 C_1 N^{-1/2}$.

Off-diagonal entries. For $i < j$ let $x := N^{1/2} h_{ij}$, so that $\mathbb{E} x = 0$ and $\mathbb{E} x^2 = 1$, and let $A := \mathbb{E} x^3$ and $B := \mathbb{E} x^4$. Then $|A| \leq \mathbb{E}|x|^3 \leq C_1$ and $B \leq C_1 \log N$, and $B \geq A^2 + 1$ by Remark 5.3. Lemma 5.2 with $C = C_1$ gives a random variable $\widetilde{x}$ with $\mathbb{E}\widetilde{x}^k = \mathbb{E} x^k$ for $k = 1, \ldots, 4$ and $|\widetilde{x}| \leq (2C_1+4)B \leq (2C_1+4)C_1 \log N$ almost surely. Let $\widetilde{h}_{ij} := N^{-1/2} \widetilde{x}$. Then $|\widetilde{h}_{ij}| \leq \widetilde{q}^{-1}$ and $\mathbb{E}\widetilde{h}_{ij}^k = \mathbb{E} h_{ij}^k$ for $k = 1, \ldots, 4$.

Diagonal entries. Let $\widetilde{h}_{ii} := \sigma_i \epsilon_i$, where $\sigma_i := (\mathbb{E} h_{ii}^2)^{1/2}$ and $\mathbb{P}(\epsilon_i = 1) = \mathbb{P}(\epsilon_i = -1) = 1/2$. Then $\mathbb{E}\widetilde{h}_{ii} = 0 = \mathbb{E} h_{ii}$, $\mathbb{E}\widetilde{h}_{ii}^2 = \mathbb{E} h_{ii}^2$, and $|\widetilde{h}_{ii}| = \sigma_i \leq (C_1/N)^{1/2} \leq C_1 N^{-1/2} \leq \widetilde{q}^{-1}$.

The last claim follows from $\mathbb{E}\widetilde{h}_{ij}^2 = N^{-1}$ for $i \neq j$ and $0 \leq \mathbb{E}\widetilde{h}_{ii}^2 = \mathbb{E} h_{ii}^2 \leq C_1/N$. □

Lemma 5.2. *For any $C \geq 0$, if $|A| \leq C$ and $B \geq A^2 + 1$, there exists a random variable x such that*

$$\mathbb{E}(x) = 0, \quad \mathbb{E}(x^2) = 1, \quad \mathbb{E}(x^3) = A, \quad \mathbb{E}(x^4) = B, \tag{5.2}$$

and

$$\operatorname{supp}(x) \subset [-DB, DB], \qquad D = 2C + 4. \tag{5.3}$$

Remark 5.3. The condition $B \geq A^2 + 1$ comes from the simple fact that if $\mathbb{E}(x) = 0$ and $\mathbb{E}(x^2) = 1$ then $\mathbb{E}(x^4) \geq 1 + (\mathbb{E}(x^3))^2$.

Proof. First we introduce a family of random variables Y_t ($t \geq 1$), whose distribution has a finite support

$$\mathrm{supp}(Y_t) \subset \left\{-t, \frac{-\sqrt{t}}{\sqrt{1+t}}, \frac{\sqrt{t}}{\sqrt{1+t}}, t\right\} \tag{5.4}$$

and satisfies

$$\mathbb{P}(Y_t = t) = \mathbb{P}(Y_t = -t) = \frac{1}{2t(-1+t+t^2)}\,, \tag{5.5}$$

$$\mathbb{P}(Y_t = \frac{\sqrt{t}}{\sqrt{1+t}}) = \mathbb{P}(Y_t = \frac{-\sqrt{t}}{\sqrt{1+t}}) = \frac{1}{2} - \frac{1}{2t(-1+t+t^2)}\,. \tag{5.6}$$

These are probabilities since $t(t^2+t-1) \geq 1$ for $t \geq 1$; for $t = 1$, Y_1 takes the values ± 1 with probability $1/2$. One can easily check that every odd moment of Y_t vanishes and

$$\mathbb{E}(Y_t)^2 = 1, \quad \mathbb{E}(Y_t)^4 = t. \tag{5.7}$$

Note that Y_t is supported in $[-t, t]$.

Next, for $b \in \mathbb{R}$ let $u < 0 < v$ be the roots of $y^2 - by - 1 = 0$, so that $u + v = b$ and $uv = -1$, and let μ_b be the law with $\mu_b(\{u\}) = v/(v-u)$ and $\mu_b(\{v\}) = -u/(v-u)$. If y has law μ_b, then $\mathbb{E}y = 0$, $\mathbb{E}y^2 = -uv = 1$, and $y^2 = by + 1$ gives $\mathbb{E}y^3 = b$ and $\mathbb{E}y^4 = b^2 + 1$. Moreover $|u|, |v| \leq (|b| + \sqrt{b^2+4})/2 \leq |b| + 1$.

We choose another random variable X, independent of Y_t, whose distribution is supported on

$$\{\sqrt{2}A - \sqrt{1+2A^2}, \sqrt{2}A + \sqrt{1+2A^2}\}$$

and satisfies

$$\mathbb{P}(X = \sqrt{2}A - \sqrt{1+2A^2}) = \frac{\sqrt{2}A + \sqrt{1+2A^2}}{2\sqrt{1+2A^2}}, \tag{5.8}$$

$$\mathbb{P}(X = \sqrt{2}A + \sqrt{1+2A^2}) = \frac{-\sqrt{2}A + \sqrt{1+2A^2}}{2\sqrt{1+2A^2}}. \tag{5.9}$$

This is the law μ_b with $b = 2\sqrt{2}A$. Hence

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad \mathbb{E}X^3 = 2\sqrt{2}A, \quad \mathbb{E}X^4 = 8A^2 + 1,$$

and $|X| \leq 2\sqrt{2}|A| + 1 \leq 2\sqrt{2}C + 1$.

Case $B \geq 2A^2 + 2$. Choose

$$t = 4B - 8A^2 - 7 \ \ (\geq 1)$$

and let

$$x = \frac{X + Y_t}{\sqrt{2}}\,.$$

Since X and Y_t are independent, $\mathbb{E}X = 0$ and the odd moments of Y_t vanish, we have $\mathbb{E}x = 0$, $\mathbb{E}x^2 = (\mathbb{E}X^2 + \mathbb{E}Y_t^2)/2 = 1$, $\mathbb{E}x^3 = \mathbb{E}X^3/2^{3/2} = A$ and

$$\mathbb{E}x^4 = \frac{\mathbb{E}X^4 + 6\,\mathbb{E}X^2\,\mathbb{E}Y_t^2 + \mathbb{E}Y_t^4}{4} = \frac{8A^2 + 7 + t}{4} = B.$$

Since $B \geq 1$ and $t \leq 4B$, $|x| \leq (|X| + t)/\sqrt{2} \leq 2C + 1 + 2\sqrt{2}B \leq (2C+4)B$.

Case $A^2 + 1 \leq B < 2A^2 + 2$. Let x_0 be the variable of the first case for $B = 2A^2 + 2$, i.e., $t = 1$. It has the moments $0, 1, A, 2A^2 + 2$ and $|x_0| \leq (2\sqrt{2}C + 2)/\sqrt{2} \leq 2C + 2$. Let

$$\theta := \frac{2A^2 + 2 - B}{A^2 + 1} \in (0, 1],$$

and let x have the law $\theta\mu_A + (1-\theta)\,\mathrm{law}(x_0)$. Both components have the moments $0, 1, A$ of orders $1, 2, 3$, so x has them too, and $\mathbb{E}x^4 = \theta(A^2+1) + (1-\theta)(2A^2+2) = B$. The atoms of μ_A are bounded by $|A| + 1 \le C + 1$, so $|x| \le 2C + 2 \le (2C+4)B$. □

Since $\widetilde{H}$ has the property we need (see Lemma 3.1), we now compare H with $\widetilde{H}$ using Green function comparison method.

To prove Theorem 3.6 and Theorem 3.7, we claim the following two lemmas first, which will be proved in the next section. We say that $(H, \widetilde{H})$ is a *matched pair* with constant C_1 and support q if H and $\widetilde{H}$ are real symmetric random matrices with independent entries h_{ij}, $\widetilde{h}_{ij}$ $(i \le j)$, all bounded by q^{-1} almost surely, such that $\mathbb{E}h_{ij} = 0$, $\mathbb{E}h_{ij}^2 = N^{-1}$ $(i \ne j)$, $\mathbb{E}h_{ii}^2 \le C_1/N$ and $\mathbb{E}h_{ij}^4 \le C_1(\log N)N^{-2}$ $(i \ne j)$, and such that the moments of orders $1, \dots, 4$ of the off-diagonal entries and of orders $1, 2$ of the diagonal entries of $\widetilde{H}$ are those of H. No bound on the third moments is required.

Lemma 5.4. *Fix C_1 and $\phi' > 0$. There is a constant $C_* > 0$, depending only on (C_1, ϕ'), such that for all $c', C' > 0$ there are constants $C > 0$ and N_0, depending only on (C_1, ϕ', c', C'), with the following property. Let $N \ge N_0$, let $(H, \widetilde{H})$ be a matched pair with constant C_1 and support $q \ge N^{\phi'}$, let $z \in \mathbf{S}(C_*)$ (see (3.1)), and let $X, Y \ge 0$ be deterministic numbers such that the event*

$$\max_{i \ne j} |\widetilde{G}_{ij}(z)| \le X, \quad |\widetilde{m}(z) - m_{sc}(z)| \le Y \tag{5.10}$$

fails with probability at most $N^{C'}e^{-c'\varphi^3}$. Then for any $p \in 2\mathbb{N}$ with $p \le \varphi$, we have

$$\mathbb{E}|m - m_{sc}|^p \le \mathbb{E}|\widetilde{m} - m_{sc}|^p + (Cp)^{Cp}\left(X^2 + Y + \varphi N^{-1}\right)^p. \tag{5.11}$$

If H satisfies the conditions of Theorem 3.6 with (C_1, ϕ) and $\widetilde{H}$ has the properties of Lemma 5.1, then $(H, \widetilde{H})$ is a matched pair with constant C_1 and support $q = N^{\min(\phi, 1/4)}$ for $N \ge N_0(C_1)$, since then $\widetilde{q} \ge N^{1/4}$. If $(H, \widetilde{H})$ is a matched pair, then so is $(H_{\gamma_0}, \widetilde{H})$ for every γ_0, where H_{γ_0} is the matrix of (3.21) built from $H^{\mathbf{v}} = H$ and $H^{\mathbf{w}} = \widetilde{H}$: each entry of H_{γ_0} has the law of the corresponding entry of H or of $\widetilde{H}$. So Lemma 5.4 also holds with H replaced by H_{γ_0}.

Lemma 5.5. *Fix $C_F > 0$, C_1 and $\phi > 0$. There exist constants $\widetilde{\epsilon} > 0$ and $C > 0$, depending only on (C_F, C_1, ϕ), such that for every $0 < \epsilon < \widetilde{\epsilon}$ there is N_0, depending only on $(C_F, C_1, \phi, \epsilon)$, with the following property. Let $N \ge N_0$, let H satisfy the conditions of Theorem 3.6 with (C_1, ϕ), let $\widetilde{H}$ have the properties of Lemma 5.1, and let $F \in C^3(\mathbb{R})$ be a real function whose derivatives satisfy*

$$\sup_x |F^{(\alpha)}(x)|(1+|x|)^{-C_F} \le C_F, \quad \alpha = 1, 2, 3. \tag{5.12}$$

Let

$$I_\epsilon := \{x : |x - 2| \le N^{-2/3+\epsilon}\}, \qquad \eta := N^{-2/3-\epsilon}. \tag{5.13}$$

Then for every $E \in I_\epsilon$,

$$\left|\mathbb{E}\,F\left(\eta^2 \sum_{i,j} G_{ij}\overline{G_{ij}}(z)\right) - \mathbb{E}\,F\left(\eta^2 \sum_{i,j} \widetilde{G}_{ij}\overline{\widetilde{G}_{ij}}(z)\right)\right| \le N^{-\phi+C\epsilon}, \quad z = E + i\eta, \tag{5.14}$$

and for all $E_1, E_2 \in I_\epsilon$, in either order ($\int_{E_1}^{E_2}$ is the oriented integral),

$$\left|\mathbb{E}F\left(N\int_{E_1}^{E_2} dy\,\mathrm{Im}\,m(y+i\eta)\right) - \mathbb{E}F\left(N\int_{E_1}^{E_2} dy\,\mathrm{Im}\,\widetilde{m}(y+i\eta)\right)\right| \le N^{-\phi+C\epsilon}. \tag{5.15}$$

Proof of Theorem 3.6. Throughout the proof, H satisfies the conditions of Theorem 3.6 with (C_1, ϕ), constants depend only on (C_1, ϕ), N is large enough, and "with high probability" is uniform over these H. Since $0 \le \mathbb{E}h_{ii}^2 \le C_1/N$, H satisfies (2.1) with $C_0 = \max(C_1, 1)$, and its entries are bounded by $N^{-\phi}$ almost surely, so the bounded support condition holds with $q = N^\phi$ and every $c > 0$. Hence

Lemma 3.1 applies to H with $q = N^\phi$, and, with (3.4), we have that for some large C, with high probability,

$$\|H\| \le 2 + \varphi^C(N^{-2/3} + N^{-2\phi}); \tag{5.16}$$

in particular $\|H\| \le 3$. First, we improve this result to

$$\lambda_N \le 2 + \varphi^C N^{-2/3} \tag{5.17}$$

with high probability.

Let $\widetilde{H}$ match H in the sense of Lemma 5.1. By Lemma 5.1, $\widetilde{H}$ satisfies (2.1) with $C_0 = \max(C_1, 1)$, and $|\widetilde{h}_{ij}| \le \widetilde{q}^{-1} \le N^{-2/5}$ almost surely, so Lemma 3.1 applies to $\widetilde{H}$ with $q = N^{2/5}$, including the items 4 and 5, at every level ζ (Remark 3.2). The pair $(H, \widetilde{H})$ is a matched pair with support $N^{\min(\phi,1/4)}$. Let C_* and C_s be the constants C_* and C of Lemma 5.4, with $\phi' = \min(\phi, 1/4)$, at the level $(c', C') = (1, 1)$. Let K be the constant of (3.7)–(3.8), and fix

$$a := 49 + 3C_s + C_* \ (> 48 = 12\zeta \text{ for } \zeta = 4).$$

Let C_o be the constant C of (3.7)–(3.8) for this a at $\zeta = 4$, and let $C_t := C_o + 2K + 3a + 2$. Then, with (3.7) and (3.8), for $z = E + i\eta$ satisfying

$$2 + \varphi^{C_t} N^{-2/3} \le E \le 3, \quad \eta = \varphi^{K+a} N^{-1} \kappa^{-1/2}, \tag{5.18}$$

we have, with 4-high probability and simultaneously for all such E,

$$|\widetilde{m} - m_{sc}| \le \frac{1}{\varphi^a N\eta}, \quad \operatorname{Im} \widetilde{m}(z) \le \frac{1}{\varphi^a N\eta}, \quad \max_{i \ne j} |\widetilde{G}_{ij}| \le \frac{1}{\varphi^{a/3} N\eta}. \tag{5.19}$$

Let c'', C'' be the constants c, C of the 4-high probability of (5.19). The complement of (5.19) has probability at most $N^{C''} e^{-c''\varphi^4} \le Ne^{-\varphi^3}$ for N large. Hence (5.10) with $X = (\varphi^{a/3} N\eta)^{-1}$ and $Y = (\varphi^a N\eta)^{-1}$ fails with probability at most $Ne^{-\varphi^3}$, which is the level $(c', C') = (1, 1)$ of Lemma 5.4. The constants C_* and C_s do not depend on a, so the choice of a is not circular. (At $\zeta = 3$ the level of (5.19) would depend on a, and then so would C_s.)

Fix E as in (5.18), so that $\kappa = E - 2 \in [\varphi^{C_t} N^{-2/3}, 1]$. Then $N\eta = \varphi^{K+a}\kappa^{-1/2} \ge \varphi^{K+a} > \varphi^{C_*}$, so $z \in \mathbf{S}(C_*)$; $\eta \le \varphi^{K+a-C_t/2} N^{-2/3} \le N^{-1/2}$; and $\eta \le \kappa$, since $\kappa^{3/2} \ge \varphi^{3C_t/2} N^{-1} \ge \varphi^{K+a} N^{-1}$. With the property of m_{sc} in (2.12) for $\kappa \ge \eta$, we have $\operatorname{Im} m_{sc}(z) \le C\eta/\sqrt{\kappa}$, and $C\varphi N\eta^2\kappa^{-1/2} = C\varphi^{1+2(K+a)} N^{-1}\kappa^{-3/2} \le C\varphi^{1+2(K+a)-3C_t/2} \le 1$, since $3C_t/2 \ge 2(K+a)+2$. This implies

$$\operatorname{Im} m_{sc}(z) \le \frac{1}{\varphi N\eta}. \tag{5.20}$$

Now, we apply Lemma 5.4 on H and $\widetilde{H}$ with z in (5.18), the even integer $p = 2\lfloor \varphi/2 \rfloor$ (so that $\varphi - 2 \le p \le \varphi$), $X = (\varphi^{a/3} N\eta)^{-1}$ and $Y = (\varphi^a N\eta)^{-1}$. Let $T := (5N\eta)^{-1}$. Since $(C_s p)^{C_s} \le \varphi^{2C_s}$, the last term of (5.11) is at most $\big(\varphi^{2C_s}(X^2 + Y + \varphi N^{-1})\big)^p$, and, using $N\eta \ge 1$ and $\eta \le N^{-1/2}$,

$$\varphi^{2C_s} Y = \frac{\varphi^{2C_s - a}}{N\eta}, \qquad \varphi^{2C_s} X^2 \le \frac{\varphi^{2C_s - 2a/3}}{N\eta}, \qquad \varphi^{2C_s} \frac{\varphi}{N} = \frac{\varphi^{2C_s+1}\eta}{N\eta} \le \frac{\varphi^{2C_s+1} N^{-1/2}}{N\eta}.$$

Since $a \ge 3C_s + 49$, each of these is at most $\varphi^{-1}(N\eta)^{-1}$, so the last term of (5.11) is at most T^p. Moreover $|\widetilde{m} - m_{sc}| \le \eta^{-1} + 1 \le 2\eta^{-1} = 10NT$, so by (5.19)

$$\mathbb{E}|\widetilde{m} - m_{sc}|^p \le Y^p + (10N)^p T^p \, Ne^{-\varphi^3} \le 2T^p,$$

since $Y \le T$ and $(10N)^p Ne^{-\varphi^3} \le 1$ for $p \le \varphi$ and $\log N \le \varphi$. Thus $\mathbb{E}|m - m_{sc}|^p \le 3T^p$, and Markov inequality gives

$$\mathbb{P}\Big(|m(z) - m_{sc}(z)| \ge \frac{2}{5N\eta}\Big) \le 3 \cdot 2^{-p} \le 12 \cdot 2^{-\varphi} \le Ne^{-(\log 2)\varphi}. \tag{5.21}$$

Here we used that, from (5.18),

$$\varphi^{C_t/2 - K - a} N^{-1/3} \le (N\eta)^{-1} = \varphi^{-K-a}\kappa^{1/2} \le \varphi^{-K-a}. \tag{5.22}$$

Outside the event of (5.21), by (5.20) and $\varphi \geq 10$, $\operatorname{Im} m(z) \leq \operatorname{Im} m_{sc}(z) + |m(z) - m_{sc}(z)| < (\frac{1}{10} + \frac{2}{5})(N\eta)^{-1} = (2N\eta)^{-1}$. By definition,

$$\operatorname{Im} m = N^{-1} \sum_{\alpha} \eta((\lambda_\alpha - E)^2 + \eta^2)^{-1}.$$

If $|\lambda_\alpha - E| \leq \eta$ for some α, the corresponding term is at least $(2\eta)^{-1}$, so $\operatorname{Im} m(z) \geq (2N\eta)^{-1}$. Hence $\operatorname{Im} m(E + i\eta) < (2N\eta)^{-1}$ implies that there are no eigenvalues in the interval $[E - \eta, E + \eta]$. Let $b := 2 + \varphi^{C_t} N^{-2/3}$, and apply this at the points $E_k := b + k/N$, $0 \leq k < N$, with $E_k \leq 3$; at each of them $\eta \geq N^{-1}$. By a union bound over at most N points, with probability at least $1 - N^2 e^{-(\log 2)\varphi}$ no eigenvalue lies within N^{-1} of any E_k. Every point of $[b, 3]$ lies within N^{-1} of some E_k, so with high probability there are no eigenvalues in $[b, 3]$. Together with $\|H\| \leq 3$ from (5.16), we obtain (5.17) with $C = C_t$.

The class of Theorem 3.6 is invariant under the reflection $H \mapsto -H$: the entries $-h_{ij}$ are independent, have mean zero, the same moments of orders 2 and 4, the same $\mathbb{E}|h_{ij}|^3$ and the same almost sure bound. The eigenvalues of $-H$ are $-\lambda_\alpha$, so (5.17) for $-H$ gives $\lambda_1 \geq -2 - \varphi^C N^{-2/3}$ with high probability. Hence we have, with high probability,

$$\|H\| \leq 2 + \varphi^C N^{-2/3}. \tag{5.23}$$

Next, we prove the local law for H. Let C_l be the constant C of Lemma 3.1 at $\zeta = 3$ (with $C_0 = \max(C_1, 1)$ and $c = 1$), and let (c_l, C_l') be the constants of Definition 2.3 there. Since $\widetilde{q} \geq \varphi^{C_l}$ for N large, (3.2) and (3.3) hold for $\widetilde{H}$ with $q = \widetilde{q}$, simultaneously for all $z \in \mathbf{S}(C_l)$, outside an event of probability at most $N^{C_l'} e^{-c_l \varphi^3}$. Let C_* and C_s' be the constants C_* and C of Lemma 5.4, with $\phi' = \min(\phi, 1/4)$, at the level $(c', C') = (c_l, C_l')$; they are fixed before X and Y are chosen. For $z \in \mathbf{S}(\max(C_*, C_l))$ let

$$X = \varphi^{C_l}\Big(\frac{1}{\widetilde{q}} + \sqrt{\frac{\operatorname{Im} m_{sc}}{N\eta}} + \frac{1}{N\eta}\Big), \quad Y = \varphi^{C_l}\Big(\frac{1}{\widetilde{q}^2\sqrt{\kappa + \eta}} + \frac{1}{N\eta}\Big),$$

where X, Y follow from (3.2)-(3.3) (in (3.2) we bound the minimum by its first term). With $D := (2C_1 + 4)C_1$ we have $\widetilde{q}^{-2} = D^2(\log N)^2 N^{-1}$, and since $N\eta \geq 1$, $\eta \leq 10$ and $\operatorname{Im} m_{sc} \leq 1$,

$$\frac{1}{\widetilde{q}^2} \leq \frac{10D^2(\log N)^2}{N\eta}, \qquad \frac{1}{\widetilde{q}^2\sqrt{\kappa + \eta}} \leq \frac{D^2(\log N)^2\eta^{1/2}}{N\eta} \leq \frac{\sqrt{10}\, D^2(\log N)^2}{N\eta}.$$

Hence $X^2 + Y + \varphi N^{-1} \leq \varphi^{2C_l+4}(N\eta)^{-1}$ for N large. (The other term $\widetilde{q}^{-1}$ of the minimum in (3.2) is of order $N^{-1/2}\log N$; it is not bounded by $\varphi^C(N\eta)^{-1}$ when η is of order 1.) Let $p = 2\lfloor \varphi/2 \rfloor$ and $T := \varphi^{2C_s'+2C_l+4}(N\eta)^{-1}$. As above, the last term of (5.11) is at most T^p; and $|\widetilde{m} - m_{sc}| \leq \eta^{-1} + 1 \leq N + 1$, so $\mathbb{E}|\widetilde{m} - m_{sc}|^p \leq Y^p + (N+1)^p N^{C_l'} e^{-c_l\varphi^3} \leq 2T^p$. With (5.11) and Markov inequality, for each $z \in \mathbf{S}(\max(C_*, C_l))$,

$$\mathbb{P}\big(|m(z) - m_{sc}(z)| \geq 2T\big) \leq 3 \cdot 2^{-p} \leq Ne^{-(\log 2)\varphi}.$$

To obtain this for all z at once, let $\mathcal{L}$ be the set of the points $w = (a' + ib')N^{-2}$ with $a', b' \in \mathbb{Z}$, $|a'| \leq 5N^2$, $1 \leq b' \leq 10N^2$ and $\operatorname{Im} w > \varphi^{\max(C_*, C_l)}/N$; it has at most $(10N^2 + 1)10N^2 \leq N^5$ points. By a union bound, with probability at least $1 - N^6 e^{-(\log 2)\varphi}$ the bound $|m(w) - m_{sc}(w)| < 2T(w)$ holds for all $w \in \mathcal{L}$. Every $z \in \mathbf{S}(\max(C_*, C_l))$ has a point $w \in \mathcal{L}$ with $|z - w| \leq 2N^{-2}$ and $\operatorname{Im} w \geq \operatorname{Im} z$, so that $T(w) \leq T(z)$. Since m and m_{sc} are Stieltjes transforms of probability measures, $|m(z) - m(w)|$ and $|m_{sc}(z) - m_{sc}(w)|$ are at most $|z - w|/(\operatorname{Im} z \operatorname{Im} w) \leq 2(N\eta)^{-2} \leq 2(N\eta)^{-1}$. Hence, with $C := \max(C_*, C_l, 2C_s' + 2C_l + 5)$ and $\varphi \geq 6$, we have that

$$|m(z) - m_{sc}(z)| \leq \varphi^C(N\eta)^{-1} \quad \text{for all } z \in \mathbf{S}(C) \tag{5.24}$$

holds with high probability. Since $(H_{\gamma_0}, \widetilde{H})$ is again a matched pair, the same argument gives (5.24), with the same constants, for every matrix H_{γ_0} of (3.21) built from $H^{\mathbf{v}} = H$ and $H^{\mathbf{w}} = \widetilde{H}$; this is used in Section 6.

Finally, we derive (3.12) and (3.13) from (5.23) and (5.24), following the argument of section 5 in [26] with explicit constants. Let C_a and C_b be the constants C of (5.23) and (5.24), and work on the intersection of these two events. Let $U := \varphi^{C_b}$ and $y_0 := 2U/N$, so that $N^{-1} \le y_0 \le 1$, and $\|H\| \le 3$. For $|x| \le 4$ and $y_0 \le y \le 1$, the point $x + iy$ lies in $\mathbf{S}(C_b)$, so $|m(x+iy) - m_{sc}(x+iy)| \le U/(Ny)$. The argument of (5.4)–(5.11) in [26], with $K = 3$, the window $|x| \le 4$ and $E_1 = -3.5$ in the notation there, then gives for every $E \in \mathbb{R}$

$$\big|\mathfrak{n}(E) - n_{sc}(E)\big| \le \frac{100\,(U + Ny_0)\log N}{N} = \frac{300\,\varphi^{C_b}\log N}{N} \le \frac{\varphi^{C_b+2}}{N}.$$

This is (3.13), for all $E \in \mathbb{R}$. Let $t := \varphi^{\max(C_b+2, C_a)}$. Then $\sup_E |\mathfrak{n}(E) - n_{sc}(E)| \le t/N$ and, by (5.23), $\|H\| \le 2 + tN^{-2/3}$. The argument of (5.12)–(5.15) in [26] then gives, with an absolute constant C_e and for all j,

$$|\lambda_j - \gamma_j| \le C_e\, t^2 \big[\min(j, N-j+1)\big]^{-1/3} N^{-2/3} \le \varphi^{2\max(C_b+2,C_a)+1} \big[\min(j, N-j+1)\big]^{-1/3} N^{-2/3}.$$

This is (3.12), for all j simultaneously, with the loss t^2 written out. Both bounds hold on an event of high probability, which completes the proof of Theorem 3.6. □

Proof of Theorem 3.7. We first state the following lemma, whose proof is that of the edge universality in [26]. Here $\mathcal{N}(a,b)$ denotes the number of eigenvalues in $[a,b]$, and $\mathcal{N}(a,\infty)$ the number of eigenvalues in $[a,\infty)$.

Lemma 5.6. *Let $\kappa > 0$ and $C_0 > 0$. For each N, let $\mathcal{P}_N$ be a set of pairs $(H^{\mathbf{v}}, H^{\mathbf{w}})$ of real symmetric $N \times N$ random matrices with independent entries h_{ij}, $1 \le i \le j \le N$, whose laws are arbitrary and may depend on N. Assume that:*

(1) *for every $C \ge C_0$, each of $H^{\mathbf{v}}$ and $H^{\mathbf{w}}$ satisfies with high probability, uniformly over $\mathcal{P}_N$ (the constants of Definition 2.3 do not depend on the pair): $\|H\| \le 3$, (5.24) for all $z \in \mathbf{S}(C)$ with $\eta \le N^{-2/3}$,*

$$|N^{2/3}(\lambda_N - 2)| \le \varphi^C \tag{5.25}$$

and the number of eigenvalues in $[2 - 2\varphi^C N^{-2/3}, 2 + 2\varphi^C N^{-2/3}]$ is bounded as follows:

$$\mathcal{N}\left(2 - \frac{2\varphi^C}{N^{2/3}}, 2 + \frac{2\varphi^C}{N^{2/3}}\right) \le \varphi^{2C}; \tag{5.26}$$

(2) *for every $C_F > 0$ there are $\widetilde{\epsilon} > 0$ and $C' > 0$ such that for every $0 < \epsilon < \widetilde{\epsilon}$ there is N_0 such that (5.15), with m and $\widetilde{m}$ the Stieltjes transforms of $H^{\mathbf{v}}$ and $H^{\mathbf{w}}$, holds with right side $N^{-\kappa+C'\epsilon}$ for all $N \ge N_0$, all pairs in $\mathcal{P}_N$, all $F \in C^4(\mathbb{R})$ satisfying (5.12) for $\alpha = 1,2,3,4$ with constant C_F, and all $E_1, E_2 \in I_\epsilon$.*

Then there exist $\delta > 0$ and N_0, depending only on κ, C_0 and the constants in (1) and (2), such that for all $N \ge N_0$, all pairs in $\mathcal{P}_N$ and all $s \in \mathbb{R}$,

$$\mathbb{P}^{\mathbf{w}}\Big(N^{2/3}(\lambda_N - 2) \le s - N^{-\delta}\Big) - N^{-\delta} \le \mathbb{P}^{\mathbf{v}}\big(N^{2/3}(\lambda_N - 2) \le s\big) \le \mathbb{P}^{\mathbf{w}}\Big(N^{2/3}(\lambda_N - 2) \le s + N^{-\delta}\Big) + N^{-\delta}. \tag{5.27}$$

Proof of Lemma 5.6. Though this lemma is not explicitly stated in [26], this is the basic structure of proving the edge universality, Theorem 2.4, in [26]; it is also the structure of the proof of Proposition 6.4 of [16]. The argument of section 6 of [26] uses only the hypotheses (1) and (2); it uses neither the independence of the entries nor the generalized Wigner structure. Hypothesis (2) plays the role of Theorem 6.3 of [26], there with test functions with four derivatives; the conditions (5.12)–(5.15) are not literally those of Theorem 6.3 of [26]. We give the steps with the parameters.

Let $C \ge C_0$, and for a matrix H of a pair let Ω_C denote the event of (1) with the constant C. Let ψ be a smooth function with $0 \le \psi \le 1$, $\psi(x) = 1$ for $|x| \le 1/9$ and $\psi(x) = 0$ for $|x| \ge 2/9$, and let

C_F be such that $F(x) := \psi(x/\pi)$ satisfies (5.12) for $\alpha = 1, \dots, 4$. Let $\widetilde{\epsilon}$ and C' be given by (2) for this C_F, and set

$$\epsilon := \min\Big(\frac{1}{20}, \frac{\widetilde{\epsilon}}{18}, \frac{\kappa}{36C'}, \frac{\kappa}{2}\Big), \quad \eta := N^{-2/3-9\epsilon}, \quad \ell_1 := N^{-2/3-3\epsilon}, \quad \ell := \frac{1}{2}N^{-2/3-\epsilon}, \quad \delta := \frac{\epsilon}{2}.$$

Let $E_L := 2 + 2\varphi^{C+1}N^{-2/3}$, $\theta_\eta(x) := \eta/(\pi(x^2+\eta^2))$, and, for $a \le E_L$,

$$\operatorname{Tr}(\chi_a * \theta_\eta)(H) := \sum_\alpha \int_a^{E_L} \theta_\eta(\lambda_\alpha - y)\, dy = \frac{N}{\pi}\int_a^{E_L} \operatorname{Im} m(y + i\eta)\, dy.$$

Step 1 (Lemma 6.1 of [26]). On Ω_{C+1}, for $|E-2| \le \frac{3}{2}\varphi^{C+1}N^{-2/3}$ and N large, $|\mathcal{N}(E, E_L) - \operatorname{Tr}(\chi_E * \theta_\eta)(H)| \le 10\big(N^{-2\epsilon} + \mathcal{N}(E-\ell_1, E+\ell_1)\big)$. This is (6.10)–(6.18) of [26]: the number of eigenvalues in $[E_L - \ell_1, \infty)$ vanishes by (5.25); the term $(\eta/\ell_1)\mathcal{N}(E, E_L)$ is at most $N^{-2\epsilon}$ by (5.26); the bound on $\operatorname{Im} m(\cdot + i\ell_1)$ near E uses (5.24) at $\eta = \ell_1 \le N^{-2/3}$ and (2.12); and the region $|E-y| > 5$ in (6.17) of [26] is controlled by $\|H\| \le 3$, which is needed there.

Step 2 (Corollary 6.2 of [26]). Averaging Step 1 over the left endpoint in $[E-\ell, E+\ell]$, and using (5.26) for the number of eigenvalues in this window, one obtains on Ω_{C+1}, for $|E-2| \le \varphi^{C+1}N^{-2/3}$ and N large,

$$\operatorname{Tr}(\chi_{E+\ell} * \theta_\eta)(H) - N^{-\epsilon} \le \mathcal{N}(E, \infty) \le \operatorname{Tr}(\chi_{E-\ell} * \theta_\eta)(H) + N^{-\epsilon}.$$

Step 3 ((6.21) of [26]). $\operatorname{Tr}(\chi_a * \theta_\eta)(H)$ is nonnegative and nonincreasing in a, and $\mathcal{N}(E, \infty) = 0$ exactly when $\lambda_N < E$. Since $N^{-\epsilon} \le 1/9$, Step 2 gives on Ω_{C+1}, for $|E-2| \le \varphi^{C+1}N^{-2/3}$,

$$\psi\big(\operatorname{Tr}(\chi_{E-\ell} * \theta_\eta)(H)\big) \le \mathbf{1}(\lambda_N < E) \le \psi\big(\operatorname{Tr}(\chi_{E+\ell} * \theta_\eta)(H)\big).$$

Taking expectations, with the probability of the bad event on both sides,

$$\mathbb{E}\,\psi\big(\operatorname{Tr}(\chi_{E-\ell} * \theta_\eta)(H)\big) - \mathbb{P}(\Omega_{C+1}^c) \le \mathbb{P}(\lambda_N < E) \le \mathbb{E}\,\psi\big(\operatorname{Tr}(\chi_{E+\ell} * \theta_\eta)(H)\big) + \mathbb{P}(\Omega_{C+1}^c).$$

In (6.21) of [26] the term $\mathbb{P}(\Omega^c)$ is missing in the lower bound.

Step 4 (comparison). We have $\psi(\operatorname{Tr}(\chi_a * \theta_\eta)(H)) = F\big(N\int_a^{E_L} \operatorname{Im} m(y+i\eta)\, dy\big)$. For $|a-2| \le \varphi^{C+1}N^{-2/3}$, both a and E_L lie in $I_{9\epsilon}$ for N large, and $\eta = N^{-2/3-9\epsilon}$. Since $9\epsilon < \widetilde{\epsilon}$, (2) at the level 9ϵ gives

$$\Big|\mathbb{E}^{\mathbf{v}}\psi\big(\operatorname{Tr}(\chi_a * \theta_\eta)(H^{\mathbf{v}})\big) - \mathbb{E}^{\mathbf{w}}\psi\big(\operatorname{Tr}(\chi_a * \theta_\eta)(H^{\mathbf{w}})\big)\Big| \le N^{-\kappa+9C'\epsilon} \le \frac{1}{2}N^{-\delta},$$

since $36C'\epsilon \le \kappa$ and $\epsilon \le \kappa/2$ give $-\kappa + 9C'\epsilon \le -\epsilon/2 - \kappa/4$.

Step 5 (the window $|s| \le \varphi^C$). Let $X^{\#} := N^{2/3}(\lambda_N(H^{\#}) - 2)$ for $\# \in \{\mathbf{v}, \mathbf{w}\}$. By (1), $\mathbb{P}(\Omega_C^c) + \mathbb{P}(\Omega_{C+1}^c) \le \frac{1}{4}N^{-\delta}$ for both matrices of every pair and N large, and $2N^{-\epsilon} \le N^{-\delta}$ for $N \ge 4^{1/\epsilon}$. Let $|s| \le \varphi^C$ and $E := 2 + (s + N^{-\epsilon})N^{-2/3}$, so that $E + 2\ell = 2 + (s + 2N^{-\epsilon})N^{-2/3}$ and all the points below lie in $|a-2| \le \varphi^{C+1}N^{-2/3}$. By Step 3 for $H^{\mathbf{v}}$ at E, Step 4 at $a = E + \ell$, and Step 3 for $H^{\mathbf{w}}$ at $E + 2\ell$,

$$\mathbb{P}^{\mathbf{v}}(X^{\mathbf{v}} \le s) \le \mathbb{P}^{\mathbf{v}}(\lambda_N < E) \le \mathbb{P}^{\mathbf{w}}(\lambda_N < E + 2\ell) + \frac{1}{2}N^{-\delta} + \frac{1}{2}N^{-\delta} \le \mathbb{P}^{\mathbf{w}}(X^{\mathbf{w}} \le s + N^{-\delta}) + N^{-\delta}.$$

The lower bound of (5.27) follows in the same way, at $E := 2 + sN^{-2/3}$ and $E - 2\ell$, since $\{X^{\mathbf{w}} \le s - N^{-\delta}\} \subset \{\lambda_N < E - 2\ell\}$.

Step 6 (the tails). On Ω_C, $|X| \le \varphi^C$ by (5.25). If $s > \varphi^C$, then $\mathbb{P}^{\mathbf{v}}(X^{\mathbf{v}} \le s) \le 1 \le \mathbb{P}^{\mathbf{w}}(X^{\mathbf{w}} \le s + N^{-\delta}) + \mathbb{P}(\Omega_C^c)$ and $\mathbb{P}^{\mathbf{w}}(X^{\mathbf{w}} \le s - N^{-\delta}) - N^{-\delta} \le 1 - \mathbb{P}(\Omega_C^c) \le \mathbb{P}^{\mathbf{v}}(X^{\mathbf{v}} \le s)$. If $s < -\varphi^C$, then $\mathbb{P}^{\mathbf{v}}(X^{\mathbf{v}} \le s) \le \mathbb{P}(\Omega_C^c)$ and $\mathbb{P}^{\mathbf{w}}(X^{\mathbf{w}} \le s - N^{-\delta}) \le \mathbb{P}(\Omega_C^c)$. Hence (5.27) holds for all $s \in \mathbb{R}$. □

Now, we return to prove Theorem 3.7. Let $H^{\mathbf{v}}$ satisfy the conditions of Theorem 3.6 with (C_1, ϕ), and for $N \ge 3$ let $\widetilde{H}$ be a matrix with the properties of Lemma 5.1 for $H = H^{\mathbf{v}}$. Let $\mathcal{P}_N$ be the set of all such pairs $(H^{\mathbf{v}}, \widetilde{H})$. We apply Lemma 5.6 to $\mathcal{P}_N$ with $\kappa = \phi$; all the constants below depend only on (C_1, ϕ).

Hypothesis (1) for $H^{\mathbf{v}}$. By (5.23), $\|H^{\mathbf{v}}\| \le 3$ with high probability, and (5.24) holds with high probability by the proof of Theorem 3.6. (5.25) is (3.12) at $j = N$, since $\gamma_N = 2$ and $\min(N, 1) = 1$. For (5.26), let C_r be the constant of (3.12)–(3.13), $C \ge C_r$ and $w := 2\varphi^C N^{-2/3}$. By (3.13), the number of eigenvalues in $[2-w, 2+w]$ is at most $N(\mathfrak{n}(2+w) - \mathfrak{n}(2-2w)) \le N(1 - n_{sc}(2-2w)) + 2\varphi^C$. Since $\varrho_{sc}(2-t) \le \sqrt{t}/\pi$, we have $1 - n_{sc}(2-x) \le \frac{2}{3\pi}x^{3/2}$, so this number is at most $\frac{16}{3\pi}\varphi^{3C/2} + 2\varphi^C \le \varphi^{2C}$ once $\varphi^C \ge 28$. These bounds hold with high probability for every $C \ge C_0$, for a suitable C_0.

Hypothesis (1) for $\widetilde{H}$. $\widetilde{H}$ is not in the class of Theorem 3.6: Lemma 5.1 gives no bound $\mathbb{E}|\widetilde{h}_{ij}|^3 \le C_1 N^{-3/2}$, only $\mathbb{E}|\widetilde{h}_{ij}|^3 \le (\mathbb{E}\widetilde{h}_{ij}^2\, \mathbb{E}\widetilde{h}_{ij}^4)^{1/2} \le (C_1 \log N)^{1/2} N^{-3/2}$. Instead, for $N \ge N_0(C_1)$, $\widetilde{H}$ satisfies (2.1) with $C_0 = \max(C_1, 1)$ and $|\widetilde{h}_{ij}| \le \widetilde{q}^{-1} \le N^{-2/5}$ almost surely, so Lemma 3.1 applies to $\widetilde{H}$ with $q = N^{2/5}$. Then (3.4) gives $\|\widetilde{H}\| \le 3$; (3.2) gives (5.24) for $\eta \le N^{-2/3}$, because $q^{-2}(\kappa+\eta)^{-1/2} \le N^{-4/5}\eta^{-1/2} = N^{1/5}\eta^{1/2}(N\eta)^{-1} \le (N\eta)^{-1}$ (equivalently, (5.24) for $\widetilde{H} = H_0$ is the case $\gamma_0 = 0$ of (5.24) for H_{γ_0}); and the rigidity of item 4 of Lemma 3.1, with $q^{-2} \le N^{-2/3}$, gives, with its constant C_r', $|\lambda_j - \gamma_j| \le 2\varphi^{C_r'} N^{-2/3}$ for all j. At $j = N$ this gives (5.25) for $C \ge C_r' + 1$. For (5.26), with $C \ge C_r'$ and $w = 2\varphi^C N^{-2/3}$ as above: if $\lambda_j \ge 2 - w$, then $\gamma_j \ge 2 - 2w$, so $N - j = N(1 - n_{sc}(\gamma_j)) \le \frac{16}{3\pi}\varphi^{3C/2}$, and there are at most $\frac{16}{3\pi}\varphi^{3C/2} + 1 \le \varphi^{2C}$ such j.

Hypothesis (2). A function $F \in C^4(\mathbb{R})$ satisfying (5.12) for $\alpha = 1, \dots, 4$ satisfies it for $\alpha = 1, 2, 3$, so Lemma 5.5 gives (2) with $\kappa = \phi$.

Then, with Lemma 5.6, there are $\delta_1 > 0$ and N_1 such that for $N \ge N_1$, all such $H^{\mathbf{v}}$ and all $s \in \mathbb{R}$,

$$\mathbb{P}\Big(N^{2/3}(\widetilde{\lambda}_N - 2) \le s - N^{-\delta_1}\Big) - N^{-\delta_1} \le \mathbb{P}\big(N^{2/3}(\lambda_N - 2) \le s\big) \le \mathbb{P}\Big(N^{2/3}(\widetilde{\lambda}_N - 2) \le s + N^{-\delta_1}\Big) + N^{-\delta_1}, \tag{5.28}$$

where λ_N and $\widetilde{\lambda}_N$ denote the largest eigenvalues of $H^{\mathbf{v}}$ and $\widetilde{H}$. Furthermore, for $N \ge N_0(C_1)$, $\widetilde{H}$ satisfies the conditions of Lemma 3.5 with $\phi = 2/5$ and $C_0 = \max(C_1, 1)$, so there are $\delta_2 > 0$ and N_2, depending only on C_1, such that (3.10) holds with $\widetilde{H}$ in place of $H^{\mathbf{v}}$ for $N \ge N_2$ and all $s \in \mathbb{R}$. Let $\delta_0 := \min(\delta_1, \delta_2)$ and $\delta := \delta_0/2$. For $N \ge 4^{1/\delta_0}$, $N^{-\delta_1} + N^{-\delta_2} \le 2N^{-\delta_0} \le N^{-\delta}$. Combining the two bounds, with the monotonicity of $s \mapsto \mathbb{P}^{\mathbf{w}}(N^{2/3}(\lambda_N - 2) \le s)$, we obtain (3.14) for all $N \ge \max(N_0(C_1), N_1, N_2, 4^{1/\delta_0})$, all $H^{\mathbf{v}}$ in the class and all $s \in \mathbb{R}$, with δ and the threshold depending only on (C_1, ϕ). This completes the proof of Theorem 3.7. □

6. Proof of Lemma 5.4, Lemma 5.5, and Lemma 3.8

To prove Lemmas 5.4 and 5.5, we will again use Green function comparison method. Recall the notations in (3.21)-(3.25) with $\widetilde{H} = H_0 = H^{\mathbf{w}}$ and $H = H_{\gamma_{\max}} = H^{\mathbf{v}}$. We let $S = G^{\gamma} = (H_\gamma - z)^{-1}$, $R = (Q - z)^{-1}$, and $T = G^{\gamma-1} = (H_{\gamma-1} - z)^{-1}$, where Q depends on $\gamma = \Phi(a, b)$. All H_γ are realized on one probability space, on which the families $(h_{ij}^{\mathbf{v}})_{i \le j}$ and $(h_{ij}^{\mathbf{w}})_{i \le j}$ are independent.

Throughout this section the pair $(H^{\mathbf{v}}, H^{\mathbf{w}})$ has the following properties, for constants $C_1 > 0$ and $\phi > 0$ and some $q \ge N^{\phi}$. The entries $h_{ij} = h_{ij}^{\mathbf{v}}$, $i \le j$, are independent, and so are the entries $h_{ij}^{\mathbf{w}}$; $\mathbb{E}h_{ij} = 0$, $\mathbb{E}h_{ij}^2 = N^{-1}$ and $\mathbb{E}h_{ij}^4 \le C_1(\log N)N^{-2}$ for $i \ne j$, and $\mathbb{E}h_{ii}^2 \le C_1 N^{-1}$; the first four moments of the off-diagonal entries and the first two moments of the diagonal entries of $H^{\mathbf{v}}$ and $H^{\mathbf{w}}$ coincide; and $|h_{ij}^{\mathbf{v}}| \le q^{-1}$ and $|h_{ij}^{\mathbf{w}}| \le q^{-1}$ almost surely. Hence $|h_{ij}| \le q^{-1} \le N^{-\phi}$ almost surely for all entries of every H_γ, $0 \le \gamma \le \gamma_{\max}$, and every H_γ is a generalized Wigner matrix satisfying (2.1) with $\max(C_1, 1)$. For H as in Theorem 3.6 and $\widetilde{H}$ as in Lemma 5.1, these properties hold with $q = \min(N^{\phi}, \widetilde{q})$, where $\widetilde{q} \sim N^{1/2}/\log N$ is the support parameter of $\widetilde{H}$; since $\widetilde{q} \ge N^{1/4}$ for $N \ge N_0(C_1)$, they hold with ϕ replaced by $\min(\phi, 1/4)$. All constants below depend only on C_1, ϕ and the parameters fixed in the statements, and not on the pair or on γ.

There is a constant $C_0 > 0$ such that for every $1 \le \gamma \le \gamma_{\max}$, with 3-high probability,

$$\max_{z \in S(C_0)} \max_{i,j} \max\{|S_{ij}(z)|, |R_{ij}(z)|, |T_{ij}(z)|\} \le 2, \tag{6.1}$$

where the constants of Definition 2.3 do not depend on γ or on the pair. Indeed, every H_γ satisfies the assumptions of Lemma 3.1 with the same constants (with $\max(C_1, 1)$ in (2.1), and with $q \geq N^\phi \geq \varphi^C$ for $N \geq N_0$). Let C be the constant of Lemma 3.1 for $\zeta = 3$, and let $C_0 := 2C + 2$; any larger C_0 works as well, since $\mathbf{S}(C_0)$ shrinks as C_0 grows. For $z \in \mathbf{S}(C_0)$ we have $N\eta \geq \varphi^{2C+2}$, so the right side of (3.3) is at most $\varphi^C q^{-1} + \varphi^{-1} + \varphi^{-C-2} \leq 1/4$ for $N \geq N_0$. With $|m_{sc}| \leq 1$, on the event of (3.3) we obtain $|S_{ij}(z)| \leq 5/4$ for all i, j and all $z \in \mathbf{S}(C_0)$. The bound for R then holds deterministically. Since $H_\gamma = Q + V$, where V has at most two nonzero entries, each of modulus at most $q^{-1} \leq 1/10$, the resolvent identity $R = S + SVR$ gives $\max_{k,l} |R_{kl}| \leq \frac{5}{4} + 2 \cdot \frac{5}{4} \cdot \frac{1}{10} \max_{k,l} |R_{kl}|$, that is, $\max_{k,l} |R_{kl}| \leq 5/3$. The same argument bounds the resolvent of H_γ with any other entry (and its transpose) replaced by 0. Finally, T is the matrix S of $H_{\gamma-1}$, so the bound for T is the same statement for $\gamma - 1$. No union bound over γ is used. On the other hand, S, R, and T satisfy the following trivial bound that always holds:

$$\max_{z \in S(C_0)} \max_{\gamma} \max_{i,j} \max\{|S_{ij}(z)|, |R_{ij}(z)|, |T_{ij}(z)|\} \leq \eta^{-1} \leq N. \tag{6.2}$$

In this section, for simplicity, we use the notation $|\mathbf{k}| = \|\mathbf{k}\|_1$ for any vector $\mathbf{k} \in \mathbb{N}^n$, $n \in \mathbb{N}$.

To illustrate the idea of Green function comparison method, we first consider the following simple example of finding a bound on $\mathbb{E}G_{ij}$ from an a priori bound on $\widetilde{G}_{ij}$:

Example 6.1. *Suppose that the bound*

$$\max_{i \neq j} |\widetilde{G}_{ij}| \leq N^{-1/3} \tag{6.3}$$

holds. Then, we have that $|\mathbb{E}G_{ij}| \leq CN^{-1/3}$.

Applying the replacement strategy, we obtain

$$\mathbb{E}G_{ij} = \mathbb{E}\widetilde{G}_{ij} + \sum_{\gamma=1}^{\gamma_{\max}} \left(\mathbb{E}G_{ij}^{\gamma} - \mathbb{E}G_{ij}^{\gamma-1} \right). \tag{6.4}$$

Note that the bound (6.3) implies that $|\mathbb{E}\widetilde{G}_{ij}| \leq N^{-1/3}$. Thus, if we can prove that for $1 \leq \gamma \leq \gamma_{\max}$

$$|\mathbb{E}S_{ij} - \mathbb{E}T_{ij}| \leq CN^{-7/3}, \tag{6.5}$$

then this will show the desired estimate on $|\mathbb{E}G_{ij}|$.

We now expand S_{ij} in terms of R and V, as in (3.28), using the resolvent expansion

$$S = R - RVR + (RV)^2R - (RV)^3R + (RV)^4R + \cdots + (-1)^m(RV)^mR + (-1)^{m+1}(RV)^{m+1}S,$$

with $m := \lceil 1/\phi \rceil$. Since each element of V is bounded by $N^{-\phi}$ almost surely, from (6.1), we find that the last term in the expansion is $O(N^{-1})$. Taking expectation, we find that

$$\mathbb{E}S_{ij} = \mathbb{E}\sum_{k=0}^{m} [(RV)^k R]_{ij} + O(N^{-1}). \tag{6.6}$$

Note that R is independent of V. We can decouple R and V by taking partial expectation $\mathbb{E}_{ab}$ with respect to V_{ab}, which gives

$$\mathbb{E}S_{ij} = \mathbb{E}\sum_{k=0}^{m} A_k \mathbb{E}_{ab}(V_{ab}^k) + O(N^{-1}), \tag{6.7}$$

where A_k depends only on R. For example, A_5 contains a term such as $R_{ia}R_{bb}R_{aa}R_{bb}R_{aa}R_{bj}$.

From the moment matching condition, we know that the first four moments of the off-diagonal entries of H and $\widetilde{H}$ coincide (in this example (a, b) is off-diagonal), thus the terms with $k = 0, 1, \cdots, 4$ will vanish when we estimate $\mathbb{E}S_{ij} - \mathbb{E}T_{ij}$. Moreover, since we know that $|\mathbb{E}(V_{ab}^k)| = O(N^{-2-\phi/2})$, it

suffices to prove the bound $|\mathbb{E}A_k| \le CN^{-1/3+\phi/2}$ for $5 \le k \le m$. Now using an expansion such as (see (6.24))

$$\mathbb{E}\prod_{t=1}^{s} R_{i_t j_t} = \mathbb{E}\prod_{t=1}^{s} S_{i_t j_t} - \mathbb{E}\sum_{l=1}^{m} \widetilde{A}_l \mathbb{E}_{ab}(V_{ab}^l) + O(N^{-1}),$$

where $\widetilde{A}_l$ are products of S, we write the expectation of the product of the elements of R's (like A_k) as the sum of the expectation of the product of the elements of S's. Hence, we can convert the problem into showing

$$|\mathbb{E}B_k| \le CN^{-1/3+\phi/2}, \quad k \ge 5, \tag{6.8}$$

where B_k is a sum of the product of the elements of S. For example, B_5 contains a term such as $S_{ia}S_{bb}S_{aa}S_{bb}S_{aa}S_{bj}$. Note here we effectively gain a factor of $N^{-\phi/2}$ in the required bound.

We now repeat the replacement argument with the terms in B_k, since the matrix H_γ also satisfies the same bounded support condition as H. We consider a telescopic series, as in (6.4),

$$\mathbb{E}B_k = \mathbb{E}B_k(S \to \widetilde{G}) + \sum_{\gamma'=1}^{\gamma} \left(\mathbb{E}B_k(S \to G^{\gamma'}) - \mathbb{E}B_k(S \to G^{\gamma'-1}) \right), \tag{6.9}$$

where the notation $B_k(S \to G^{\gamma'})$ means that we consider the product of $G^{\gamma'} = (H_{\gamma'} - z)^{-1}$ instead of S while keeping the indices the same. The first term in the r.h.s. of (6.9) will be the sum of the products such as $\mathbb{E}\widetilde{G}_{ia}\widetilde{G}_{bb}\widetilde{G}_{aa}\widetilde{G}_{bb}\widetilde{G}_{aa}\widetilde{G}_{bj}$, and in a generic case, it contains at least one off-diagonal term of $\widetilde{G}$. In particular, we have $|\mathbb{E}B_k(S \to \widetilde{G})| \le CN^{-1/3}$. Hence, we are left to estimate the telescopic sum, where we use the argument above, which involves the resolvent expansion and partial expectation, again. Note that, each time we repeat the procedure, we effectively gain a factor of $N^{-\phi/2}$ in the required bound. Therefore, after repeating the procedure sufficiently many times (i.e., $O(1/\phi)$-times), we find that it suffices to prove that $|\mathbb{E}\widetilde{B}| \le C$, where $\widetilde{B}$ is a sum of the products of the elements of G^γ, $1 \le \gamma \le \gamma_{\max}$. Since this is trivial from the bound (6.1), we find that the bound $|\mathbb{E}G_{ij}| \le CN^{-1/3}$ holds.

We now prove the lemmas using the ideas explained above. We first introduce some notations for simplifying the expressions, which will helps us study the expectation of the product of S_{ij}'s.

Definition 6.2 (Matrix operators $*_\gamma$ and $*$)**.** For a γ with $\Phi(a,b) = \gamma$, we define $A *_\gamma B$ as

$$(A *_\gamma B)_{ij} = (A\mathcal{I}_\gamma B)_{ij}, \qquad (\mathcal{I}_\gamma)_{ij} = \mathbf{1}_{\{i,j\}=\{a,b\}} \tag{6.10}$$

When $a \ne b$, it satisfies

$$(A *_\gamma B)_{ij} = A_{ia}B_{bj} + A_{ib}B_{aj}. \tag{6.11}$$

We will often drop the subscript γ for convenience as $A * B$. For simplicity, we denote the k-th power of A under $*-$ product by A^{*k}, i.e.,

$$A * A * A * A \cdots * A = A^{*k} \tag{6.12}$$

Definition 6.3 ($\mathcal{P}_{\gamma,\mathbf{k}}$ and $\mathcal{P}_{\gamma,k}$)**.** For $k \in \mathbb{N}$ and $\mathbf{k} = (k_1, k_2, \cdots, k_s) \in \mathbb{N}^s$, $\gamma = \Phi(a,b)$, we define

$$\mathcal{P}_{\gamma,k}G_{ij} := G_{ij}^{*_\gamma(k+1)} \tag{6.13}$$

and

$$\mathcal{P}_{\gamma,\mathbf{k}}\left(\prod_{t=1}^{s} G_{i_t j_t}\right) := \prod_{t} \left(\mathcal{P}_{\gamma,k_t} G_{i_t j_t}\right) = \prod_{t=1}^{s} G_{i_t j_t}^{*_\gamma(k_t+1)}. \tag{6.14}$$

If $\mathcal{G}_1$ and $\mathcal{G}_2$ are products of matrix entries of G's as above, then we define

$$\mathcal{P}_{\gamma,\mathbf{k}}(\mathcal{G}_1 + \mathcal{G}_2) := \mathcal{P}_{\gamma,\mathbf{k}}\,\mathcal{G}_1 + \mathcal{P}_{\gamma,\mathbf{k}}\,\mathcal{G}_2 \tag{6.15}$$

Note that $\mathcal{P}_{\gamma,\mathbf{k}}$ and $\mathcal{P}_{\gamma,k}$ are not linear operators but just notations we use for simplification. Similarly, for the product of the entries of the matrix $G - m_{sc} = G - m_{sc}I$, we define

$$\widetilde{\mathcal{P}}_{\gamma,\mathbf{k}}\left(\prod_{t=1}^{s}[G-m_{sc}]_{i_tj_t}\right) := \prod_{t}\left(\widetilde{\mathcal{P}}_{\gamma,k_t}[G-m_{sc}]_{i_tj_t}\right), \tag{6.16}$$

where

$$\widetilde{\mathcal{P}}_{\gamma,k}(G_{ij}-m_{sc})_{ij} = \begin{cases}(G_{ij}-m_{sc})_{ij} & \text{if} \quad i=j \quad \text{and} \quad k=0,\\ \mathcal{P}_{\gamma,k}G_{ij} = G_{ij}^{*_\gamma(k+1)} & \text{otherwise.}\end{cases} \tag{6.17}$$

Using Definition 6.3, we may write, for example,

$$\mathcal{P}_{\gamma,\mathbf{k}}\left(\prod_{t=1}^{s}G_{i_tj_t}^{\gamma}\right) = \prod_{t=1}^{s}S_{i_tj_t}^{*_\gamma(k_t+1)}, \quad \mathcal{P}_{\gamma,\mathbf{k}}\left(\prod_{t=1}^{s}G_{i_tj_t}^{\gamma-1}\right) = \prod_{t=1}^{s}T_{i_tj_t}^{*_\gamma(k_t+1)}. \tag{6.18}$$

With the fact that $G^{*s}\mathcal{I}_\gamma G^{*t} = G^{*(s+t)}$, one can easily see that for $k \in \mathbb{N}$ and $\mathbf{k} \in \mathbb{N}^{k+1}$,

$$\mathcal{P}_{\gamma,\mathbf{k}}(\mathcal{P}_{\gamma,k}G_{ij}) = \mathcal{P}_{\gamma,k+|\mathbf{k}|}G_{ij} \tag{6.19}$$

Here, note that $(\mathcal{P}_{\gamma,k}G_{ij})$ is the sum of the products of the matrix entries of G, where each product contains $k+1$ matrix entries of G.

Using the definitions above, we have the following lemma from the bound (6.1). Recall that R, S, and T depend on γ.

Lemma 6.4. *For any $\mathbf{k} \in \mathbb{N}^{\mathbf{s}}$, γ, γ' and $i_1, j_1, \cdots, i_s, j_s$, we have with 3-high probability that*

$$\left|\mathcal{P}_{\gamma',\mathbf{k}}\left(\prod_{t=1}^{s}A_{i_tj_t}\right)\right|, \quad \left|\widetilde{\mathcal{P}}_{\gamma',\mathbf{k}}\left(\prod_{t=1}^{s}(A-m_{sc})_{i_tj_t}\right)\right| \le 4^{s+|\mathbf{k}|+1}, \tag{6.20}$$

where A can be R, S, or T.

The following lemma shows how we expand the expectation of the term $S_{i_1j_1}S_{i_2j_2}\cdots S_{i_sj_s}$. In it, $H_{\gamma'}$ is any of the matrices $H_0, \ldots, H_{\gamma_{\max}}$, and the entry (a,b) is arbitrary; in the applications, $\gamma' \in \{\gamma, \gamma-1\}$ with $\gamma = \Phi(a,b)$.

Lemma 6.5. *Fix C_1, $\phi > 0$ and $K_s > 0$, and let C_0 be as in (6.1). There is N_0 such that the following holds for all $N \ge N_0$. Let $0 \le \gamma' \le \gamma_{\max}$, let $a \le b$ and $\gamma = \Phi(a,b)$, and let $S = (H_{\gamma'} - z)^{-1}$, $h_{ab} = (H_{\gamma'})_{ab}$ and $R = (Q-z)^{-1}$, where Q is obtained from $H_{\gamma'}$ by replacing the entries (a,b) and (b,a) by 0. Let $m_0 := 4$ if $a \ne b$ and $m_0 := 2$ if $a = b$. Let $1 \le \zeta \le \varphi^2$, let $s \in \mathbb{N}$ with $s \le K_s\varphi^2$, and let $\xi := \lceil 2\zeta/\phi \rceil$. Then there are real numbers $\mathcal{A}_\alpha$, $\alpha = (\alpha_1, \alpha_2, \cdots, \alpha_s) \in \mathbb{N}^s$, $|\alpha| = \sum \alpha_t$, which depend on N, $H_{\gamma'}$, (a,b), ζ and s, but not on z or on the indices i_t, j_t, such that for all $z \in \mathbf{S}(C_0)$ and all $i_1, j_1, \ldots, i_s, j_s$,*

$$\begin{aligned}\Big|\mathbb{E}\prod_{t=1}^{s}S_{i_tj_t} - \sum_{k=0}^{m_0}\mathbb{E}[(-h_{ab})^k] \sum_{\alpha\in\mathbb{N}^s:\,|\alpha|=k}\mathbb{E}\,\mathcal{P}_{\gamma,\alpha}\prod_{t=1}^{s}R_{i_tj_t}\\ - \sum_{\alpha\in\mathbb{N}^s:\,m_0<|\alpha|\le\xi}\mathcal{A}_\alpha\,\mathbb{E}\,\mathcal{P}_{\gamma,\alpha}\prod_{t=1}^{s}S_{i_tj_t}\Big| \le 4^sN^{-\zeta},\end{aligned} \tag{6.21}$$

and

$$|\mathcal{A}_\alpha| \le N^{-2-|\alpha|\phi/10} \quad (a \ne b), \qquad |\mathcal{A}_\alpha| \le N^{-1-|\alpha|\phi/4} \quad (a = b). \tag{6.22}$$

Here $\mathcal{A}_\alpha := 0$ outside the window $m_0 < |\alpha| \le \xi$, which is part of the statement, so that (6.22) *holds for all $\alpha \in \mathbb{N}^s$. Similarly, with the same $\mathcal{A}_\alpha$,*

$$\Big|\mathbb{E}\prod_{t=1}^{s}(S-m_{sc})_{i_tj_t} - \sum_{k=0}^{m_0}\mathbb{E}[(-h_{ab})^k]\sum_{|\alpha|=k}\mathbb{E}\,\widetilde{\mathcal{P}}_{\gamma,\alpha}\prod_{t=1}^{s}(R-m_{sc})_{i_tj_t} - \sum_{m_0<|\alpha|\le\xi}\mathcal{A}_\alpha\,\mathbb{E}\,\widetilde{\mathcal{P}}_{\gamma,\alpha}\prod_{t=1}^{s}(S-m_{sc})_{i_tj_t}\Big| \le 4^s N^{-\zeta}. \tag{6.23}$$

Both estimates also hold for products in which some factors are replaced by their complex conjugates, with $\mathcal{P}_{\gamma,k}\overline{G_{ij}} := \overline{\mathcal{P}_{\gamma,k}G_{ij}}$ and $\widetilde{\mathcal{P}}_{\gamma,k}\overline{(G-m_{sc})_{ij}} := \overline{\widetilde{\mathcal{P}}_{\gamma,k}(G-m_{sc})_{ij}}$, and for products that mix factors S_{ij} and $(S-m_{sc})_{ij}$.

Furthermore, there are real numbers $\widetilde{\mathcal{A}}_\alpha$, with the same dependence, such that

$$\Big|\mathbb{E}\prod_{t=1}^{s}S_{i_tj_t} - \mathbb{E}\prod_{t=1}^{s}R_{i_tj_t} - \sum_{\alpha\in\mathbb{N}^s:\,1\le|\alpha|\le\xi}\widetilde{\mathcal{A}}_\alpha\,\mathbb{E}\,\mathcal{P}_{\gamma,\alpha}\prod_{t=1}^{s}S_{i_tj_t}\Big| \le 4^s N^{-\zeta}, \tag{6.24}$$

where, both for $a \ne b$ and for $a = b$,

$$|\widetilde{\mathcal{A}}_\alpha| \le N^{-|\alpha|\phi/10}. \tag{6.25}$$

Here $\widetilde{\mathcal{A}}_\alpha := 0$ outside the window $1 \le |\alpha| \le \xi$, which is part of the statement. The same remark on conjugate and centred factors applies to (6.24).

We prove this lemma later in the section.

The iteration. A *word* of length s is a product $\omega = \prod_{t=1}^{s} g_t$ in which every factor g_t is one of G_{xy}, $\overline{G_{xy}}$, $(G-m_{sc})_{xy}$ and $\overline{(G-m_{sc})_{xy}}$; we write $\omega(G)$ for its value at a matrix G. The operators $\mathcal{P}_{\gamma,\mathbf{k}}$ act on words factor by factor as in (6.14) and (6.16) (for centred factors $\mathcal{P}_{\gamma,k}$ means $\widetilde{\mathcal{P}}_{\gamma,k}$), with the conjugation rule of Lemma 6.5. A *formal sum* $\mathcal{F} = \sum_c f_c\omega_c$ is a finite linear combination of words with weights $f_c \ge 0$; its total weight is $\|\mathcal{F}\| := \sum_c f_c$, and $\mathcal{P}_{\gamma,\mathbf{k}}\mathcal{F} := \sum_c f_c\,\mathcal{P}_{\gamma,\mathbf{k}}\omega_c$. If ω has length s' and $|\mathbf{k}| = m$, then $\mathcal{P}_{\gamma,\mathbf{k}}\omega$ is a sum of at most 2^m words of length $s'+m$, since by (6.13) each $*_\gamma$-product doubles the number of terms if $a \ne b$ and does not change it if $a = b$. For data $d = ((\gamma_1,\mathbf{k}_1),\dots,(\gamma_n,\mathbf{k}_n))$ we write $\mathcal{P}_d := \mathcal{P}_{\gamma_n,\mathbf{k}_n}\cdots\mathcal{P}_{\gamma_1,\mathbf{k}_1}$ (so $\mathcal{P}_{\gamma_1,\mathbf{k}_1}$ acts first) and $|d| := \sum_t|\mathbf{k}_t|$; for the empty data $d = \emptyset$, $\mathcal{P}_\emptyset$ is the identity and $|\emptyset| = 0$. We write $m_0(\gamma) := 2$ and $W(\gamma) := N^{-1}$ if $\gamma = \Phi(a,a)$, $m_0(\gamma) := 4$ and $W(\gamma) := N^{-2}$ otherwise, and

$$w(d) := \prod_{t=1}^{n} W(\gamma_t)\,N^{-(\phi/20)|\mathbf{k}_t|}, \qquad w(\emptyset) = 1.$$

Fix C_1, $\phi > 0$ and $K_s > 0$, and let $K' := K_s + (6/\phi+1)(10/\phi+1)$. Let C_0 be at least the constants of (6.1) and of Lemma 6.5 with K_s replaced by K'. We show that there is N_0 such that the following holds for all $N \ge N_0$, all pairs $(H^{\mathbf{v}}, H^{\mathbf{w}})$ as above with $q \ge N^\phi$, all $0 \le \gamma_0 \le \gamma_{\max}$, all ζ with $1 \le \zeta \le \varphi$, all $s \in \mathbb{N}$ with $s \le K_s\zeta$, all $z \in \mathbf{S}(C_0)$, and every formal sum $\mathcal{F}$ of words of length s with total weight $\|\mathcal{F}\| = 1$ (that is, $\mathcal{F} = \sum_{I,J} f(I,J)\,\omega_{I,J}$ with f as in (6.36) below): the sum form (6.33) and the max forms (6.35), (6.37) and (6.38) hold, with

$$n_* := \lceil 6\zeta/\phi\rceil, \qquad \xi_* := \lceil 2(2\zeta+3)/\phi\rceil.$$

All words that occur below have length at most $s + n_*\xi_* \le K'\varphi^2$, since $s \le K_s\zeta$, $1 \le \zeta \le \varphi$ and $n_*\xi_* \le (6\zeta/\phi+1)(10\zeta/\phi+1)$. We take N_0 so large that $\varphi \ge 3$ (so that $2\zeta+3 \le \varphi^2$), $\rho := N^{\phi/20} \ge 8$ (so that $\theta := N^{-\phi/5} = \rho^{-4} \le 1/2$), $8(K'\varphi^2)^5 \le N^{\phi/20}$, $(N+1)^{K'\varphi^2}N^C e^{-c\varphi^3} \le 1$ for the constants c, C of the 3-high probability in (6.1), and $16\cdot 4^{K_s} \le N^{1/5}$. We write $\mathbb{E}\,\mathcal{F}(G^\gamma)$ for the expectation of $\mathcal{F}$ evaluated at $G^\gamma = (H_\gamma - z)^{-1}$; in particular $G^0 = \widetilde{G}$.

Step 1: one swap. Fix $\gamma = \Phi(a,b)$ and a word ω of length $s' \le K'\varphi^2$. Lemma 6.5 with ζ replaced by $2\zeta+3$ and K_s by K', applied to H_γ ($\gamma'=\gamma$, $h_{ab}=v_{ab}$) and to $H_{\gamma-1}$ ($\gamma'=\gamma-1$, $h_{ab}=w_{ab}$), gives coefficients $\mathcal{A}_\alpha^{\gamma,0}$ and $\mathcal{A}_\alpha^{\gamma,1}$ satisfying (6.22), with the same matrix R in both cases, such that

$$\mathbb{E}\,\omega(S) = \sum_{k=0}^{m_0(\gamma)} \mathbb{E}[(-v_{ab})^k] \sum_{|\alpha|=k} \mathbb{E}\,\mathcal{P}_{\gamma,\alpha}\omega(R) + \sum_{m_0(\gamma)<|\alpha|\le\xi_*} \mathcal{A}_\alpha^{\gamma,0}\mathbb{E}\,\mathcal{P}_{\gamma,\alpha}\omega(S) + \epsilon^{\gamma,0}(\omega), \tag{6.26}$$

$$\mathbb{E}\,\omega(T) = \sum_{k=0}^{m_0(\gamma)} \mathbb{E}[(-w_{ab})^k] \sum_{|\alpha|=k} \mathbb{E}\,\mathcal{P}_{\gamma,\alpha}\omega(R) + \sum_{m_0(\gamma)<|\alpha|\le\xi_*} \mathcal{A}_\alpha^{\gamma,1}\mathbb{E}\,\mathcal{P}_{\gamma,\alpha}\omega(T) + \epsilon^{\gamma,1}(\omega), \tag{6.27}$$

with $|\epsilon^{\gamma,a}(\omega)| \le 4^{s'}N^{-2\zeta-3}$. For words with centred factors one uses (6.23). Since $\mathbb{E}(v_{ab})^k = \mathbb{E}(w_{ab})^k$ for $k \le m_0(\gamma)$ (four matched moments off the diagonal, two on it), the first sums coincide. As the coefficients do not depend on the word, for every formal sum $\mathcal{F}'$ of words of length s' we obtain

$$\begin{aligned}\Big|\mathbb{E}\,\mathcal{F}'(G^\gamma) - \mathbb{E}\,\mathcal{F}'(G^{\gamma-1}) - \sum_{m_0(\gamma)<|\alpha|\le\xi_*} \Big(\mathcal{A}_\alpha^{\gamma,0}\,\mathbb{E}\,\mathcal{P}_{\gamma,\alpha}\mathcal{F}'(G^\gamma) - \mathcal{A}_\alpha^{\gamma,1}\,\mathbb{E}\,\mathcal{P}_{\gamma,\alpha}\mathcal{F}'(G^{\gamma-1})\Big)\Big| \\ \le 2\cdot 4^{s'}N^{-2\zeta-3}\|\mathcal{F}'\|.\end{aligned} \tag{6.28}$$

Summing over the at most $\gamma_{\max} \le N^2$ swaps, we obtain for every $0 \le \gamma' \le \gamma_{\max}$

$$\Big|\mathbb{E}\,\mathcal{F}'(G^{\gamma'})\Big| \le \big|\mathbb{E}\,\mathcal{F}'(G^0)\big| + \sum_{\gamma=1}^{\gamma'}\sum_{a=0,1}\sum_{m_0(\gamma)<|\alpha|\le\xi_*} |\mathcal{A}_\alpha^{\gamma,a}|\,\big|\mathbb{E}\,\mathcal{P}_{\gamma,\alpha}\mathcal{F}'(G^{\gamma-a})\big| + 2N^2 4^{s'}N^{-2\zeta-3}\|\mathcal{F}'\|. \tag{6.29}$$

Step 2: the coefficients. Let $t \ge 1$ with $8t^5 \le N^{\phi/20}$, and let $s' \le t$. We claim that

$$\sum_{\gamma=1}^{\gamma_{\max}}\sum_{a=0,1}\sum_{\alpha\in\mathbb{N}^{s'}:\, m_0(\gamma)<|\alpha|\le\xi_*} |\mathcal{A}_\alpha^{\gamma,a}|\,N^{(\phi/20)|\alpha|} \le N^{-\phi/5}. \tag{6.30}$$

Indeed, the number of $\alpha \in \mathbb{N}^{s'}$ with $|\alpha| = m$ is at most t^m. Let $x := N^{-\phi/20}$, so that $tx \le 1/8$. By (6.22), an off-diagonal γ contributes at most $2N^{-2}\sum_{m\ge5}(tx)^m \le 4N^{-2}(tx)^5$, and a diagonal γ at most $2N^{-1}\sum_{m\ge3}(tx^4)^m \le 4N^{-1}(tx^4)^3$. With at most $N^2/2$ off-diagonal and N diagonal γ, the left side of (6.30) is at most $2t^5N^{-\phi/4} + 4t^3N^{-3\phi/5} \le N^{-\phi/5}(2t^5N^{-\phi/20} + 4t^5N^{-\phi/20}) \le N^{-\phi/5}$. Moreover, for $|\alpha| = m \ge 1$ we have $2N^{-2-m\phi/10} \le N^{-2}N^{-(\phi/20)m}$ and $2N^{-1-m\phi/4} \le N^{-1}N^{-(\phi/20)m}$, since $N^{\phi/20} \ge 8$. Hence

$$|\mathcal{A}_\alpha^{\gamma,0}| + |\mathcal{A}_\alpha^{\gamma,1}| \le W(\gamma)\,N^{-(\phi/20)|\alpha|}. \tag{6.31}$$

Step 3: one word. For every word ω of length $s' \le K'\varphi^2$ and every $0 \le \gamma' \le \gamma_{\max}$,

$$|\mathbb{E}\,\omega(G^{\gamma'})| \le 2\cdot 4^{s'+1}.$$

Indeed, on the event of (6.1) for $H_{\gamma'}$, (6.20) with $\mathbf{k}=0$ gives $|\omega(G^{\gamma'})| \le 4^{s'+1}$; outside it, $|\omega(G^{\gamma'})| \le (\eta^{-1}+1)^{s'} \le (N+1)^{K'\varphi^2}$, and the event has probability at most $N^C e^{-c\varphi^3}$. Consequently, $|\mathbb{E}\,\mathcal{F}'(G^{\gamma'})| \le 2\cdot 4^{s'+1}\|\mathcal{F}'\|$ for every formal sum $\mathcal{F}'$ of words of length s'.

Step 4: induction. Let $E_* := 4N^2\,4^sN^{-2\zeta-3}$. For $s' \in \mathbb{N}$ and $j \in \mathbb{N}$, let $\mathcal{D}_j(s')$ be the set consisting of the empty data and of the data $d = ((\gamma_1,\mathbf{k}_1),\dots,(\gamma_n,\mathbf{k}_n))$ with $1 \le n \le j$ that satisfy (6.34) below with s' in place of s. We prove by induction on j: for all $o, j \in \mathbb{N}$ with $s' := s+o$ and $s'+j\xi_* \le s+n_*\xi_*$, every formal sum $\mathcal{F}'$ of words of length s' with $\|\mathcal{F}'\| \le 2^o$, and every $0 \le \gamma' \le \gamma_{\max}$,

$$\Big|\mathbb{E}\,\mathcal{F}'(G^{\gamma'})\Big| \le \sum_{d\in\mathcal{D}_j(s')} w(d)\,\big|\mathbb{E}\,\mathcal{P}_d\mathcal{F}'(G^0)\big| + \rho^o\left(\theta^j\,8\cdot4^s + 2E_*\right). \tag{6.32}$$

For $j=0$, $\mathcal{D}_0(s') = \{\emptyset\}$, and Step 3 gives $|\mathbb{E}\,\mathcal{F}'(G^{\gamma'})| \le 2^o\cdot2\cdot4^{s+o+1} = 8^o\cdot8\cdot4^s \le \rho^o\,8\cdot4^s$, which implies (6.32). Assume (6.32) for j, and let $s'+(j+1)\xi_* \le s+n_*\xi_*$. We apply (6.29); its last term is

at most $2N^2\,4^{s+o}N^{-2\zeta-3}2^o \le 8^oE_* \le \rho^oE_*$. For each (γ,α) in (6.29), $\mathcal{F}'' := \mathcal{P}_{\gamma,\alpha}\mathcal{F}'$ is a formal sum of words of length $s'+|\alpha|$ with $\|\mathcal{F}''\| \le 2^{o+|\alpha|}$, and $s'+|\alpha|+j\xi_* \le s+n_*\xi_*$. By (6.32) for j,

$$|\mathbb{E}\,\mathcal{F}''(G^{\gamma-a})| \le \Sigma(\gamma,\alpha) + \rho^{o+|\alpha|}\left(\theta^j\,8\cdot 4^s + 2E_*\right),$$

where

$$\Sigma(\gamma,\alpha) := \sum_{d'\in\mathcal{D}_j(s'+|\alpha|)} w(d')\left|\mathbb{E}\,\mathcal{P}_{d'}\mathcal{P}_{\gamma,\alpha}\mathcal{F}'(G^0)\right|.$$

We multiply by $|\mathcal{A}_\alpha^{\gamma,a}|$ and sum over a, γ and α; since all terms are nonnegative, the sum over $\gamma\le\gamma'$ may be extended to all γ. By (6.31), the first parts give at most $\sum_{\gamma,\alpha} W(\gamma)N^{-(\phi/20)|\alpha|}\Sigma(\gamma,\alpha)$, and this is the sum in (6.32) for $j+1$ over the nonempty data, which are exactly the data $d=((\gamma,\alpha),d')$ with $d'\in\mathcal{D}_j(s'+|\alpha|)$; note that $w(d)=W(\gamma)N^{-(\phi/20)|\alpha|}w(d')$ and $\mathcal{P}_d=\mathcal{P}_{d'}\mathcal{P}_{\gamma,\alpha}$. The term $|\mathbb{E}\,\mathcal{F}'(G^0)|$ of (6.29) is the term $d=\emptyset$. By (6.30) with $t=K'\varphi^2$, the second parts give at most $\theta\rho^o(\theta^j\,8\cdot4^s+2E_*)\le \rho^o(\theta^{j+1}\,8\cdot4^s+E_*)$, since $2\theta\le1$. Adding the swap term ρ^oE_* gives (6.32) for $j+1$.

Step 5: conclusion. We apply (6.32) with $o=0$, $j=n_*$, $\mathcal{F}'=\mathcal{F}$ and $\gamma'=\gamma_0$. Since $\theta^{n_*}\le N^{-6\zeta/5}$ and $2E_* = 8\cdot4^sN^{-2\zeta-1}\le 8\cdot4^sN^{-6\zeta/5}$, the remainder is at most $16\cdot4^sN^{-6\zeta/5}\le N^{-\zeta}$, because $16\cdot4^s\le(16\cdot4^{K_s})^\zeta\le N^{\zeta/5}$ (by $s\le K_s\zeta$ and $\zeta\ge1$). This replaces the counting argument based on $\sum_i|\mathbf{k}_i|\ge5n$. We obtain the sum form

$$\begin{aligned}|\mathbb{E}\,\mathcal{F}(G^{\gamma_0})| &\le \sum_d w(d)\left|\mathbb{E}\,\mathcal{P}_d\mathcal{F}(G^0)\right| + N^{-\zeta}\\ &= \left|\mathbb{E}\,\mathcal{F}(G^0)\right| + \sum_{d\ne\emptyset}\prod_{t=1}^{n}\left(W(\gamma_t)N^{-\frac{\phi}{20}|\mathbf{k}_t|}\right)\\ &\qquad\times\left|\mathbb{E}\,\mathcal{P}_{\gamma_n,\mathbf{k}_n}\cdots\mathcal{P}_{\gamma_1,\mathbf{k}_1}\mathcal{F}(G^0)\right| + N^{-\zeta},\end{aligned} \tag{6.33}$$

where d runs over the empty data and all data $d=((\gamma_1,\mathbf{k}_1),\dots,(\gamma_n,\mathbf{k}_n))$, $1\le\gamma_t\le\gamma_{\max}$, with

$$1\le n\le n_*,\quad \mathbf{k_1}\in\mathbb{N}^s,\quad \mathbf{k_2}\in\mathbb{N}^{s+|\mathbf{k_1}|},\quad \mathbf{k_3}\in\mathbb{N}^{s+|\mathbf{k_1}|+|\mathbf{k_2}|},\quad \text{etc.,}\quad\text{and}\quad m_0(\gamma_t)<|\mathbf{k}_t|\le\xi_*. \tag{6.34}$$

The left side of (6.33) involves H_{γ_0}, and the right side involves only $\widetilde{H}=H_0$.

The max forms. For every word ω of length s,

$$|\mathbb{E}\,\omega(G^{\gamma_0})| \le \left|\mathbb{E}\,\omega(G^0)\right| + \max_{d\ne\emptyset} N^{-(\phi/20)|d|}\left|\mathbb{E}\,\mathcal{P}_d\,\omega(G^0)\right| + N^{-\zeta}, \tag{6.35}$$

where the maximum is over the nonempty data satisfying (6.34). This is the case of a single word in the following statement. Since the coefficients $\mathcal{A}$ and the operators $\mathcal{P}$ do not depend on the indices i_t and j_t $(1\le t\le s)$, we may consider linear combinations with a coefficient function $f(I,J)$,

$$\sum_{I,J} f(I,J)=1,\quad f(I,J)\ge0,\quad I=(i_1,i_2,\cdots,i_s),\quad J=(j_1,j_2,\cdots,j_s), \tag{6.36}$$

that is, formal sums $\mathcal{F}=\sum_{I,J}f(I,J)\,\omega_{I,J}$, where the words $\omega_{I,J}$ of length s have the indices I,J. For such $\mathcal{F}$,

$$\begin{aligned}\left|\mathbb{E}\sum_{I,J}f(I,J)\,\omega_{I,J}(G^{\gamma_0})\right| &\le \left|\mathbb{E}\sum_{I,J}f(I,J)\,\omega_{I,J}(G^0)\right|\\ &\quad+\max_{\mathbf{k},n,\gamma}(N^{-\phi/20})^{\sum_i|\mathbf{k_i}|}\left|\mathbb{E}\sum_{I,J}f(I,J)\mathcal{P}_{\gamma_n,\mathbf{k}_n}\cdots\mathcal{P}_{\gamma_1,\mathbf{k}_1}\omega_{I,J}(G^0)\right| + N^{-\zeta},\end{aligned} \tag{6.37}$$

the maximum being over n, $\gamma_1,\dots,\gamma_n$ and $\mathbf{k}_1,\dots,\mathbf{k}_n$ satisfying (6.34). Equivalently: if $B\ge0$ and $N^{-(\phi/20)|d|}|\mathbb{E}\,\mathcal{P}_d\mathcal{F}(G^0)|\le B$ for every nonempty d satisfying (6.34), then $|\mathbb{E}\,\mathcal{F}(G^{\gamma_0})|\le|\mathbb{E}\,\mathcal{F}(G^0)|+B+N^{-\zeta}$.

The max forms are not consequences of (6.33): bounding each term of (6.33) by $BN^{(\phi/20)|d|}$ leaves $B\sum_{d\neq\emptyset}\prod_t W(\gamma_t)$, which is not bounded by B: already $\sum_\gamma W(\gamma) = 1 + (N-1)/(2N) > 1$. They are proved by the same induction, with the uniform bound B in place of the sums. Namely, for $0 \le j \le n_*$, every d of length $n_* - j$ satisfying (6.34) (or $d = \emptyset$ if $j = n_*$) and every $0 \le \gamma' \le \gamma_{\max}$,

$$\left|\mathbb{E}\,\mathcal{P}_d\mathcal{F}(G^{\gamma'})\right| \le \left|\mathbb{E}\,\mathcal{P}_d\mathcal{F}(G^0)\right| + \rho^{|d|}\left(B + \theta^j\, 8\cdot 4^s + 2E_*\right).$$

For $j = 0$ this follows from Step 3, as $\|\mathcal{P}_d\mathcal{F}\| \le 2^{|d|}$. For the step from $j-1$ to j, we apply (6.29) to $\mathcal{F}' = \mathcal{P}_d\mathcal{F}$. For each (γ,α), the data d' obtained by appending (γ,α) to d is nonempty, has length $n_* - j + 1$ and satisfies (6.34), and $\mathcal{P}_{\gamma,\alpha}\mathcal{P}_d = \mathcal{P}_{d'}$. By the induction hypothesis and the assumption on B, $|\mathbb{E}\,\mathcal{P}_{d'}\mathcal{F}(G^{\gamma-a})| \le \rho^{|d'|}(2B + \theta^{j-1}\,8\cdot 4^s + 2E_*)$. By (6.30), the sum over (γ,a,α) is at most $\theta\rho^{|d|}(2B + \theta^{j-1}\,8\cdot 4^s + 2E_*) \le \rho^{|d|}(B + \theta^j\,8\cdot 4^s + E_*)$. Adding the swap term $\rho^{|d|}E_*$ completes the step. For $d = \emptyset$ and $j = n_*$, Step 5 gives (6.37).

In application, we let $H = H^{\gamma_{\max}}$, $\widetilde{H} = H^0$.

For words whose factors are centred, (6.37) reads as follows; here Step 1 uses (6.23), not (6.24):

$$\begin{aligned}
&\left|\mathbb{E}\sum_{I,J} f(I,J)\prod_{t=1}^{s}((G-m_{sc})_{i_tj_t})\right| \\
&\le \left|\mathbb{E}\sum_{I,J} f(I,J)\prod_{t=1}^{s}((\widetilde{G}-m_{sc})_{i_tj_t})\right| \qquad (6.38)\\
&\quad + \max_{\gamma,n,\mathbf{k}}(N^{-\phi/20})^{\sum_i|\mathbf{k}_i|}\left|\mathbb{E}\sum_{I,J} f(I,J)\widetilde{\mathcal{P}}_{\gamma_n,\mathbf{k}_n}\cdots\widetilde{\mathcal{P}}_{\gamma_1,\mathbf{k}_1}\left(\prod_{t=1}^{s}(\widetilde{G}-m_{sc})_{i_tj_t}\right)\right| + N^{-\zeta}.
\end{aligned}$$

The same holds when some of the factors are replaced by their complex conjugates. We note that, in the case $f = N^{-s}\prod\delta_{i_tj_t}$ (with $f(I,J) = 0$ unless $I = J$), the left hand side equals $|\mathbb{E}(m - m_{sc})^s|$ and the first term in the right hand side equals $|\mathbb{E}(\widetilde{m} - m_{sc})^s|$; with conjugates on half of the factors, s even, they equal $\mathbb{E}|m - m_{sc}|^s$ and $\mathbb{E}|\widetilde{m} - m_{sc}|^s$.

Now, we first use (6.33) to prove Lemma 3.8. The following lemma bounds the iterated words that occur.

Lemma 6.6. *Let H be a random real symmetric matrix, $G = (H-z)^{-1}$ with $\eta > 0$, $i \neq j$ and $0 \le \epsilon \le 1$, and let Ω be an event on which $|G_{xy} - \delta_{xy}m_{sc}| \le \epsilon$ for all x, y. Let $n_0, \xi \in \mathbb{N}$, and let $d = ((\gamma_1,\mathbf{k}_1),\dots,(\gamma_n,\mathbf{k}_n))$ with $n \le n_0$, $\mathbf{k}_t \in \mathbb{N}^{2+|\mathbf{k}_1|+\cdots+|\mathbf{k}_{t-1}|}$ and $m_0(\gamma_t) < |\mathbf{k}_t| \le \xi$. Let $\Phi(a_t,b_t) = \gamma_t$. Then*

$$\left|\mathbb{E}\,\mathcal{P}_d\big(G_{ij}\overline{G_{ij}}\big)\right| \le 4^{|d|}\Big(c_d + (\eta^{-1}+1)^{2+n_0\xi}\,\mathbb{P}(\Omega^c)\Big),$$

where $c_d = 4$ if $\{a_t,b_t\}\cap\{i,j\} \neq \emptyset$ for some t, and $c_d = \epsilon^2$ otherwise.

Proof. By the remark at the beginning of the iteration, $\mathcal{P}_d(G_{ij}\overline{G_{ij}})$ is a sum of at most $2^{|d|}$ words of length $2+|d| \le 2+n_0\xi$, whose factors are entries of G or $\overline{G}$. On Ω every factor is at most $|m_{sc}|+\epsilon \le 2$, so every word is at most $2^{2+|d|}$, which gives the bound with $c_d = 4$. Off Ω every factor is at most η^{-1}, which gives the term with $\mathbb{P}(\Omega^c)$. Now let $\{a_t,b_t\}\cap\{i,j\} = \emptyset$ for all t. The operator $\mathcal{P}_{\gamma_t,k}$ with $k \ge 1$ replaces a factor G_{xy} by chains $G_{xu_1}G_{v_1u_2}\cdots G_{v_ky}$ with $u_r, v_r \in \{a_t,b_t\}$; it keeps the end indices x, y and creates only indices in $\{a_t,b_t\}$. Hence the index i appears in every word exactly twice, as the first index of the first factor of the chain coming from G_{ij} and of the chain coming from $\overline{G_{ij}}$. The other index of each of these two factors is j or some a_t or b_t, so it differs from i. Thus every word contains two off-diagonal factors, which are at most ϵ on Ω, and is at most $\epsilon^2 2^{|d|}$ on Ω. This gives the bound with $c_d = \epsilon^2$. □

Proof of Lemma 3.8. Let H satisfy the conditions of Lemma 3.8; in particular $|h_{ij}| \le N^{-\phi}$ almost surely. Fix $0 < c < 1$, let $z = E + i\eta$ with $|E| \le 5$ and $\eta \ge N^{-1+c}$, and let $i \ne j$. All constants below depend only on C_1, ϕ and c. We write

$$\Psi := \frac{\operatorname{Im} m_{sc}(z)}{N\eta} + \frac{1}{(N\eta)^2}, \qquad \operatorname{Im} m_{sc}(z) = \int \frac{\eta\,\rho_{sc}(x)\,dx}{(x-E)^2+\eta^2} \ge \frac{\eta}{49+\eta^2},$$

where the inequality holds since $|x - E| \le 7$ on the support of ρ_{sc}. In particular, for $0 < \eta \le 10$ we have $49 + \eta^2 \le 149$, hence $N^{-1} \le 149 \operatorname{Im} m_{sc}/(N\eta) \le 149\,\Psi$.

The case $\eta > 10$. Let $G^{(i)}$ and $G^{(ij)}$ be the resolvents of the matrices obtained from H by removing the row and the column i, and the rows and the columns i and j, respectively. By the Schur complement formula,

$$G_{ij} = -G_{ii}\,G^{(i)}_{jj}\Big(h_{ij} - Z\Big), \qquad Z := \sum_{k,l\notin\{i,j\}} h_{ik} G^{(ij)}_{kl} h_{lj}.$$

Since $|G_{ii}|, |G^{(i)}_{jj}| \le \eta^{-1}$, we get $|G_{ij}|^2 \le \eta^{-4}(2h_{ij}^2 + 2|Z|^2)$. Let $Y_k := \sum_{l\notin\{i,j\}} G^{(ij)}_{kl} h_{lj}$. The entries h_{ik}, $k \notin \{i,j\}$, are independent of each other and of $(Y_k)_k$, centred, with variance N^{-1}; hence $\mathbb{E}|Z|^2 = N^{-1}\sum_k \mathbb{E}|Y_k|^2$. In the same way, $\mathbb{E}|Y_k|^2 = N^{-1}\sum_{l\notin\{i,j\}} \mathbb{E}|G^{(ij)}_{kl}|^2 \le N^{-1}\eta^{-2}$, since $\sum_l |G^{(ij)}_{kl}|^2 \le \|G^{(ij)}\|^2 \le \eta^{-2}$. Therefore $\mathbb{E}|Z|^2 \le N^{-1}\eta^{-2}$ and

$$\mathbb{E}|G_{ij}|^2 \le \eta^{-4}\big(2N^{-1} + 2N^{-1}\eta^{-2}\big) \le \frac{3}{N\eta^4} \le \frac{\operatorname{Im} m_{sc}}{N\eta},$$

where the last inequality follows from $\operatorname{Im} m_{sc} \ge \eta/(49+\eta^2)$ and $3(49+\eta^2) \le \eta^4$ for $\eta \ge 10$.

The case $\eta \le 10$. Let $\widetilde{H}$ match H as in Lemma 5.1. Its support parameter is $\widetilde{q} = N^{1/2}/(K\log N)$ with $K := (2C_1+4)C_1$, so $\widetilde{q} \sim N^{1/2}/\log N$ and $|\widetilde{h}_{ij}| \le \widetilde{q}^{-1}$ almost surely. The pair $(H, \widetilde{H})$ has the properties stated at the beginning of this section with $q = \min(N^{\phi}, \widetilde{q}) \ge N^{\phi_0}$, $\phi_0 := \min(\phi, 1/4)$, for $N \ge N_0$. For $N \ge N_0$ we also have $z \in \mathbf{S}(C')$ for any fixed C', since $\eta \ge N^{-1+c} > \varphi^{C'}N^{-1}$. By Lemma 3.1 with $\zeta = 1$, applied to $\widetilde{H}$, there are constants C_L, c_L, C'_L and an event Ω with $\mathbb{P}(\Omega^c) \le N^{C'_L} e^{-c_L\varphi}$ on which

$$|\widetilde{G}_{xy} - \delta_{xy} m_{sc}| \le \epsilon := \varphi^{C_L}\left(\frac{1}{\widetilde{q}} + \sqrt{\frac{\operatorname{Im} m_{sc}}{N\eta}} + \frac{1}{N\eta}\right) \quad \text{for all } x, y. \tag{6.39}$$

The term $\widetilde{q}^{-1}$ is kept. Let $c' := \min(c, 1/2)$. For $N \ge N_0$, $\varphi^{C_L} \le N^{c'/4}$ and each of the three terms in the parenthesis is at most $N^{-c'/2}$, so $\epsilon \le 3N^{-c'/4} \le 1$; moreover $\epsilon^2 \le 3\varphi^{2C_L}(\widetilde{q}^{-2} + \operatorname{Im} m_{sc}/(N\eta) + (N\eta)^{-2})$.

We apply (6.33) with ϕ_0 in place of ϕ, $K_s = 3$, $\zeta = 1$, $s = 2 \le K_s\zeta$, $\gamma_0 = \gamma_{\max}$ and the single word $G_{ij}\overline{G_{ij}}$. Since $\mathbb{E}\,G_{ij}\overline{G_{ij}} = \mathbb{E}|G_{ij}|^2$, we obtain

$$\mathbb{E}|G_{ij}|^2 \le \sum_d w(d)\left|\mathbb{E}\,\mathcal{P}_d\big(\widetilde{G}_{ij}\overline{\widetilde{G}_{ij}}\big)\right| + N^{-1}, \tag{6.40}$$

the sum being over the data of (6.34) with $s = 2$, $n_* = \lceil 6/\phi_0 \rceil$ and $\xi_* = \lceil 10/\phi_0 \rceil$. By Lemma 6.6 with $n_0 = n_*$, $\xi = \xi_*$ and $\eta^{-1} \le N$, each term is at most $w(d)4^{|d|}(c_d + (N+1)^{2+n_*\xi_*}\mathbb{P}(\Omega^c))$.

We now bound the weights. Note that $w(d)4^{|d|} = \prod_t W(\gamma_t) y^{|\mathbf{k}_t|}$ with $y := 4N^{-\phi_0/20}$. Let $t_* := 2 + n_*\xi_*$, so that every word length s' that occurs satisfies $s' \le t_*$, and let N_0 be so large that $t_* y \le 1/8$. The number of $\mathbf{k} \in \mathbb{N}^{s'}$ with $|\mathbf{k}| = m$ is at most t_*^m. Hence, for each step,

$$\sum_{\gamma} W(\gamma) \sum_{m_0(\gamma) < |\mathbf{k}| \le \xi_*} y^{|\mathbf{k}|} \le \frac{N^2}{2} N^{-2} \cdot 2(t_* y)^5 + N \cdot N^{-1} \cdot 2(t_* y)^3 \le \frac{1}{4}.$$

Restricted to the $\gamma = \Phi(a,b)$ with $\{a,b\} \cap \{i,j\} \ne \emptyset$ (at most $2N$ off-diagonal and 2 diagonal ones), the same sum is at most $2N \cdot N^{-2} \cdot 2(t_* y)^5 + 2N^{-1} \cdot 2(t_* y)^3 \le 1/(16N)$. Summing over the steps one at a time, we get $\sum_d w(d)4^{|d|} \le \sum_{n\ge 0} 4^{-n} \le 2$. For the data that meet $\{i,j\}$, at least one of the n

steps is restricted, so their total weight is at most $\sum_{n\ge1} n4^{-(n-1)}/(16N) = 1/(9N)$. This is the gain N^{-1} of the touching iterates; in particular it is at most $\lceil 6/\phi_0\rceil/N$. Therefore

$$\mathbb{E}|G_{ij}|^2 \le 2\epsilon^2 + 2(N+1)^{2+n_*\xi_*}N^{C_L'}e^{-c_L\varphi} + \frac{4}{9N} + \frac{1}{N}. \tag{6.41}$$

We bound the four terms by multiples of Ψ, using $N^{-1} \le 149\,\Psi$. First, $\widetilde{q}^{-2} = K^2(\log N)^2N^{-1} \le 149K^2(\log N)^2\Psi$, so $2\epsilon^2 \le 6\varphi^{2C_L}(149K^2(\log N)^2+1)\Psi$. Second, for $N \ge N_0$ we have

$$2(N+1)^{2+n_*\xi_*}N^{C_L'}e^{-c_L\varphi} \le (149N)^{-1} \le \Psi,$$

since $\varphi \ge (\log N)^3$. Third, $4/(9N) + 1/N \le 298\,\Psi$. Hence

$$\mathbb{E}|G_{ij}|^2 \le \varphi^{2C_L}\Big(6\big(149K^2(\log N)^2+1\big)+299\Big)\Psi \le \varphi^{2C_L+1}\Psi$$

for $N \ge N_0$, since $(\log N)^3 \le \varphi$. This is (3.15) with $C = 2C_L+1$, and it completes the proof of Lemma 3.8. □

Next, we use (6.38) to prove Lemma 5.4.

Proof of Lemma 5.4. We prove (5.11) directly, with $C = 2$. Let C_0 be the constant of (6.1) and of (6.38) for $K_s = 1$, let $z \in \mathbf{S}(C_0)$, let $X, Y \ge 0$, and assume that (5.10) holds for $\widetilde{H}$ outside an event of probability at most $N^{C'}e^{-c'\varphi^3}$. The threshold N_0 depends on c' and C'. Let $p = 2h \le \varphi$ with $h \ge 1$ (for $p = 0$ there is nothing to prove). Since $|m - m_{sc}|^p = (m-m_{sc})^h\,\overline{(m-m_{sc})}^h$,

$$|m - m_{sc}|^p = \frac{1}{N^p}\sum_{i_1,\dots,i_p}\prod_{t=1}^{h}(G-m_{sc})_{i_ti_t}\prod_{t=h+1}^{p}\overline{(G-m_{sc})_{i_ti_t}} =: \mathcal{F}(G), \tag{6.42}$$

a formal sum of words of length p with centred factors, with weights $f(I,I) = N^{-p}$ and $f(I,J) = 0$ for $I \ne J$, which satisfy (6.36). We apply (6.38), with conjugates on half of the factors, in the equivalent form with a bound B, with $s = \zeta = p$ and $K_s = 1$; then $1 \le \zeta \le \varphi$ and $s \le K_s\zeta$. If $B \ge 0$ bounds $N^{-(\phi/20)|d|}|\mathbb{E}\,\widetilde{\mathcal{P}}_d\mathcal{F}(\widetilde{G})|$ for every nonempty d satisfying (6.34), then

$$\mathbb{E}|m - m_{sc}|^p \le \mathbb{E}|\widetilde{m} - m_{sc}|^p + B + N^{-p}. \tag{6.43}$$

Fix such d and $I = (i_1,\dots,i_p)$. The expression

$$\widetilde{\mathcal{P}}_{\gamma_n,\mathbf{k}_n}\cdots\widetilde{\mathcal{P}}_{\gamma_1,\mathbf{k}_1}\left(\prod_{t=1}^{h}(\widetilde{G}-m_{sc})_{i_ti_t}\prod_{t=h+1}^{p}\overline{(\widetilde{G}-m_{sc})_{i_ti_t}}\right) \tag{6.44}$$

is a sum of at most $2^{|d|}$ products $\beta_1[i_1]\cdots\beta_p[i_p]$ of total length $p+|d|$. Here the block $\beta_t[i_t]$ contains all factors that carry the index i_t, and it is either (a) $(\widetilde{G}-m_{sc})_{i_ti_t}$ or its conjugate, or (b) a chain $\widetilde{G}_{i_tx}\cdots\widetilde{G}_{yi_t}$ or its conjugate, whose inner factors and end indices x, y come from the γ_u and do not depend on I. Indeed, $\widetilde{\mathcal{P}}_{\gamma,k}$ with $k \ge 1$ replaces $(\widetilde{G}-m_{sc})_{ii}$ by chains $\widetilde{G}_{iu_1}\cdots\widetilde{G}_{v_ki}$ with $u_r, v_r \in \{a,b\}$, $\Phi(a,b) = \gamma$, and it replaces a factor of a chain by chains with the same end indices. So the shapes (a) and (b) are preserved, and the blocks do not depend on I.

Let $\widetilde{\Omega}$ be the intersection of the event (6.1) for $\widetilde{H}$ with the event (5.10); then $\mathbb{P}(\widetilde{\Omega}^c) \le N^{C''}e^{-c''\varphi^3}$. On $\widetilde{\Omega}$ every factor is at most 3, since $|m_{sc}| \le 1$. Let $Q := X^2 + Y + 18N^{-1}$. For every block β of length $|\beta|$, on $\widetilde{\Omega}$,

$$\Big|\frac{1}{N}\sum_{i=1}^{N}\beta[i]\Big| \le 3^{|\beta|}Q.$$

In case (a) the left side is $|\widetilde{m} - m_{sc}| \le Y$. In case (b), for $i \notin \{x,y\}$ both end factors are off-diagonal and at most X; the two values $i \in \{x,y\}$ contribute at most $9+9$; and the inner factors are at most 3.

So the left side is at most $3^{|\beta|-2}(X^2+18N^{-1}) \le 3^{|\beta|}Q$. Since the blocks of one product carry distinct summation indices, $N^{-p}\sum_{i_1,\dots,i_p}\prod_t \beta_t[i_t] = \prod_t N^{-1}\sum_i \beta_t[i]$. Hence, on $\widetilde{\Omega}$,

$$\Big|\frac{1}{N^p}\sum_{i_1,\dots,i_p}\widetilde{\mathcal{P}}_d\Big(\prod_{t=1}^{h}(\widetilde{G}-m_{sc})_{i_ti_t}\prod_{t=h+1}^{p}\overline{(\widetilde{G}-m_{sc})_{i_ti_t}}\Big)\Big| \le 2^{|d|}\,3^{p+|d|}Q^p = 6^{|d|}3^pQ^p. \tag{6.45}$$

Off $\widetilde{\Omega}$, the factors are at most $\eta^{-1}+1 \le N+1$ and the words have length $p+|d| \le K'\varphi^2$. So this part of the expectation is at most $2^{|d|}(N+1)^{K'\varphi^2}N^{C''}e^{-c''\varphi^3} \le N^{-p}$ for $N \ge N_0$, since $\varphi/\log N \to \infty$. Since $N^{\phi/20} \ge 6$ for $N \ge N_0$, the premise of (6.43) holds with $B := 3^pQ^p + N^{-p}$, and (6.43) gives

$$\mathbb{E}|m-m_{sc}|^p \le \mathbb{E}|\widetilde{m}-m_{sc}|^p + 3^pQ^p + 2N^{-p}.$$

For $N \ge N_0$ we have $\varphi \ge 18$, so $Q \le X^2+Y+\varphi N^{-1}$ and $N^{-p} \le (X^2+Y+\varphi N^{-1})^p$. Since $3^p+2 \le (2p)^{2p}$ for $p \ge 2$,

$$\mathbb{E}|m-m_{sc}|^p \le \mathbb{E}|\widetilde{m}-m_{sc}|^p + (2p)^{2p}\left(X^2+Y+\varphi N^{-1}\right)^p,$$

which is (5.11) with $C=2$. This completes the proof of Lemma 5.4. □

Proof of Lemma 6.5. Throughout the proof, $\gamma = \Phi(a,b)$, and m_0 is the number of moments matched at the entry (a,b), i.e., $m_0 = 4$ if $a \ne b$ and $m_0 = 2$ if $a = b$. Every entry h of $H_{\gamma'}$ and of Q satisfies $|h| \le q^{-1} \le N^{-\phi}$ almost surely, where $q \ge N^\phi$ is the almost sure bound fixed at the beginning of this section. Choose

$$\xi := \lceil 2\zeta/\phi \rceil, \quad \text{so that} \quad \xi\phi \ge 2\zeta. \tag{6.46}$$

We apply the expansion

$$S = R - RVR + (RV)^2R - \cdots + (-1)^\xi(RV)^\xi R + (-1)^{\xi+1}(RV)^{\xi+1}S. \tag{6.47}$$

With the condition $|v_{ab}| \le N^{-\phi}$, we note that the last term in this expansion is $O(N^{-\zeta})$. Thus,

$$S = R - RVR + (RV)^2R - \cdots + (-1)^\xi(RV)^\xi R + O(N^{-\xi\phi}). \tag{6.48}$$

Since $v_{kl} = 0$ if $\{k,l\} \ne \{a,b\}$, we have

$$([-RV]^mR)_{i_tj_t} = \sum_{(a_i,b_i)\in\{(a,b),(b,a)\};\,1\le i\le m} R_{i_ta_1}V_{a_1b_1}R_{b_1a_2}\cdots V_{a_mb_m}R_{b_mj_t}\,. \tag{6.49}$$

We note that $V_{ab} = V_{ba} = v_{ab} = h_{ab}$ (for $a=b$, the only nonzero entry of V is $V_{aa} = h_{aa}$). Using Definition 6.2 and Definition 6.3, we have

$$[-RV]^kR = R^{*(k+1)}(-h_{ab})^k, \quad S = \sum_{0\le k\le\xi}(R^{*(k+1)})(-h_{ab})^k + O(N^{-\xi\phi}). \tag{6.50}$$

Similarly,

$$R_{ij} = \sum_{0\le k\le\xi}(\mathcal{P}_{\gamma,k}S_{ij})(h_{ab})^k + O(N^{-\xi\phi}), \quad (R-m_{sc})_{ij} = \sum_{0\le k\le\xi}(\widetilde{\mathcal{P}}_{\gamma,k}S_{ij})(h_{ab})^k + O(N^{-\xi\phi}). \tag{6.51}$$

The proof below is written for $\prod_t S_{i_tj_t}$; it applies without change to the products with conjugate factors, centred factors $(S-m_{sc})_{ij}$, or both, that are covered by Lemma 6.5, because it uses only an expansion of each factor separately. Since h_{ab} is real, conjugating (6.50) term by term gives

$$\overline{S_{ij}} = \sum_{0\le k\le\xi}\overline{(R^{*(k+1)})_{ij}}\,(-h_{ab})^k + O(N^{-\xi\phi}),$$

and subtracting $m_{sc}\delta_{ij}$ from the term $k=0$ of (6.50) gives

$$\begin{aligned}(S-m_{sc})_{ij} &= (R-m_{sc})_{ij} + \sum_{1\le k\le \xi} (R^{*(k+1)})_{ij}(-h_{ab})^k + O(N^{-\xi\phi}) \\ &= \sum_{0\le k\le \xi} \big(\widetilde{\mathcal{P}}_{\gamma,k}(R-m_{sc})_{ij}\big)(-h_{ab})^k + O(N^{-\xi\phi}),\end{aligned}$$

and its conjugate in the same way (for $i\neq j$, $(R-m_{sc})_{ij}=R_{ij}$). Thus each factor of each of the four kinds expands with the operator of its kind, $\mathcal{P}_{\gamma,k}$, $\overline{\mathcal{P}_{\gamma,k}}$, $\widetilde{\mathcal{P}}_{\gamma,k}$ or $\overline{\widetilde{\mathcal{P}}_{\gamma,k}}$. For every k these operators produce the same number of products as $\mathcal{P}_{\gamma,k}$, since $\widetilde{\mathcal{P}}_{\gamma,0}$ and $\mathcal{P}_{\gamma,0}$ both give one entry and $\widetilde{\mathcal{P}}_{\gamma,k}=\mathcal{P}_{\gamma,k}$ for $k\ge 1$. The composition rule (6.19) holds factor by factor for all four kinds: on a centred factor $\widetilde{\mathcal{P}}_{\gamma,k'}\widetilde{\mathcal{P}}_{\gamma,0}=\widetilde{\mathcal{P}}_{\gamma,k'}$, and for $k\ge 1$ the operator $\widetilde{\mathcal{P}}_{\gamma,k}$ produces uncentred entries, to which (6.19) applies; conjugation commutes with all these operations. The bound (6.20) holds for the conjugates as well, since $|\overline{w}|=|w|$. The rest of the proof uses only these per-factor expansions, the bound (6.20) and the moments $\mathbb{E}h_{ab}^k$, and it uses (6.19) and the counts $|I^a_{b,c}|$ in the same way for every kind of factor. Hence the coefficients $\mathcal{A}_\alpha$ and $\widetilde{\mathcal{A}}_\alpha$ constructed below depend only on α, s, ξ, (a,b) and the moments of h_{ab}, and not on the kinds of the factors, and the proof gives (6.21)–(6.24) for all such products with the same coefficients.

For this reason, we only show the proof of (6.21). The proof of (6.23) is the same, with the second identity above in place of the first and with $\widetilde{\mathcal{P}}$ applied to $\prod_t (S-m_{sc})_{i_tj_t}$ and to $\prod_t (R-m_{sc})_{i_tj_t}$ in place of $\mathcal{P}$. The proof of (6.24) and (6.25) is given at the end of this proof.

Using (6.50) and (6.1), with 3-high probability, we have that (with definition 6.3)

$$\prod_{t=1}^s S_{i_tj_t} = \sum_{0\le k\le \xi s;\, \mathbf{k}\in I^s_{\xi,k}} \left(\mathcal{P}_{\gamma,\mathbf{k}} \prod_{t=1}^s R_{i_tj_t}\right)(-h_{ab})^{k_t} + O(2^s N^{-\xi\phi}) \tag{6.52}$$

where

$$\mathbf{k} := (k_1,k_2,\cdots,k_s), \quad I^a_{b,c} := \{\mathbf{k}\in\mathbb{N}^a : 0\le k_i\le b, \sum k_i = c\} \tag{6.53}$$

Note that, from the above definition,

$$|I^a_{b,c}| \le a^c\,. \tag{6.54}$$

We note that the term in (6.52) belonging to $k=0$ is $\prod_{t=1}^s R_{i_tj_t}$. For the terms belonging to $k\ge\xi$, using (6.20), we know that with 3-high probability they are bounded by

$$\sum_{k\ge\xi} s^k 4^{k+s} N^{-k\phi} \le O(s^\xi 4^{\xi+s} N^{-\xi\phi}), \tag{6.55}$$

where s^k comes from $\sum_{\mathbf{k}\in I^s_{\xi,k}}$. Hence,

$$\prod_{t=1}^s S_{i_tj_t} = \prod_{t=1}^s R_{i_tj_t} + \sum_{1\le k\le\xi} \left(\sum_{\mathbf{k}\in I^s_{\xi,k}} \mathcal{P}_{\gamma,\mathbf{k}} \prod_{t=1}^s R_{i_tj_t}\right)(-h_{ab})^k + O(s^\xi 4^{\xi+s} N^{-\xi\phi}). \tag{6.56}$$

Recall that $(S^{*s})_{ij}$ is a sum of terms as in Definition 6.2. As a special case, consider a term in $(S^{*s})_{ij}=\mathcal{P}_{\gamma,s-1}S_{ij}$ and rewrite it as $S_{i_1j_1}S_{i_2j_2}\cdots S_{i_sj_s}$. We have

$$(S^{*s})_{ij} = (R^{*s})_{ij} + \sum_{1\le k\le\xi} \left(\sum_{\mathbf{k}\in I^s_{\xi,k}} (\mathcal{P}_{\gamma,\mathbf{k}}(R^{*s})_{ij})\right)(-h_{ab})^k + O(s^\xi 4^{\xi+s} N^{-\xi\phi}). \tag{6.57}$$

Then, with (6.19), we have (here we replaced s with $s+1$ for simplicity)

$$\mathcal{P}_{\gamma,s}S_{ij} = \mathcal{P}_{\gamma,s}R_{ij} + \sum_{1\le k\le\xi} |I^{s+1}_{\xi,k}|(\mathcal{P}_{\gamma,s+k}R)_{ij}(-h_{ab})^k + O(s^\xi 4^{\xi+s} N^{-\xi\phi}), \tag{6.58}$$

Define, for $0 \le k \le m_0$,

$$A_k := \sum_{\mathbf{k}\in I^s_{\xi,k}} \mathcal{P}_{\gamma,\mathbf{k}} \prod_{t=1}^{s} R_{i_t j_t} = \sum_{\mathbf{k}\in I^s_{\xi,k}} \prod_{t=1}^{s} R^{*(k_t+1)}_{i_t j_t} \tag{6.59}$$

Clearly, they depend only on R. Thus, as in (6.56), with 3-high probability,

$$\prod_{t=1}^{s} S_{i_t j_t} = \sum_{k=0}^{m_0} A_k (-h_{ab})^k + \sum_{m_0<k\le\xi} \left(\sum_{\mathbf{k}\in I^s_{\xi,k}} \mathcal{P}_{\gamma,\mathbf{k}} \prod_{t=1}^{s} R_{i_t j_t} \right) (-h_{ab})^k + O(s^\xi 4^{\xi+s} N^{-\xi\phi}). \tag{6.60}$$

We take the expectation $\mathbb{E}$ in the both sides of the equation. Recall that the good event holds with 3-high probability and the entries of S and R are bounded by $\eta^{-1} \le N$ (see (6.2)). Furthermore, in this proof, no product has more than $s+\xi \le (K_s + 2/\phi + 1)\varphi^2$ entries of S or R, and the number of products is at most $N^{C(s+\xi)}$. Hence the contribution of the bad event to the expectation is at most $N^{C(s+\xi)} e^{-c\varphi^3} \le N^{-\zeta}$, since $(s+\xi)\log N + \zeta \log N \le C\varphi^2 \log N$ is much smaller than φ^3. Thus, when taking the expectation $\mathbb{E}$, we can simply ignore the set of bad event. Since R is independent of h_{ab}, the expectation of each term factorizes, e.g., $\mathbb{E}[A_k(-h_{ab})^k] = \mathbb{E}A_k \cdot \mathbb{E}(-h_{ab})^k$.

To simplify the notation, we define

$$n_k := \mathbb{E}(-h_{ab})^k. \tag{6.61}$$

Then, we get

$$\mathbb{E}\prod_{t=1}^{s} S_{i_t j_t} = \sum_{k=0}^{m_0} n_k \mathbb{E}A_k + \sum_{m_0<k\le\xi} n_k \left(\sum_{\mathbf{k}\in I^s_{\xi,k}} \mathbb{E}\mathcal{P}_{\gamma,\mathbf{k}} \prod_{t=1}^{s} R_{i_t j_t} \right) + O(s^\xi 4^{\xi+s} N^{-\xi\phi}) \tag{6.62}$$

To estimate $\mathbb{E}\mathcal{P}_{\gamma,\mathbf{k}} \prod_{t=1}^{s} R_{i_t j_t} = \mathbb{E}\prod_t \mathcal{P}_{\gamma,k_t} R_{i_t j_t}$, we first use (6.58) and obtain

$$\mathcal{P}_{\gamma,k_t} S_{i_t j_t} = \mathcal{P}_{\gamma,k_t} R_{i_t j_t} + \sum_{1\le l_t\le\xi-k} |I^{k_t+1}_{\xi-k,l_t}| \mathcal{P}_{\gamma,(k_t+l_t)} R_{i_t j_t} (-h_{ab})^{l_t} + O((4k_t+4)^{\xi-k} 4^{k_t+1} N^{-\xi\phi+k\phi}). \tag{6.63}$$

Note that from (6.54) we have a bound

$$|I^{k_t+1}_{\xi-k,l_t}| \le (k_t+1)^{l_t}.$$

From that $|\mathcal{P}_{\gamma,k_t} S_{i_t j_t}| \le 4^{k_t+2}$ (see (6.20)), for $0 \le k_1, k_2, \cdots, k_s \le \zeta/\phi$, $\sum k_i = k$ and $k \le \zeta/\phi$, we obtain that

$$\begin{aligned} \mathcal{P}_{\gamma,\mathbf{k}} \prod_{t=1}^{s} S_{i_t j_t} =& \mathcal{P}_{\gamma,\mathbf{k}} \prod_{t=1}^{s} R_{i_t j_t} + \sum_{1\le l\le(\xi-k)s} \left(\sum_{\mathbf{l}\in I^s_{\xi-k,l}} \prod_t |I^{k_t+1}_{\xi-k,l_t}| \mathcal{P}_{\gamma,k_t+l_t} R_{i_t j_t} \right) (-h_{ab})^l \\ &+ O((4k+4)^{\xi-k} 4^{k+s} N^{-\xi\phi+k\phi}). \end{aligned} \tag{6.64}$$

For the terms that belong to $l \ge \xi - k$, from (6.20) and (6.54), they are bounded by

$$\sum_{l\ge\xi-k} s^l (k+1)^l 4^{k+l+s} N^{-l\phi} \le (sk+s)^{\xi-k} 4^{\xi+s} N^{-\xi\phi+k\phi}.$$

Then, the upper bound of l in (6.64) can be reduced to $\xi - k$ as follows:

$$\begin{aligned} \mathcal{P}_{\gamma,\mathbf{k}} \prod_{t=1}^{s} S_{i_t j_t} =& \mathcal{P}_{\gamma,\mathbf{k}} \prod_{t=1}^{s} R_{i_t j_t} + \sum_{1\le l\le\xi-k} \left(\sum_{\mathbf{l}\in I^s_{\xi-k,l}} \prod_t |I^{k_t+1}_{\xi-k,l_t}| \mathcal{P}_{\gamma,k_t+l_t} R_{i_t j_t} \right) (-h_{ab})^l \\ &+ O((sk+s)^{\xi-k} 4^{\xi+s} N^{-\xi\phi+k\phi}) \end{aligned} \tag{6.65}$$

We observe that

$$\prod_t |I^{k_t+1}_{\xi-k,l_t}|\mathcal{P}_{\gamma,k_t+l_t}R_{i_tj_t} = \left(\prod_t |I^{k_t+1}_{\xi-k,l_t}|\right)\left(\mathcal{P}_{\gamma,\mathbf{k}+\mathbf{l}}\prod_t R_{i_tj_t}\right), \quad \mathbf{l}=(l_1,l_2,\cdots,l_s). \tag{6.66}$$

Inserting it into (6.62), with $|h_{ab}|\le N^{-\phi}$ and $|I^s_{\xi,k}|\le s^k$, $k\le\xi$, we get

$$\begin{aligned}\mathbb{E}\prod_{t=1}^s S_{i_tj_t} =& \sum_{k=0}^{m_0} n_k \mathbb{E}A_k + \sum_{m_0<k\le\xi} n_k\left(\sum_{\mathbf{k}\in I^s_{\xi,k}}\mathbb{E}\mathcal{P}_{\gamma,\mathbf{k}}\prod_{t=1}^s S_{i_tj_t}\right) \\ &- \sum_{k=m_0+1}^{\xi}\sum_{l=1}^{\xi-k} n_k n_l\left(\sum_{\mathbf{k}\in I^s_{\xi,k}}\sum_{\mathbf{l}\in I^s_{\xi-k,l}}\left(\prod_t |I^{k_t+1}_{\xi-k,l_t}|\right)\mathbb{E}\left(\mathcal{P}_{\gamma,\mathbf{k}+\mathbf{l}}\prod_t R_{i_tj_t}\right)\right) \\ &+ O(\xi(s\xi+s)^\xi 4^{s+\xi}N^{-\xi\phi}),\end{aligned} \tag{6.67}$$

where the single ξ factor in the last term comes from $\sum_k$. For $k>m_0$, define

$$A_k = \sum_{\mathbf{k}\in I^s_{\xi,k}}\mathcal{P}_{\gamma,\mathbf{k}}\prod_{t=1}^s S_{i_tj_t}\,. \tag{6.68}$$

Clearly, it has at most s^k terms having the form of $\mathcal{P}\prod S$. Applying (6.65) again on $\mathbb{E}\mathcal{P}_{\gamma,\mathbf{k}+\mathbf{l}}\prod_t R_{i_tj_t}$, as in (6.65), we have

$$\begin{aligned}&\mathcal{P}_{\gamma,\mathbf{k}+\mathbf{l}}\prod_t R_{i_tj_t} \\ =&\mathcal{P}_{\gamma,\mathbf{k}+\mathbf{l}}\prod_t S_{i_tj_t} - \sum_{1\le o\le\xi-k-l}\left(\sum_{\mathbf{o}\in I^s_{\xi-k-l,o}}\left(\prod_t |I^{k_t+l_t+1}_{\xi-k-l,o_t}|\right)\mathcal{P}_{\gamma,\mathbf{k}+\mathbf{l}+\mathbf{o}}\prod_t R_{i_tj_t}\right)(-h_{ab})^o \\ &+ O((s(k+l)+s)^{\xi-k-l}4^{\xi+l+s}N^{(-\xi+k+l)\phi})\end{aligned} \tag{6.69}$$

We now insert it back to (6.67), replacing the notation k, l, o with k_1, k_2, k_3 and k_t, l_t, o_t with $k_1(t)$, $k_2(t)$, $k_3(t)$, respectively. Using

$$|n_k n_l|\sum_{\mathbf{k}\in I^s_{\xi,k}}\sum_{\mathbf{l}\in I^s_{\xi-k,l}}\prod_{t=1}^s |I^{k_t+1}_{\xi-k,l_t}| \le N^{-(k+l)\phi}s^k s^l(k+1)^l, \tag{6.70}$$

we obtain

$$\begin{aligned}&\mathbb{E}\prod_{t=1}^s S_{i_tj_t} \\ =&\sum_{k_1=0}^{\xi} n_{k_1}\mathbb{E}A_{k_1} - \sum_{k_1=m_0+1}^{\xi}\sum_{k_2=1}^{\xi-k_1} n_{k_1}n_{k_2}\left(\sum_{\mathbf{k_1}\in I^s_{\xi,k_1}}\sum_{\mathbf{k_2}\in I^s_{\xi-k_1,k_2}}\left(\prod_{t=1}^s |I^{k_1(t)+1}_{\xi-k_1,k_2(t)}|\right)\mathbb{E}\mathcal{P}_{\gamma,\mathbf{k_1}+\mathbf{k_2}}\prod_t S_{i_tj_t}\right) \\ &+ \sum_{k_1=m_0+1}^{\xi}\sum_{k_2=1}^{\xi-k_1}\sum_{k_3=1}^{\xi-k_1-k_2} n_{k_1}n_{k_2}n_{k_3}\times \\ &\left(\sum_{\mathbf{k_1}\in I^s_{\xi,k_1}}\sum_{\mathbf{k_2}\in I^s_{\xi-k_1,k_2}}\sum_{\mathbf{k_3}\in I^s_{\xi-k_1-k_2,k_3}}\left(\prod_{t=1}^s |I^{k_1(t)+1}_{\xi-k_1,k_2(t)}||I^{k_1(t)+k_2(t)+1}_{\xi-k_1-k_2,k_3(t)}|\right)\mathbb{E}\mathcal{P}_{\gamma,\mathbf{k_1}+\mathbf{k_2}+\mathbf{k_3}}\prod_t R_{i_tj_t}\right) \\ &+ O(\xi^2(s\xi+s)^\xi 4^{s+\xi}N^{-\xi\phi}),\end{aligned} \tag{6.71}$$

where the factor ξ^2 comes from $\sum_{k_1}\sum_{k_2}$. Define

$$A_{k_1,k_2} := -\sum_{\mathbf{k_1}\in I^s_{\xi,k_1}}\sum_{\mathbf{k_2}\in I^s_{\xi-k_1,k_2}}\left(\prod_{t=1}^{s}|I^{k_1(t)+1}_{\xi-k_1,k_2(t)}|\right)\mathbb{E}\mathcal{P}_{\gamma,\mathbf{k_1}+\mathbf{k_2}}\prod_t S_{i_tj_t}. \tag{6.72}$$

Clearly, letting $k := k_1+k_2$, we find that A_{k_1,k_2} has at most

$$s^{k_1+k_2}\prod_{t=1}^{s}|I^{k_1(t)+1}_{\xi-k_1,k_2(t)}| \le s^{k_1+k_2}(k_1+1)^{k_2} \le s^k(k+1)^k \le C(sk)^k$$

terms of the form $\mathcal{P}\prod S$.

We repeat the previous procedure ξ times. Recall that $k_i := \sum_t k_i(t)$. Let

$$\widetilde{k}_i := k_1+k_2+\cdots+k_i, \quad \widetilde{k}_i(t) := k_1(t)+k_2(t)+\cdots+k_i(t). \tag{6.73}$$

Define

$$\begin{aligned}&A_{k_1,k_2,k_3\cdots,k_n}\\ &= (-1)^{n-1}\sum_{\mathbf{k_1}\in I^s_{\xi,k_1}}\sum_{\mathbf{k_2}\in I^s_{\xi-\widetilde{k}_1,k_2}}\sum_{\mathbf{k_3}\in I^s_{\xi-\widetilde{k}_2,k_3}}\cdots\sum_{\mathbf{k_n}\in I^s_{\xi-\widetilde{k}_{n-1},k_n}}\\ &\qquad\prod_{t=1}^{s}\left(\prod_{m=1}^{n-1}|I^{\widetilde{k}_m(t)+1}_{\xi-\widetilde{k}_m,k_{m+1}(t)}|\right)\mathbb{E}\,\mathcal{P}_{\gamma,(\sum_{i=1}^n\mathbf{k_i})}\prod_t S_{i_tj_t}.\end{aligned} \tag{6.74}$$

Let $\#A_{k_1,k_2,k_3\cdots,k_n}$ be the number of the terms of the form $\mathcal{P}\prod S$ in $A_{k_1,k_2,k_3\cdots,k_n}$. Clearly, with $k := \sum_i k_i$,

$$\#A_{k_1,k_2,k_3\cdots,k_n} \le s^k(k+1)^k \le C(sk)^k. \tag{6.75}$$

Thus, we obtain that

$$\begin{aligned}\mathbb{E}\prod_{t=1}^{s}S_{i_tj_t} &= \sum_{k_1=0}^{\xi}n_{k_1}\mathbb{E}A_{k_1}+\sum_{k_1=m_0+1}^{\xi}\sum_{k_2=1}^{\xi-k_1}n_{k_1}n_{k_2}A_{k_1,k_2}+\sum_{k_1=m_0+1}^{\xi}\sum_{k_2=1}^{\xi-k_1}\sum_{k_3=1}^{\xi-k_1-k_2}n_{k_1}n_{k_2}n_{k_3}A_{k_1,k_2,k_3}\\ &\quad+\cdots\\ &\quad+\sum_{k_1,k_2,\cdots,k_\xi;}\left(\prod_{i=1}^{\xi}n_{k_i}\right)A_{k_1,k_2,\cdots,k_\xi}+O(\xi^\xi(s\xi+s)^\xi 4^{s+\xi}N^{-\xi\phi}),\end{aligned} \tag{6.76}$$

where we sum up $k_1,k_2,\cdots,k_\xi$ under the condition $k_1>m_0, k_i\ge 1, 1\le\sum k_i\le\xi$. The factor ξ^ξ comes from $\sum_{k_1,k_2,\cdots,k_\xi;}$.

The equation (6.76) implies (6.21). Indeed, by (6.74), $\prod_i n_{k_i}A_{k_1,\ldots,k_n}$ is a linear combination of the terms $\mathbb{E}\,\mathcal{P}_{\gamma,\alpha}\prod_t S_{i_tj_t}$ with $\alpha=\sum_{i=1}^n\mathbf{k}_i$. Collecting, for each $\alpha\in\mathbb{N}^s$, the coefficients of $\mathbb{E}\,\mathcal{P}_{\gamma,\alpha}\prod_t S_{i_tj_t}$ in all terms of (6.76) with $k_1>m_0$ gives (6.21) with

$$\mathcal{A}_\alpha=\sum_{n\ge1}\sum_{(\mathbf{k}_1,\ldots,\mathbf{k}_n)}(-1)^{n-1}\Big(\prod_{i=1}^{n}n_{k_i}\Big)\prod_{t=1}^{s}\prod_{m=1}^{n-1}|I^{\widetilde{k}_m(t)+1}_{\xi-\widetilde{k}_m,k_{m+1}(t)}|, \qquad m_0<|\alpha|\le\xi,$$

where the second sum is over $\mathbf{k}_1\in I^s_{\xi,k_1}$, $\mathbf{k}_i\in I^s_{\xi-\widetilde{k}_{i-1},k_i}$ $(2\le i\le n)$ with $k_1>m_0$, $k_i\ge1$ $(i\ge2)$ and $\sum_i\mathbf{k}_i=\alpha$, and with $\mathcal{A}_\alpha:=0$ outside the window $m_0<|\alpha|\le\xi$. The coefficients $\mathcal{A}_\alpha$ depend only on α, s, ξ and the moments n_k of h_{ab}; in particular they do not depend on (i_t,j_t) or on z. The terms of (6.76) with $k_1\le m_0$ are $n_{k_1}\mathbb{E}A_{k_1}$, where A_{k_1} depends only on R. For the error term of (6.76), recall

that $s \le K_s\varphi^2$ and $\xi \le 2\varphi^2/\phi + 1$, so that $\log(4\xi(s+1)(\xi+1)) \le C(\log\log N)^2 \le (\phi/4)\log N$ for N large enough, depending only on (ϕ, K_s). With $\xi\phi \ge 2\zeta$, we get

$$\xi^\xi (s\xi+s)^\xi 4^{s+\xi} N^{-\xi\phi} \le 4^s \exp\Big(\xi\big[\log(4\xi(s+1)(\xi+1)) - \phi\log N\big]\Big) \le 4^s N^{-3\xi\phi/4} \le 4^s N^{-\zeta} N^{-\xi\phi/4},$$

so that the error terms, including the implicit constants and the contribution of the bad event, are at most $4^s N^{-\zeta}$ for N large enough, depending only on (C_1, ϕ, K_s).

Now we are ready to prove (6.22). Since α plays the role of $(\sum_{i=1}^n \mathbf{k_i})$ in (6.74), we have that

$$|\alpha| = \sum_t \alpha_t = \sum_i k_i \tag{6.77}$$

Then, since there are at most $2^{|\alpha|}$ sequences $(k_1, \dots, k_n)$ of positive integers with $\sum_i k_i = |\alpha|$, we obtain from (6.75) that

$$|\mathcal{A}_\alpha| \le \sum_{n\ge1} \sum_{k_1>m_0,\, k_i\ge1,\, \sum k_i=|\alpha|} \#A_{k_1,k_2,\cdots,k_n} \prod_{i=1}^n |n_{k_i}| \le C\,(4s\xi)^{|\alpha|} \max_{k_1>m_0,\, k_i\ge1,\, \sum k_i=|\alpha|} \prod_{i=1}^n |n_{k_i}|.$$

Every entry h of H_γ has the law of the corresponding entry of $H^{\mathbf{v}}$ or of $H^{\mathbf{w}}$, and $|h| \le q^{-1}$ almost surely. For $a \ne b$, $\mathbb{E}h_{ab}^4 \le C_1(\log N)N^{-2}$, and for the diagonal entries $\mathbb{E}h_{aa}^2 \le C_1 N^{-1}$; for the entries of $H^{\mathbf{w}}$ this follows from the moment matching, since these moments are among the matched ones. Since $|\mathbb{E}h^k| \le q^{-(k-m)}\mathbb{E}h^m$ for even $m \le k$, every entry of every H_γ satisfies

$$\begin{aligned} &|\mathbb{E}(h_{ab})^k| \le C_1(\log N)N^{-2}q^{-(k-4)} \le C_1(\log N)N^{-2-(k-4)\phi}, \quad a \ne b, \quad k \ge 4, \\ &|\mathbb{E}(h_{aa})^k| \le C_1 N^{-1}q^{-(k-2)} \le C_1 N^{-1-(k-2)\phi}, \quad k \ge 2. \end{aligned} \tag{6.78}$$

Moreover, $n_1 = 0$, and $N^{-1} = \mathbb{E}h_{12}^2 \le q^{-2}$, so that, with $M := \max(C_1, 1)$, $|n_k| \le Mq^{-k} \le MN^{-k\phi}$ for all $k \ge 2$, for $a \ne b$ and for $a = b$.

If $a \ne b$, then $m_0 = 4$, $k_1 \ge 5$, and the two bounds give $\prod_i |n_{k_i}| \le C_1(\log N)M^{|\alpha|}N^{-2-(|\alpha|-4)\phi}$. Since $|\alpha| - 4 \ge |\alpha|/5$ for $|\alpha| \ge 5$,

$$|\mathcal{A}_\alpha| \le C\,C_1(\log N)(4Ms\xi)^{|\alpha|}N^{-2-|\alpha|\phi/5} \le N^{-2-|\alpha|\phi/10},$$

because $s\xi \le C\varphi^4$, so that $C\,C_1(\log N)(4Ms\xi)^{|\alpha|} \le N^{|\alpha|\phi/10}$ for N large enough. If $a = b$, then $m_0 = 2$, $k_1 \ge 3$, and $\prod_i |n_{k_i}| \le C_1 M^{|\alpha|}N^{-1-(|\alpha|-2)\phi}$. Since $|\alpha| - 2 \ge |\alpha|/3$ for $|\alpha| \ge 3$,

$$|\mathcal{A}_\alpha| \le C\,C_1(4Ms\xi)^{|\alpha|}N^{-1-|\alpha|\phi/3} \le N^{-1-|\alpha|\phi/4}.$$

This proves (6.22) in both cases.

Finally, we prove (6.24) and (6.25). Taking the expectation of (6.56), as in (6.62), we get

$$\mathbb{E}\prod_{t=1}^s S_{i_tj_t} = \mathbb{E}\prod_{t=1}^s R_{i_tj_t} + \sum_{1\le k\le\xi} n_k \left(\sum_{\mathbf{k}\in I^s_{\xi,k}} \mathbb{E}\mathcal{P}_{\gamma,\mathbf{k}} \prod_{t=1}^s R_{i_tj_t} \right) + O(s^\xi 4^{\xi+s} N^{-\xi\phi}), \tag{6.79}$$

i.e., the terms $k = 1, \dots, m_0$ are moved from the first sum of (6.62) to the second one, and the first sum keeps only $k = 0$. We now convert each $\mathbb{E}\,\mathcal{P}_{\gamma,\mathbf{k}} \prod_t R_{i_tj_t}$, $|\mathbf{k}| \ge 1$, into terms $\mathbb{E}\,\mathcal{P}_{\gamma,\alpha} \prod_t S_{i_tj_t}$ by (6.65), and repeat the procedure (6.67)–(6.76) without change, except that the condition $k_1 > m_0$ is replaced by $k_1 \ge 1$. The error terms are bounded as above. Collecting the coefficients as above gives (6.24) with

$$\widetilde{\mathcal{A}}_\alpha = \sum_{n\ge1} \sum_{(\mathbf{k}_1,\dots,\mathbf{k}_n)} (-1)^{n-1} \Big(\prod_{i=1}^n n_{k_i} \Big) \prod_{t=1}^s \prod_{m=1}^{n-1} |I^{\widetilde{k}_{m(t)+1}}_{\xi-\widetilde{k}_m, k_{m+1}(t)}|, \qquad 1 \le |\alpha| \le \xi,$$

where the second sum is over the same vectors as for $\mathcal{A}_\alpha$, but with $k_1 \ge 1$ in place of $k_1 > m_0$. Again, $\widetilde{\mathcal{A}}_\alpha$ depends only on α, s, ξ and the moments n_k of h_{ab}, and not on (i_t, j_t) or on z. Since $|n_k| \le MN^{-k\phi}$

for all $k \geq 1$, we have $\prod_i |n_{k_i}| \leq M^{|\alpha|} N^{-|\alpha|\phi}$ for every sequence with $\sum_i k_i = |\alpha|$, and, as above,

$$|\widetilde{\mathcal{A}}_\alpha| \leq C(4Ms\xi)^{|\alpha|} N^{-|\alpha|\phi} \leq N^{-|\alpha|\phi/10}$$

for N large enough. This proves (6.24) and (6.25). In all the steps above, the constants and the threshold for N depend only on (C_1, ϕ, K_s); they do not depend on ζ, s, γ, (a,b), $z \in \mathbf{S}(C_0)$ or the indices (i_t, j_t). □

Last, we prove Lemma 5.5.

Proof of Lemma 5.5. For simplicity, we only prove (5.14). The proof for (5.15) is similar.

Recall that $\eta = N^{-2/3-\epsilon}$. Define

$$x^S := \eta \operatorname{Im} \operatorname{Tr} S = \eta^2 \sum_{i,j} S_{ij}\overline{S_{ij}}, \quad x^R := \eta \operatorname{Im} \operatorname{Tr} R = \eta^2 \sum_{i,j} R_{ij}\overline{R_{ij}}. \tag{6.80}$$

Recall also that $S = (H_\gamma - z)^{-1}$ and $R = (Q - z)^{-1}$, where all the entries of H_γ and Q are the same except the (a,b) entries. Since H satisfies the conditions of Theorem 3.6, $N^{-1} = \mathbb{E}h_{12}^2 \leq N^{-2\phi}$, so $\phi \leq 1/2$. Let $q_0 := \min(N^\phi, \widetilde{q})$, with $\widetilde{q}$ of Lemma 5.1. Then $|h_{ab}| \leq q_0^{-1}$ and $|\widetilde{h}_{ab}| \leq q_0^{-1}$ almost surely, for all a, b, and $q_0 \geq N^{\phi'}$ with $\phi' := \min(\phi, 1/4)$ for $N \geq N_0(C_1)$, since $\widetilde{q} \geq N^{1/4}$. Hence Lemma 6.5, (6.1) and the iteration (6.37) apply to every H_γ with the exponent ϕ' in place of ϕ. Since $H_\gamma - Q$ has rank at most 2, we have

$$|\operatorname{Tr} S - \operatorname{Tr} R| \leq 2\eta^{-1}. \tag{6.81}$$

Indeed, let A be real symmetric, $\tau \in \mathbb{R}$, $u \in \mathbb{R}^N \setminus \{0\}$, $G = (A - z)^{-1}$ and $G' = (A + \tau uu^T - z)^{-1}$. By the Sherman–Morrison formula, $\operatorname{Tr} G' - \operatorname{Tr} G = -\tau\, u^T G^2 u/(1 + \tau u^T G u)$. Here $|u^T G^2 u| \leq \|Gu\|^2 = \eta^{-1}\operatorname{Im}\, u^T G u$ and $|1 + \tau u^T G u| \geq |\tau| \operatorname{Im}\, u^T G u > 0$, so $|\operatorname{Tr} G' - \operatorname{Tr} G| \leq \eta^{-1}$. If $a = b$, then $H_\gamma - Q = h_{aa} e_a e_a^T$; if $a \neq b$, then $H_\gamma - Q = \frac{h_{ab}}{2}\big[(e_a + e_b)(e_a + e_b)^T - (e_a - e_b)(e_a - e_b)^T\big]$. In general, $|\operatorname{Tr} S - \operatorname{Tr} R| \leq \operatorname{rank}(H_\gamma - Q)/\eta$.

Together with (5.24) and (2.12), with high probability,

$$\max_\gamma \{|x^S| + |x^R|\} \leq N^{C\epsilon}. \tag{6.82}$$

From (3.3) and (3.28), with $q_0^{-1} \leq N^{-\phi+\epsilon}$ and $(N\eta)^{-1} = N^{-1/3+\epsilon}$, we find that

$$\max_\gamma \left(|R_{ij}| + |S_{ij}|\right) \leq N^{-\min(\phi, 1/3)+C\epsilon} + C\delta_{ij} \tag{6.83}$$

with high probability. We also have the trivial bounds

$$x^S = \eta^2 \sum_{i,j} S_{ij}\overline{S_{ij}} \leq \eta^2 N^2 \eta^{-2} = N^2, \quad x^R \leq N^2, \quad |S|, |R| \leq \eta^{-1} \leq N. \tag{6.84}$$

Since the bad event is so small in probability space, in this proof, we basically ignore the bad set. Using the definitions we used in (3.21) - (3.25), we get a telescopic series

$$\mathbb{E}\, F\left(\eta^2 \sum_{i,j} G_{ij}\overline{G_{ij}}\right) - \mathbb{E}\, F\left(\eta^2 \sum_{i,j} \widetilde{G}_{ij}\overline{\widetilde{G}_{ij}}\right) = \sum_{\gamma=1}^{\gamma_{\max}} \left[\mathbb{E}\, F\left(x^S\right) - \mathbb{E}\, F\left(x^T\right)\right]. \tag{6.85}$$

From the Taylor expansion, we have

$$\begin{aligned} F(x^S) - F(x^R) &= \sum_{s=1}^{2} \frac{1}{s!} F^{(s)}(x^R)(x^S - x^R)^s + \Omega_S, \\ |\Omega_S| &\leq \frac{C_F}{6}\big(1 + \max(|x^S|, |x^R|)\big)^{C_F} |x^S - x^R|^3. \end{aligned} \tag{6.86}$$

The bound on Ω_S follows from the Lagrange form of the remainder and (5.12), since every point between x^S and x^R has absolute value at most $\max(|x^S|, |x^R|)$; it is used in place of the intermediate

point, which need not depend measurably on H_γ. We obtain a similar formula for $F(x^T) - F(x^R)$, with a remainder Ω_T.

We estimate the terms of (6.85) one swap at a time. For $\gamma = \Phi(a,b)$, let $w(a,b) := N^{-2}$ if $a \neq b$ and $w(a,a) := N^{-1}$, and let $m_0 := 4$ if $a \neq b$ and $m_0 := 2$ if $a = b$. We will show that there are C and N_0, independent of γ, such that for $N \geq N_0$,

$$\left|\mathbb{E}F(x^S) - \mathbb{E}F(x^T)\right| \leq w(a,b)\, N^{-\phi+C\epsilon}, \qquad 1 \leq \gamma \leq \gamma_{\max}.$$

Since $\sum_\gamma w(a,b) = N \cdot N^{-1} + \frac{N(N-1)}{2} N^{-2} \leq \frac{3}{2}$, this and (6.85) give (5.14) with $C+1$ in place of C, once $N^\epsilon \geq 3/2$. The N diagonal swaps contribute $N \cdot N^{-1-\phi+C\epsilon} = N^{-\phi+C\epsilon}$, the same order as the off-diagonal ones.

We now expand the term $S_{ij}\overline{S_{ij}}$ using (6.56), where the terms with the complex conjugate are treated in the same manner. Let $\xi := \lceil 3/\phi \rceil$ in this proof. Taking this ξ and $s = 2$ in (6.56), with the almost sure bound $|h_{ab}| \leq q_0^{-1}$ in place of $N^{-\phi}$ (the remainder of the resolvent expansion is $(-RV)^{\xi+1}S$, of order $|h_{ab}|^{\xi+1}$), we can see that

$$S_{ij}\overline{S_{ij}} = R_{ij}\overline{R_{ij}} + \sum_{1\leq k\leq \xi} \left(\sum_{\mathbf{k}\in I^2_{\xi,k}} \mathcal{P}_{\gamma,\mathbf{k}}(R_{ij}\overline{R_{ij}}) \right) (-h_{ab})^k + O\big(q_0^{-\xi-1}\big) \tag{6.87}$$

holds with high probability. Averaging over i,j and multiplying η^2 to both sides, we obtain

$$x^S = x^R + \sum_{1\leq k\leq \xi} \left(\sum_{\mathbf{k}\in I^2_{\xi,k}} \eta^2 \sum_{i,j} \mathcal{P}_{\gamma,\mathbf{k}}\,(R_{ij}\overline{R_{ij}}) \right) (-h_{ab})^k + O\big(N^{-7/3-\phi}\big). \tag{6.88}$$

Here the error is $\eta^2 N^2 \cdot O(q_0^{-\xi-1}) = O(N^{-7/3-\phi})$. If $\phi < 1/2$, then $q_0 = N^\phi$ for N large enough and $(\xi+1)\phi \geq 3+\phi$, so $\eta^2 N^2 q_0^{-\xi-1} \leq N^{2/3-2\epsilon} N^{-3-\phi}$. If $\phi = 1/2$, then $\xi = 6$, $q_0 = \widetilde{q}$, and $\eta^2 N^2 \widetilde{q}^{-7} = N^{-7/3-\phi}\, N^{-2\epsilon}((2C_1+4)C_1 \log N)^7 \leq N^{-7/3-\phi}$ for N large enough.

Now, we claim that there is a constant C, depending only on (C_1, ϕ), such that for every $\mathbf{k} \in \mathbb{N}^2$ with $1 \leq |\mathbf{k}| \leq \xi$ and every even $p \geq 2$, for $N \geq N_0(C_1, \phi, \epsilon, p)$, uniformly in γ and $E \in I_\epsilon$,

$$\mathbb{E}\left| \sum_{i,j} \mathcal{P}_{\gamma,\mathbf{k}}\,(R_{ij}\overline{R_{ij}}) \right|^p \leq (N^{1+C\epsilon})^p. \tag{6.89}$$

Assuming the claim (6.89), with Markov inequality, we find that, for any $D > 0$, for $N \geq N_0(D)$ and every $\mathbf{k} \in \mathbb{N}^2$ with $1 \leq |\mathbf{k}| \leq \xi$,

$$|\mathcal{P}_{\gamma,\mathbf{k}}\, x^R| := |\eta^2 \sum_{i,j} \mathcal{P}_{\gamma,\mathbf{k}}\,(R_{ij}\overline{R_{ij}})| \leq N^{-1/3+C\epsilon} \tag{6.90}$$

holds with probability at least $1 - N^{-D}$, with $C+1$ in place of the C of (6.89). Indeed, $\eta^2 N^{1+C\epsilon} = N^{-1/3+(C-2)\epsilon}$, so by (6.89) the probability that (6.90) fails is at most $(N^{-3\epsilon})^p \leq N^{-D}$ for an even $p \geq D/(3\epsilon)$.

We derive (6.89) from a bound of the form (6.91). Let $Y := \sum_{i,j} \mathcal{P}_{\gamma,\mathbf{k}}(R_{ij}\overline{R_{ij}})$, $\mathbf{k} = (k_1, k_2)$, and $\mathbf{k}' := (k_2, k_1)$. Since $\mathcal{I}_\gamma$ is real, $\overline{\mathcal{P}_{\gamma,\mathbf{k}}(R_{ij}\overline{R_{ij}})} = \mathcal{P}_{\gamma,\mathbf{k}'}(R_{ij}\overline{R_{ij}})$. Hence, for $p = 2r$, $|Y|^p = Y^r \overline{Y}^r = \sum_{I,J} \prod_{t=1}^p \mathcal{P}_{\gamma,\mathbf{k}^{(t)}}(R_{i_t j_t}\overline{R_{i_t j_t}})$, with $I = (i_1, \ldots, i_p)$, $J = (j_1, \ldots, j_p)$, $\mathbf{k}^{(t)} = \mathbf{k}$ for $t \leq r$ and $\mathbf{k}^{(t)} = \mathbf{k}'$ for $t > r$. We prove that for every choice of $\mathbf{k}^{(t)} \in \{\mathbf{k}, \mathbf{k}'\}$,

$$\left| \mathbb{E} \sum_{I,J} \prod_{t=1}^p \mathcal{P}_{\gamma,\mathbf{k}^{(t)}}\,(R_{i_t j_t}\overline{R_{i_t j_t}}) \right| \leq (N^{1+C\epsilon})^p. \tag{6.91}$$

Since $\mathbb{E}|Y|^p \geq 0$, this gives (6.89). Let $s := (2+|\mathbf{k}|)p$ and $\zeta := 2p$. Each product $\prod_t \mathcal{P}_{\gamma,\mathbf{k}^{(t)}}(R_{i_tj_t}\overline{R_{i_tj_t}})$ is a sum of at most $2^{|\mathbf{k}|p}$ products of s entries of R and $\overline{R}$. Using (6.24) with these s and ζ, with ϕ' in place of ϕ, we get

$$\mathbb{E}\prod_{t=1}^{p}\mathcal{P}_{\gamma,\mathbf{k}^{(t)}}\left(R_{i_tj_t}\overline{R_{i_tj_t}}\right) = \mathbb{E}\prod_{t=1}^{p}\mathcal{P}_{\gamma,\mathbf{k}^{(t)}}\left(S_{i_tj_t}\overline{S_{i_tj_t}}\right) - \sum_{\alpha\in\mathbb{N}^s}^{1\leq|\alpha|\leq\lceil 4p/\phi'\rceil}\widetilde{\mathcal{A}}_\alpha\,\mathbb{E}\,\mathcal{P}_{\gamma,\alpha}\left(\prod_{t=1}^{p}\mathcal{P}_{\gamma,\mathbf{k}^{(t)}}\left(S_{i_tj_t}\overline{S_{i_tj_t}}\right)\right) + O(4^s 2^{|\mathbf{k}|p}N^{-2p}) \tag{6.92}$$

for each (I,J), where the coefficients $\widetilde{\mathcal{A}}_\alpha$ do not depend on (I,J). The sum over (I,J) of the error terms is $O(C^pN^{2p}N^{-2p}) = O(C^p)$. By (6.25), $|\widetilde{\mathcal{A}}_\alpha| \leq 1$, and the number of α in the sum depends only on (p,ϕ). Thus, in order to show (6.91), it only remains to prove that

$$\left|\sum_{i_1,j_1\cdots,i_p,j_p}\mathbb{E}\left(\prod_{t=1}^{p}\mathcal{P}_{\gamma,\mathbf{k}^{(t)}}\left(S_{i_tj_t}\overline{S_{i_tj_t}}\right)\right)\right| \leq (N^{1+C\epsilon})^p \tag{6.93}$$

and for $\alpha\in\mathbb{N}^s$, $1\leq|\alpha|\leq\lceil 4p/\phi'\rceil$,

$$\left|\sum_{i_1,j_1\cdots,i_p,j_p}\mathbb{E}\,\mathcal{P}_{\gamma,\alpha}\left(\prod_{t=1}^{p}\mathcal{P}_{\gamma,\mathbf{k}^{(t)}}\left(S_{i_tj_t}\overline{S_{i_tj_t}}\right)\right)\right| \leq (N^{1+C\epsilon})^p. \tag{6.94}$$

We give the proof for (6.93). The proof of (6.94) is the same except that the products are longer by $|\alpha|$ entries. Using (6.37) at the exponent ϕ', with $\zeta = 2p$, $K_s := p(2+\lceil 3/\phi\rceil) + \lceil 4p/\phi'\rceil$ (so that all the products have at most K_s entries),

$$f(I,J) = N^{-2p}, \quad I = (i_1,\cdots,i_p), \quad J = (j_1,\cdots,j_p),$$

and $\prod_{t=1}^{p}\mathcal{P}_{\gamma,\mathbf{k}^{(t)}}\left(S_{i_tj_t}\overline{S_{i_tj_t}}\right)$ playing the role of $\prod_{t=1}^{s}G^{\gamma_{\max}}_{i_tj_t}$ in (6.37), $\widetilde{G}$ being G^0, we have

$$\begin{aligned}
&\left|\mathbb{E}\sum_{I,J}N^{-2p}\prod_{t=1}^{p}\mathcal{P}_{\gamma,\mathbf{k}^{(t)}}\left(S_{i_tj_t}\overline{S_{i_tj_t}}\right)\right| \\
&\leq \left|\mathbb{E}\sum_{I,J}N^{-2p}\prod_{t=1}^{p}\mathcal{P}_{\gamma,\mathbf{k}^{(t)}}\left(\widetilde{G}_{i_tj_t}\overline{\widetilde{G}_{i_tj_t}}\right)\right| \\
&\quad + \max_{\mathbf{k}_i,n,\gamma_i}(N^{-\phi'/20})^{\sum_i|\mathbf{k}_i|}\left|\mathbb{E}\sum_{I,J}N^{-2p}\mathcal{P}_{\gamma_n,\mathbf{k}_n}\cdots\mathcal{P}_{\gamma_1,\mathbf{k}_1}\left(\prod_{t=1}^{p}\mathcal{P}_{\gamma,\mathbf{k}^{(t)}}\left(\widetilde{G}_{i_tj_t}\overline{\widetilde{G}_{i_tj_t}}\right)\right)\right| + O(N^{-\zeta})
\end{aligned} \tag{6.95}$$

where the maximum is over the iterates of (6.34), with

$$\mathbf{k}_1\in\mathbb{N}^s, \quad \mathbf{k}_2\in\mathbb{N}^{s+|\mathbf{k}_1|}, \quad \mathbf{k}_3\in\mathbb{N}^{s+|\mathbf{k}_1|+|\mathbf{k}_2|}, \quad \cdots. \tag{6.96}$$

Since $(N^{-\phi'/20})^{\sum_i|\mathbf{k}_i|}\leq 1$, it suffices to bound the expectation in the second term by $(N^{-1+C\epsilon})^p$ for every such iterate. By the local law (3.3) for $\widetilde{H}$, which applies with $q=\widetilde{q}\geq N^{1/3}$, at $z=E+i\eta$ with $E\in I_\epsilon$ (see (5.13)) and $\epsilon<1/12$, with high probability,

$$|\widetilde{G}_{ij}| \leq N^{-1/3+2\epsilon} + 2\delta_{ij}. \tag{6.97}$$

Indeed, $\widetilde{q}^{-1}\leq N^{-1/3}$, $(N\eta)^{-1} = N^{-1/3+\epsilon}$, and, by (2.12), $\operatorname{Im} m_{sc}(z)\leq C\sqrt{\kappa+\eta}\leq CN^{-1/3+\epsilon/2}$, so that $(\operatorname{Im} m_{sc}/(N\eta))^{1/2}\leq CN^{-1/3+3\epsilon/4}$; the factor φ^C of (3.3) is at most $N^{\epsilon/4}$, and $|m_{sc}|\leq 1$.

Now, we estimate the term

$$\mathcal{P}_{\gamma_n,\mathbf{k}_n}\cdots\mathcal{P}_{\gamma_1,\mathbf{k}_1}\left(\prod_{t=1}^{p}\mathcal{P}_{\gamma,\mathbf{k}^{(t)}}\,(\widetilde{G}_{i_tj_t}\overline{\widetilde{G}_{i_tj_t}})\right) \tag{6.98}$$

as in (6.44). First, it is the sum of at most $C^{\sum|\mathbf{k}_i|+|\mathbf{k}|\mathbf{p}}=O(1)$ products of $\widetilde{G}_{ij}$ (possibly $i=j$), where in each product the number of $\widetilde{G}_{ij}$ is $\sum|\mathbf{k}_i|+(2+|\mathbf{k}|)p=O(1)$. Since $\widetilde{G}$ satisfies a rough bound $|\widetilde{G}_{ij}(z)|\le\eta^{-1}\le N$, we know (6.98) is always bounded by $N^{O(1)}$. Since (6.97) holds with high probability, when estimating (6.98), we may neglect the event that (6.97) does not hold. For each product of above type and for any fixed t, the indices i_t and j_t only appear twice each in the entries coming from the t-th factor. Since $\mathbf{k}^{(t)}\neq 0$, they cannot attain the form $\widetilde{G}_{i_t,j_t}\overline{\widetilde{G}_{i_t,j_t}}$. Thus, they must appear as one of the following forms for some a,b,c,d, which comes from $\mathcal{P}$'s:

$$\widetilde{G}_{i_ta}\widetilde{G}_{bj_t}\overline{\widetilde{G}_{i_tj_t}},\quad \widetilde{G}_{i_tj_t}\overline{\widetilde{G}_{i_ta}\widetilde{G}_{bj_t}},\quad \widetilde{G}_{i_ta}\widetilde{G}_{bj_t}\overline{\widetilde{G}_{i_tc}\widetilde{G}_{dj_t}}. \tag{6.99}$$

For each case, after averaging over $1\le i_t,j_t\le N$, i.e., applying $N^{-2}\sum_{i_t,j_t}$, these terms are bounded by $N^{-1+C\epsilon}$: if $i_t\neq j_t$ and neither i_t nor j_t is an index of $\gamma,\gamma_1,\dots,\gamma_n$, then each form contains at least three off-diagonal entries, each bounded by $N^{-1/3+2\epsilon}$ by (6.97), and the remaining $O(N)$ pairs (i_t,j_t) contribute $O(N^{-1})$ after the averaging. Thus, so far we have proved that, for each t, the term G_{ij} with an index i_t or j_t contributes a factor $N^{-1+C\epsilon}$ to

$$\sum_{I,J}N^{-2p}\mathcal{P}_{\gamma_n,\mathbf{k}_n}\cdots\mathcal{P}_{\gamma_1,\mathbf{k}_1}\left(\prod_{t=1}^{p}\mathcal{P}_{\gamma,\mathbf{k}^{(t)}}\,(\widetilde{G}_{i_tj_t}\overline{\widetilde{G}_{i_tj_t}})\right). \tag{6.100}$$

Similarly, the G_{ij}'s with indices $i_1,j_1,\cdots,i_p,j_p$ contribute a factor $(N^{-1+C\epsilon})^p$ to (6.100). For the other G's without indices $i_1,j_1,\cdots,i_p,j_p$, we simply bound them by a constant C. Therefore, we obtain that (6.100) is bounded by $(N^{-1+C\epsilon})^p$ with high probability. Then, the expectation of (6.100) is less than $(N^{-1+C\epsilon})^p$. Analogously, we can bound the first term in the right hand side of (6.95) by $(N^{-1+C\epsilon})^p$. Thus, we proved (6.93) and (6.94), which imply (6.91) with (6.25), and hence (6.89). The constant C obtained here does not depend on p; only the threshold for N does.

Now, we return to estimate x^S-x^R in (6.88). Let $Z:=\sum_{1\le|\mathbf{k}|\le\xi}|\mathcal{P}_{\gamma,\mathbf{k}}\,x^R|$, a function of R alone, hence independent of h_{ab} and of $\widetilde{h}_{ab}$. By (6.88), $|x^S-x^R|\le|h_{ab}|Z+O(N^{-7/3-\phi})$ with high probability, and $|x^S-x^R|\le 2$ always, by the bound on $|\operatorname{Tr}S-\operatorname{Tr}R|$. By (6.90), $Z\le CN^{-1/3+C\epsilon}$ outside an event of probability at most N^{-D}, and $Z\le N^C$ always. First, we note that, for $a\neq b$,

$$\mathbb{E}|h_{ab}|^3\le(\mathbb{E}|h_{ab}|^2\mathbb{E}|h_{ab}|^4)^{1/2}\le(C_1\log N)^{1/2}N^{-3/2},$$

and $\mathbb{E}|h_{aa}|^3\le N^{-\phi}\mathbb{E}h_{aa}^2\le C_1N^{-1-\phi}$. The same bounds hold for $\widetilde{h}_{ab}$, $a\neq b$, since its second and fourth moments are those of h_{ab}, and $\mathbb{E}|\widetilde{h}_{aa}|^3\le\widetilde{q}^{-1}C_1N^{-1}\le N^{-1-\phi+\epsilon}$. Hence there exists a constant C such that

$$\mathbb{E}|x^S-x^R|^3\le N^{-5/2+C\epsilon}\quad(a\neq b),\qquad \mathbb{E}|x^S-x^R|^3\le N^{-2-\phi+C\epsilon}\quad(a=b), \tag{6.101}$$

for any sufficiently large N independent of γ, and the same holds for x^T in place of x^S. Since $\phi\le 1/2$, both bounds are at most $w(a,b)N^{-\phi+C\epsilon}$. On the event (6.82), $(1+\max(|x^S|,|x^R|))^{C_F}\le N^{C\epsilon}$; on its complement, we use the trivial bounds above and $|x^S-x^R|\le 2$. Hence, with the absolute value inside the expectation,

$$\mathbb{E}|\Omega_S|\le\frac{C_F}{6}\,\mathbb{E}\Big[\big(1+\max(|x^S|,|x^R|)\big)^{C_F}|x^S-x^R|^3\Big]\le w(a,b)\,N^{-\phi+C\epsilon}, \tag{6.102}$$

where we have used (5.12) on F. We can estimate $\mathbb{E}|\Omega_T|$ analogously.

From (6.86), it only remains to prove that, for $1\le s\le 2$,

$$\left|\mathbb{E}\left[F^{(s)}(x_R)(x^S-x^R)^s\right]-\mathbb{E}\left[F^{(s)}(x_R)(x^T-x^R)^s\right]\right|\le w(a,b)\,N^{-\phi+C\epsilon}. \tag{6.103}$$

Using (6.88) again, recalling $(H_{\gamma-1})_{ab}$ has the distribution of $\widetilde{H}_{ab}$, we have

$$x^T = x^R + \sum_{1\le k\le \xi} \left(\sum_{\mathbf{k}\in I^2_{\xi,k}} \eta^2 \sum_{ij} \mathcal{P}_{\gamma,\mathbf{k}}\left(R_{ij}\overline{R_{ij}}\right)\right)(-\widetilde{h}_{ab})^k + O\big(N^{-7/3-\phi}\big) \tag{6.104}$$

with $\mathbb{E}h_{ab}^k = \mathbb{E}\widetilde{h}_{ab}^k$, $1\le k\le m_0$. Inserting (6.88) and (6.104) into (6.103), and using that x^R and $\mathcal{P}_{\gamma,\mathbf{k}}x^R$ are independent of h_{ab} and of $\widetilde{h}_{ab}$, the terms of total order $K\le m_0$ cancel. Thus, we obtain that

$$\begin{aligned}&\left|\mathbb{E}\left[F^{(s)}(x_R)(x^S-x^R)^s\right] - \mathbb{E}\left[F^{(s)}(x_R)(x^T-x^R)^s\right]\right| \qquad (6.105)\\ &\le \sum_{K=m_0+1}^{s\xi} \sum_{\sum_{t=1}^s |\mathbf{k}_t|=K} \left|\mathbb{E}\left[F^{(s)}(x_R)\prod_{t=1}^s\left(\mathcal{P}_{\gamma,\mathbf{k}_t}\, x^R\right)\right]\right| \left(\left|\mathbb{E}(-h_{ab})^K\right| + \left|\mathbb{E}(-\widetilde{h}_{ab})^K\right|\right) + O(N^{-7/3-\phi+C\epsilon}),\end{aligned}$$

where the inner sum is over $\mathbf{k}_t\in I^2_{\xi,|\mathbf{k}_t|}$ with $|\mathbf{k}_t|\ge 1$, $1\le t\le s$. By the moment bounds in the proof of Lemma 6.5, applied with the almost sure bounds $N^{-\phi}$ for h_{ab} and $\widetilde{q}^{-1}$ for $\widetilde{h}_{ab}$: for $a\ne b$ and $K\ge 5$, $|\mathbb{E}(-h_{ab})^K|\le C_1(\log N)N^{-2-\phi}$ and $|\mathbb{E}(-\widetilde{h}_{ab})^K|\le C_1(\log N)N^{-2}\widetilde{q}^{-1}\le (C\log N)^2N^{-5/2}$; for $a=b$ and $K\ge 3$, $|\mathbb{E}(-h_{aa})^K|\le C_1N^{-1-\phi}$ and $|\mathbb{E}(-\widetilde{h}_{aa})^K|\le C_1N^{-1}\widetilde{q}^{-1}$. Since $\phi\le 1/2$, in both cases $|\mathbb{E}(-h_{ab})^K|+|\mathbb{E}(-\widetilde{h}_{ab})^K|\le w(a,b)N^{-\phi+\epsilon}$. By (6.90), (6.82) and (5.12), each expectation in the sum is at most $N^{C\epsilon}$ (on the bad events we use the trivial bounds), and the number of terms depends only on ϕ. Finally, $N^{-7/3-\phi+C\epsilon}\le w(a,b)N^{-\phi+C\epsilon}$. This proves (6.103). Together with the bounds on $\mathbb{E}|\Omega_S|$ and $\mathbb{E}|\Omega_T|$, it gives $|\mathbb{E}F(x^S)-\mathbb{E}F(x^T)|\le w(a,b)N^{-\phi+C\epsilon}$ and completes the proof. □

Appendix A. Changes with respect to the published version

The base text of this revision is arXiv:1206.2251v2. The version published in Duke Math. J. **163** (2014) is not in the repository [54], and differences between it and arXiv:1206.2251v2, if any, are not recorded here.

The numbering of sections, statements and displays is that of arXiv:1206.2251v2. The new statements (Remarks 1.5, 3.9 and 4.1, Lemma 6.6) come last in their sections, and the new displays are unnumbered or tagged, (2.4′) and (3.20′), so no number changes. Each line below gives the location in this revision and the change in short (old → new). Obvious spelling slips in the English text (such as "statics", "stretagy" and "Isotrpic") were corrected without being listed.

Formalization. Every theorem, proposition, lemma and corollary of this revised paper that the proof of Theorem 1.2 uses is formalized in [54] in the form stated here, and so are the results of [15], [16], [25, Lemma 4.2] and [26] in the forms stated here (Lemma 2.6, Lemma 3.1, Lemma 3.5, Lemma 5.6 and the steps of the proof of Theorem 3.6 that follow [26, Section 5]). The Lean names, all in the namespace `WEdgeNS`, are:

- Theorem 1.2: `E1`, `E1_not_tight`, `E1_no_limit` (necessity) and `E2` (sufficiency);
- Lemma 2.6: `norm_msc_eq_inv_and_lt_one`, `msc_bounds`;
- Lemma 3.1: `localLaw`, `norm_le_two_add`, `delocalization`, `rigidity`, `outOfSpectrum`;
- Lemma 3.5: `edge_universality_largeQ`; Theorem 3.6: `rigidity_smallQ`; Theorem 3.7: `edge_universality_smallQ`; Lemma 3.8: `offDiag_sq_le_smallQ`;
- Lemma 5.1: `exists_momentMatched`; Lemma 5.2: `exists_law_four_moments`; Remark 5.3: `sq_moment_three_add_one_le`; Lemma 5.4: `moment_comparison`; Lemma 5.5: `green_comparison_smallQ`; Lemma 5.6: `edge_universality_of_inputs`;
- Lemma 6.4: `norm_evalW_wordP_le_of_good` on the event of `highProb_goodEvent`; Lemma 6.5: `expectation_expansion`, `expectation_expansion_inv`; the iteration, (6.33) and (6.37): `comparison_iterate_sum`, `comparison_iterate`; Lemma 6.6: `norm_integral_evalF_iterP_le`.

The following are not formalized, and the proof of Theorem 1.2 does not use them: Remark 1.5 (the Tracy–Widom limit, which rests on [44, 45]); the Hermitian case and the case $k \geq 2$ mentioned in Remark 1.4; the example of Remarks 1.3 and 4.1; the sentence on the error $N^{-\delta'}$ at the end of Remark 4.1; Theorem 3.3 in its stated form, with the bounded support condition of Definition 2.4 (it is used only through Lemma 3.5); Remark 3.4 and (3.11); the probabilistic form of Lemma 3.5 mentioned after it; the form of Theorem 3.6 under the bounded support condition of Definition 2.4 mentioned after it; and the heuristic Example 6.1.

Title page, abstract and bibliography.

- Abstract: "Tracy-Widom law holds if and only if (1.4); the same criterion holds for Hermitian Wigner matrices" → for real symmetric Wigner matrices, (1.4) is necessary and sufficient for the edge universality of λ_N; under (1.4) the distribution of $N^{2/3}(\lambda_N - 2)$ is that of the GOE up to a shift $N^{-\delta}$ and an additive error $o(1)$ uniform in s, hence F_1 by [44, 45]; otherwise it is not tight; the proof is formalized in Lean 4 [54]. The Hermitian sentence is removed.
- Revision line after the title: new.
- Bibliography: [15] "arXiv:1103.1919v2 (2011)" → "Ann. Probab. **41** (2013); arXiv:1103.1919v5"; [16] "arXiv:1103.3869v2 (2011)" → "arXiv:1103.3869v3"; [26]: arXiv:1007.4652v7 added; [54]: new entry, the Lean formalization, with a breakable URL.

Section 1.

- Paragraph after (1.3), before (1.4): "the joint probability density of the k largest eigenvalues ... weakly converge to that of Gaussian ensembles" → "the rescaled largest eigenvalue of a real symmetric Wigner matrix has asymptotically the same distribution as for the GOE, and hence the Tracy–Widom distribution F_1".
- Definition 1.1: new last sentence, Theorem 1.2 concerns the real symmetric case.
- Theorem 1.2, (1.5)–(1.6), and the new paragraph after it: real symmetric case and λ_N only; sufficiency: "for any fixed k the joint distribution of the k rescaled largest eigenvalues converges to the GUE (GOE) limit" → the GOE comparison (1.5) with a shift $N^{-\delta}$ and an additive error $\epsilon_N \to 0$ uniform in s; necessity: "(1.4) does not hold" is read as "$s^4\mathbb{P}(|x_{12}| \geq s)$ does not tend to 0", and the consequences "not tight" and "no limit distribution" are stated. The paragraph after the theorem defines $\mathbb{P}^{GOE}$ and F_N^{GOE}, fixes the order of the quantifiers and derives the two consequences from (1.6).
- Remark 1.3: density "decaying as $|x|^{-5}\log|x|$" (for which (1.4) fails) → $c|x|^{-5}(\log|x|)^{-1}$ on $|x| \geq e$, normalised to variance 1, with $s^4\mathbb{P}(|x_{12}| \geq s) \to 0$ and $\mathbb{E}x_{12}^4 = \infty$; new sentence: for this law, (1.5) fails with $\epsilon_N = N^{-\delta}$ (Remark 4.1).
- After Remark 1.3, the two key observations: "entries have a bounded support of size $N^{-\epsilon}$" → "entries are almost surely bounded by $N^{-\epsilon}$"; "first four moments coincide" → "first four moments of the off-diagonal entries and first two moments of the diagonal entries coincide". The organisation paragraph ends with a pointer to this appendix.
- Remark 1.4: "we prove only the real symmetric case with $k = 1$; the general case can be proved analogously" → the Hermitian case and $k \geq 2$ are expected to follow from the same method but are not proved here.
- Remark 1.5 (new): deduction of $\mathbb{P}(N^{2/3}(\lambda_N - 2) \leq s) \to F_1(s)$ from (1.5) and [44, 45]; not formalized. After it, a new unnumbered paragraph "Formalization".

Section 2.

- Definition 2.4, (2.4) and new (2.4′): "$|h_{ij}| \leq q^{-1}$ with probability larger than $1 - e^{-N^c}$ for some $c > 0$; usually $N^\phi \leq q \leq N^{1/2}(\log N)^{-1}$" → $\mathbb{P}(|h_{ij}| > q^{-1}) \leq e^{-N^c}$ with c a fixed parameter, plus the tail-variance condition (2.4′) for $i \neq j$; the almost sure form is defined; $q \leq \sqrt{2}N^{1/2}$.

- Paragraph after (2.4′): "throughout the paper, we will not consider the case it happens" → the event $\{|h_{ij}| > q^{-1}\}$ is removed by an explicit truncation in the proof of Lemma 3.1, and it does not occur in Sections 5 and 6, where the entries are bounded almost surely.
- Definition 2.7, (2.13): new sentence, γ_j is the smallest solution, so $\gamma_N = 2$.
- Remark 2.8: new sentences on the dependence of constants on the fixed parameters and on N-dependent distributions.

Section 3.

- Lemma 3.1, (3.1)–(3.8): now "fix $\zeta > 0$, $C_0 \geq 0$, $c > 0$", with (2.1) and Definition 2.4, ζ-high probability (was 3-high probability), constants depending on (ζ, C_0, c) only; (3.2), (3.3) and (3.6) with $\bigcap$ (printed $\bigcup$); (3.5) for every unit eigenvector; parts 4 and 5 for $q \geq N^{\phi}$, $\phi > 1/3$; in part 5 the offset 3 of (3.7) becomes a constant K, $a > 12$ becomes $a > 12\zeta$, and (3.8) holds simultaneously for all E in (3.7).
- Proof of Lemma 3.1 (after (3.8)), including (3.9): rewritten in five steps: an explicit truncation reduces to entries bounded almost surely; [15] is run at level ζ with diagonal variances bounded by C_0 (no rescaling by $1 + O(N^{-1})$), extended to $2 \leq \eta \leq 10$, and made uniform in z by a lattice; delocalization and rigidity (including $j = N$) are derived from parts 1 and 2; the offset of part 5 is a constant K_0 (the printed 3 conflates two constants C_1 of [15]); (3.9): $O(q^{-1}) + (\varphi^{a/2-2}N\eta)^{-1} \to C_o(q^{-1} + \varphi^{2\zeta - a/2}(N\eta)^{-1})$; "$\eta \geq \varphi^{O(1)}N^{-2/3}$" (wrong direction) $\to N\eta \leq \varphi^{K+a-C/2}N^{1/3}$.
- Remark 3.2: "we only use $1 \leq \zeta \leq 3$" → Lemma 3.1 holds for every fixed ζ, and $\zeta \in \{1, 3, 4\}$ is used, $\zeta = 4$ for (5.19).
- Theorem 3.3, (3.10), and the new paragraph after it: cited as [16, Theorem 2.7 and Proposition 6.4, (6.15) at $f = 0$]; δ and N_0 independent of s; the variance convention of [16] is noted; in the stated form the theorem is cited and not used.
- Remark 3.4, (3.11): new sentence, not used (see (4.26)) and not formalized.
- Lemma 3.5 and the sentence after it: "the same condition except $\mathbb{E}|h_{ii}|^2 \leq C_0/N$" → (2.1) with $C_0 \geq 0$ and $|h_{ij}| \leq N^{-\phi}$ almost surely, $1/3 < \phi \leq 1/2$, δ and N_0 depending on (C_0, ϕ), all $N \geq N_0$ and all s; the probabilistic form is not proved here.
- Theorem 3.6, (3.12)–(3.13), and the new paragraph after it: bounded support with $q = N^{\phi}$ → $|h_{ij}| \leq N^{-\phi}$ almost surely; (3.12) with $\bigcap_j$ (printed $\bigcup_j$); (3.13) for all $E \in \mathbb{R}$ (was $|E| \leq 5$); the form under Definition 2.4 is not proved here.
- Theorem 3.7, (3.14): entries bounded almost surely; δ, N_0 depend only on (C_1, ϕ); all $N \geq N_0$, all $H^{\mathbf{v}}$ and all s.
- Lemma 3.8, (3.15): entries bounded almost surely; for every $0 < c < 1$ there are C and N_0 depending on (C_1, ϕ, c), for all $i \neq j$.
- Proof of Lemma 3.5, (3.16)–(3.39): rewritten. "A simple extension of [16, Theorem 2.7]" → a derivation from Lemma 5.6 with rate $N^{-\kappa+C\epsilon}$, $\kappa = \phi - 1/3$, its hypotheses checked from Lemma 3.1; the Claim with explicit constants, $F \in C^4$ (constant renamed C_F) and E_1, E_2 in either order; reduction to the off-diagonal parts by the Ward identity, new display (3.20′); the local law for every H_γ with $q' = N^{(1/3+\phi)/2}$ (the interpolating matrices do not satisfy the support condition with q); (3.26)–(3.28) with explicit constants and the entry bound $N^{\epsilon}q^{-1}$; the case $a \neq b$, (3.29), follows [16, (6.41)–(6.97)] at $f = 0$ with the changes listed there; the case $a = b$, (3.31)–(3.38), uses that the linear term has mean zero, the bad-event bound (3.36) (new label) and only an upper bound on $\mathbb{E}v^2$ (the diagonal second moments need not match); the integrated functional and the diagonal parts are treated explicitly; telescoping (3.39).
- Remark 3.9 (new): corrections to [15] and [16], see "Corrections to the cited works" below.

Section 4.

- Beginning of Section 4, proof of the necessary condition, (4.1)–(4.8): "$\lim_{s\to\infty} s^4\mathbb{P}(|x_{12}| \geq s) \neq 0$" → "$s^4\mathbb{P}(|x_{12}| \geq s)$ does not tend to 0"; $r_k \geq 4$; in (4.2) $|h_{ii}| \leq 1$

(was < 1); explicit $c_2 = c_1/1024$, $c_3 = e^{-c_2/4}$ and $N \geq \max(2, 2\mathbb{E}|x_{11}|^2)$; the independence used in (4.8) is justified through the first index pair with $|h_{ij}| \geq 4$.

- (4.9): additive error $N^{-\delta} \to \epsilon_N \to 0$, with the quantifiers and dependence of (1.5); "this is (1.5)".
- Remark 4.1 (new): for the law of Remark 1.3, (4.9) fails with $\epsilon_N = N^{-\delta}$ for every $\delta > 0$; this example is not formalized; the error $N^{-\delta'}$ under higher moments is expected and not proved here. The paragraph after it: the comparison matrix is $\widetilde{H}^S$, bounded almost surely.
- Proof of the sufficient condition, (4.10)–(4.13): "for fixed $\epsilon > 0$" → "fix $0 < \epsilon < 1/30$", with the step that needs each bound; (4.12)–(4.13) in $o(\cdot)$ form (the printed δ clashed with the δ of (4.9)) with a proof; the conditional means m, $\widetilde{m}$.
- (4.14)–(4.17): laws defined by densities shifted by $\beta \to$ conditional laws centred by m (no densities); (4.16) is an identity on a product space with the law of H_N (was "$\sim$"); $M = N^{-1/2}mJ + N^{-1/2}(\widetilde{m} - m)I$ and (4.17) bounds the shift of λ_N.
- (4.18)–(4.19): written as integrals against ν and $\widetilde{\nu}$; second moment $1 - o(N^{-1+2\epsilon})$; diagonal second moment $\leq C_1$.
- (4.20): $H^S \to \widetilde{H}^S = \sigma^{-1}H^S$, bounded almost surely by $N^{-\phi_s}$, $\phi_s = 3\epsilon/4$; Theorem 3.7 applied in the orientation of (3.14).
- (4.21)–(4.25): "$> 3/4$" with a proof; in (4.22) the indicator of $|\widetilde{H}_{ij}H^C_{ij}| \leq 3/4$ (printed without absolute value); the event SelGood and $\widetilde{H}^C$ defined explicitly; (4.24) proved; $H = H^S + E + M$ on an event of probability $1 - o(1)$.
- (4.26)–(4.37): (4.26) is stated for $\widetilde{H}^S$ and $\widetilde{E} = \sigma^{-1}E$; the derivation of (4.9) from (4.26) is written out on the scale $N^{2/3}(\lambda_N - 2)$, with $\delta = \frac{1}{2}\min(\delta_0, 1/24)$ and $\epsilon_N = N^{-\delta_0} + p_N$.
- (4.27)–(4.37): only the nonzero positions of $\widetilde{E}$ are kept, so D is invertible and V has orthonormal columns; [34, Lemma 6.1] is replaced by the matrix determinant lemma, with its proof ((4.32) and the identity after it).
- (4.38): maximum over $1 \leq i \leq N$ (was $1 \leq i \leq 2s + t$), bound $\varphi^C(N^{-\phi_s} + N^{-1/3}) \leq N^{-\epsilon/2}$.
- (4.39): maximum over the index set touched by $\widetilde{H}^C$, a function of H^C only; proved from Lemma 3.8 by Markov's inequality.
- (4.40)–(4.47): $\widetilde{\mu} := \lambda_N \pm N^{-3/4} \to$ the one-sided barrier $\lambda_N + N^{-3/4}$; (4.41) "the largest eigenvalues of GOE are separated on the scale $N^{-2/3}$" (with Remark 3.4) → $\lambda_\alpha \leq \widetilde{\mu} - N^{-3/4}$, so (4.42) holds deterministically; (4.43) from (3.12) (was "see (3.14)"); (4.44) "$=$" → "$\leq$"; (4.45) $CN^{-1/4} \to \varphi^C N^{-1/4}$; (4.46) a Gershgorin bound with $\epsilon < 1/30$; (4.47) the upper bound by continuity in γ and the lower bound by the Rayleigh quotient, $\mu_N \in [\lambda_N - \varphi^C N^{-1+5\epsilon}, \lambda_N + N^{-3/4})$.

Section 5.

- Beginning of Section 5: $\widetilde{H}$ "satisfies the bounded support condition with $q = O(N^{1/2}/\log N)$ and has the same first four moments" → four moments off the diagonal, two on it, entries bounded almost surely by $\widetilde{q}^{-1}$ with $\widetilde{q} = N^{1/2}/((2C_1 + 4)C_1 \log N)$.
- Lemma 5.1, (5.1), and its proof: hypotheses are independence and the moment conditions of Theorem 3.6 (no support condition); explicit $\widetilde{q}$; $\widetilde{H}$ satisfies (2.1) with $\max(C_1, 1)$; "the diagonal part is trivial" → an explicit construction.
- Lemma 5.2, (5.2)–(5.3), and its proof (after Remark 5.3), (5.4)–(5.9): "$C \leq |A|$" → "$|A| \leq C$"; $\mathbb{E}x^3 = C \to \mathbb{E}x^3 = A$; $D = 2C + 4$; "$x = X + Y_t$, $t = 4B - 8A^2 - 2$" (variance 2) → $x = (X + Y_t)/\sqrt{2}$, $t = 4B - 8A^2 - 7$, for $B \geq 2A^2 + 2$, and a mixture with the two-point law μ_A for $A^2 + 1 \leq B < 2A^2 + 2$ (was "it is easy to find").
- Paragraph after (5.9), before Lemma 5.4: new term "matched pair".

- Lemma 5.4, (5.10)–(5.11): explicit quantifiers (C_* depends on (C_1, ϕ'); for all c', C' there are C, N_0); matched pairs with $q \geq N^{\phi'}$; (5.10) is assumed to fail with probability at most $N^{C'} e^{-c'\varphi^3}$; (5.11) for every even $p \leq \varphi$.
- Lemma 5.5, (5.12)–(5.15): for every C_F there are $\widetilde{\epsilon}$, C; for every $\epsilon < \widetilde{\epsilon}$ there is N_0; $F \in C^3$ with (5.12); (5.14) for all $E \in I_\epsilon$ and (5.15) for all $E_1, E_2 \in I_\epsilon$ (oriented).
- Proof of Theorem 3.6, (5.16)–(5.23): level $\zeta = 4$ with $a = 49 + 3C_s + C_*$ (printed $a \geq 12$, which is not enough at this level); (5.18) with the window $E \leq 3$ (printed $E \leq 2 + q^{-1}$) and offset K; $p = 2\lfloor \varphi/2 \rfloor$ (printed $p = \varphi$); the criterion $\operatorname{Im} m < (2N\eta)^{-1}$ for no eigenvalue near E (printed "$\ll$"); (5.22) $\varphi^{C_t/2-K-a} N^{-1/3} \leq (N\eta)^{-1} \leq \varphi^{-K-a}$ (printed $\varphi^{O(1)} N^{1/3} \leq (N\eta)^{-1} \leq \varphi^{O(1)} N^{-\phi/2}$); a grid in E; (5.23) by reflection.
- (5.24) and the last step of the proof of Theorem 3.6: (5.24) from Lemma 3.1 at $\zeta = 3$ for $\widetilde{H}$ and a grid; the last step follows [26, (5.4)–(5.15)] with explicit parameters and uses the norm bound (5.23).
- Lemma 5.6, (5.25)–(5.27), and its proof: "two generalized Wigner matrices" satisfying (3.12), (3.13), (5.24), (5.25), (5.26) and (5.12)–(5.15) → pairs of matrices with independent entries satisfying (1) $\|H\| \leq 3$, (5.24) for $\eta \leq N^{-2/3}$, (5.25) and (5.26) with high probability for every $C \geq C_0$, and (2) a comparison hypothesis with test functions with four derivatives and rate $N^{-\kappa + C'\epsilon}$; (5.27) for all $s \in \mathbb{R}$ (was $s > 0$); the proof is written in six steps, with the corrections to [26] (the sentence that lists [26, (2.19), (2.25), (2.26), (6.2), (6.3)] as the hypotheses is removed).
- Proof of Theorem 3.7 after Lemma 5.6, (5.28): the hypotheses of Lemma 5.6 are checked for H^{v} and for $\widetilde{H}$, and Lemma 3.5 is applied to $\widetilde{H}$ at $\phi = 2/5$.

Section 6.

- Beginning of Section 6, (6.1)–(6.2): the class of pairs of the section is stated (independent entries, matched moments, $|h_{ij}| \leq q^{-1}$ almost surely, $q \geq N^{\phi}$; for Theorem 3.6, $q = \min(N^{\phi}, \widetilde{q})$ and ϕ replaced by $\min(\phi, 1/4)$); "bounded support condition with $q = N^{-\phi}$" removed; all H_γ on one probability space; (6.1) for each γ with constants uniform in γ, proved from Lemma 3.1 (the argument "finitely many distributions" is removed); $\mathbf{k} \in \mathbb{R}^n \to \mathbb{N}^n$.
- Example 6.1, discussion after (6.5) and (6.7): "bounded by $N^{-\phi}$ with high probability" → "almost surely"; "the first four moments of H and $\widetilde{H}$ coincide" → "the first four moments of the off-diagonal entries coincide (in this example (a, b) is off-diagonal)".
- Example 6.1, the sum in the heuristic after (6.5): $\sum_{k=0}^{4} A_k \to \sum_{k=0}^{m} A_k$ (misprint).
- Definition 6.3, (6.13)–(6.19), and Lemma 6.4, (6.20): $k \in \mathbb{R}$, $\mathbf{k} \in \mathbb{R}^s \to \mathbb{N}$, $\mathbb{N}^s$; "Definition 6.14" → "Definition 6.3".
- Lemma 6.5, (6.21)–(6.25), and the sentence before it: $H_{\gamma'}$ any of the interpolating matrices and (a, b) arbitrary; "$s = O(\varphi)$, $\zeta = O(\varphi)$" → $1 \leq \zeta \leq \varphi^2$, $s \leq K_s \varphi^2$, $\xi = \lceil 2\zeta/\phi \rceil$, with N_0 after (C_1, ϕ, K_s); the matched order $m_0 = 4$ off the diagonal and $m_0 = 2$ on it (printed 4 and 5); real coefficients independent of z and of the indices; (6.22) in two cases ($N^{-2-|\alpha|\phi/10}$ and $N^{-1-|\alpha|\phi/4}$); (6.23) with $\widetilde{\mathcal{P}}$ and the same coefficients; the estimates hold for products with conjugate and centred factors; remainder $4^s N^{-\zeta}$.
- The iteration, (6.26)–(6.38): rewritten with words, formal sums and data d; $1 \leq \zeta \leq \varphi$, $s \leq K_s \zeta$; diagonal swaps match two moments; per-step gain $N^{-\phi/5}$, absorption (6.31) (new label) and induction (6.32); the termination "$\sum |\mathbf{k}_i| \geq 5n$" → $16 \cdot 4^s N^{-6\zeta/5} \leq N^{-\zeta}$; (6.33) is the sum form with explicit weights and (6.34) the admissible data; the max forms (6.35), (6.37) with the coefficient function (6.36) (new label), (6.38) for centred words through (6.23).
- Lemma 6.6 (new): a bound on words with explicit constants.

- Proof of Lemma 3.8, (6.39)–(6.41): the case $\eta > 10$ by the Schur complement; $\widetilde{q} \sim N^{1/2}/\log N$; the proof uses the sum form (6.33) with $\min(\phi, 1/4)$ and explicit weights (the max form does not give (3.15)); (6.39) keeps $\widetilde{q}^{-1}$.
- Proof of Lemma 5.4, (6.42)–(6.45): (5.11) is proved directly for words with conjugate factors; the max form with $s = \zeta = p$; block structure (6.44); block bound $3^{|\beta|}Q$ with $Q = X^2 + Y + 18N^{-1}$.
- Proof of Lemma 6.5, first paragraph (after (6.45)) and (6.46): the matched order m_0; the entries of $H_{\gamma'}$ and Q are bounded by q^{-1} almost surely; $\xi := \lceil 2\zeta/\phi \rceil$.
- Proof of Lemma 6.5, (6.47)–(6.56): the diagonal case of V; new paragraph on products with conjugate and centred factors (each factor expands separately, and the coefficients do not depend on the kinds of the factors); "(6.23) can be proved analogously" $\to$ the proof with $\widetilde{\mathcal{P}}$; $I^a_{b,c} \subset \mathbb{N}^a$.
- Proof of Lemma 6.5, (6.57)–(6.72): 4, 5 $\to m_0$, $m_0 + 1$ in the sums; the bad event and the independence of R and h_{ab} are written out.
- (6.74): the third summation variable $\mathbf{k_2} \to \mathbf{k_3}$; $\mathbb{E}$ inserted before $\mathcal{P}$; $k_{m+1,t} \to k_{m+1}(t)$ (misprints).
- Proof of Lemma 6.5, (6.76)–(6.79): from (6.76) to (6.21) with an explicit formula for $\mathcal{A}_\alpha$; (6.78) with the constant C_1 restored and a diagonal case; (6.22) in both cases; (6.24) and (6.25) written out.
- Proof of Lemma 5.5 (after (6.79)), (6.80)–(6.105): Lemma 6.5 and (6.38) are used at $\phi' = \min(\phi, 1/4)$; (6.81) with the constant 2; (6.83) $N^{-\min(\phi,1/3)+C\epsilon}$; in (6.85) the second term $\widetilde{G}_{ij}\widetilde{G}_{ji} \to \widetilde{G}_{ij}\overline{\widetilde{G}_{ij}}$; (6.86) with an explicit remainder; the diagonal swaps add $N \cdot N^{-1-\phi+C\epsilon}$; (6.88) $\xi = \lceil 3/\phi \rceil$; in (6.89) C does not depend on p; (6.91)–(6.94) with window $\lceil 4p/\phi' \rceil$; in (6.99) the third form $G_{i_t a}G_{bj_t} \to \widetilde{G}_{i_t a}\widetilde{G}_{bj_t}$; the extra "(5.10)" sentence removed; (6.104) $R_{ij}R_{ji} \to R_{ij}\overline{R_{ij}}$; in (6.105) $F^{(s)}(x_R)$ inside $\mathbb{E}$ and $\mathbf{k}_t \in I^2_{3/\phi,|\mathbf{k}_t|}$.

Misprints. The misprints corrected above are: $\bigcup \to \bigcap$ in (3.2), (3.3), (3.6) and (3.12); the direction of the bound on η before (3.9); $\mathbb{R} \to \mathbb{N}$ in Section 6; the absolute value in (4.22); the window $E \le 3$ in (5.18); the conjugates in (6.85) and (6.104) and the position of $\mathbb{E}$ in (6.105); the tildes in (6.99); the indices and the $\mathbb{E}$ in (6.74); and the upper limit of the sum in the heuristic of Example 6.1. The level-ζ form of (3.9), the norm bound used in the last step of the proof of Theorem 3.6, the exponent $\min(\phi, 1/4)$ in the proof of Lemma 5.5, and the products with conjugate and centred factors in Lemma 6.5 are corrections of statements or arguments, listed above.

Corrections to the cited works. The corrections to [15] and [16] found in the formalization are listed in Remark 3.9. The corrections to [26] are given in the proof of Lemma 5.6. None of them changes a statement used in this paper.

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[54] Lean 4 and Mathlib formalization of Theorem 1.2 of this paper (library `WEdgeNS`), `https://github.com/JYin80/Lean-WEdgeNS_arXiv1206.2251`.

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